



# MATHEMATICS

STANDARD SIX



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## Tele MANAS Program

**Tele MANAS (Tele Mental Health Assistance and Networking Across States)**  
**(14416 / 1800-89-14416)**

Under the National Mental Health Program, the centrally sponsored scheme Tele MANAS is being implemented in the state of Maharashtra from October 2022. Under the Tele MANAS Program, free 24 × 7 integrated and comprehensive mental health services are provided through the helpline number 14416. This includes treatment and counselling by experts for various mental problems. The information provided by you after calling the Tele MANAS helpline remains completely confidential, so students, parents and teachers should feel free to call the helpline number and tell them about their problems and doubts and get guidance on it. Tell others to take help from this facility as well.

## Manodarpan

**The Manodarpan Helpline (8448440632)** is a free telephone service that has been made available since July 2020. It is provided by the Ministry of Education through the Manodarpan Cell at the National Council of Educational Research and Training (NCERT) to offer psychological guidance and support to students, parents and teachers. This service is available every day of the week from 8:00 AM to 8:00 PM. Students, parents, and teachers can call this helpline to receive the necessary guidance and counseling.

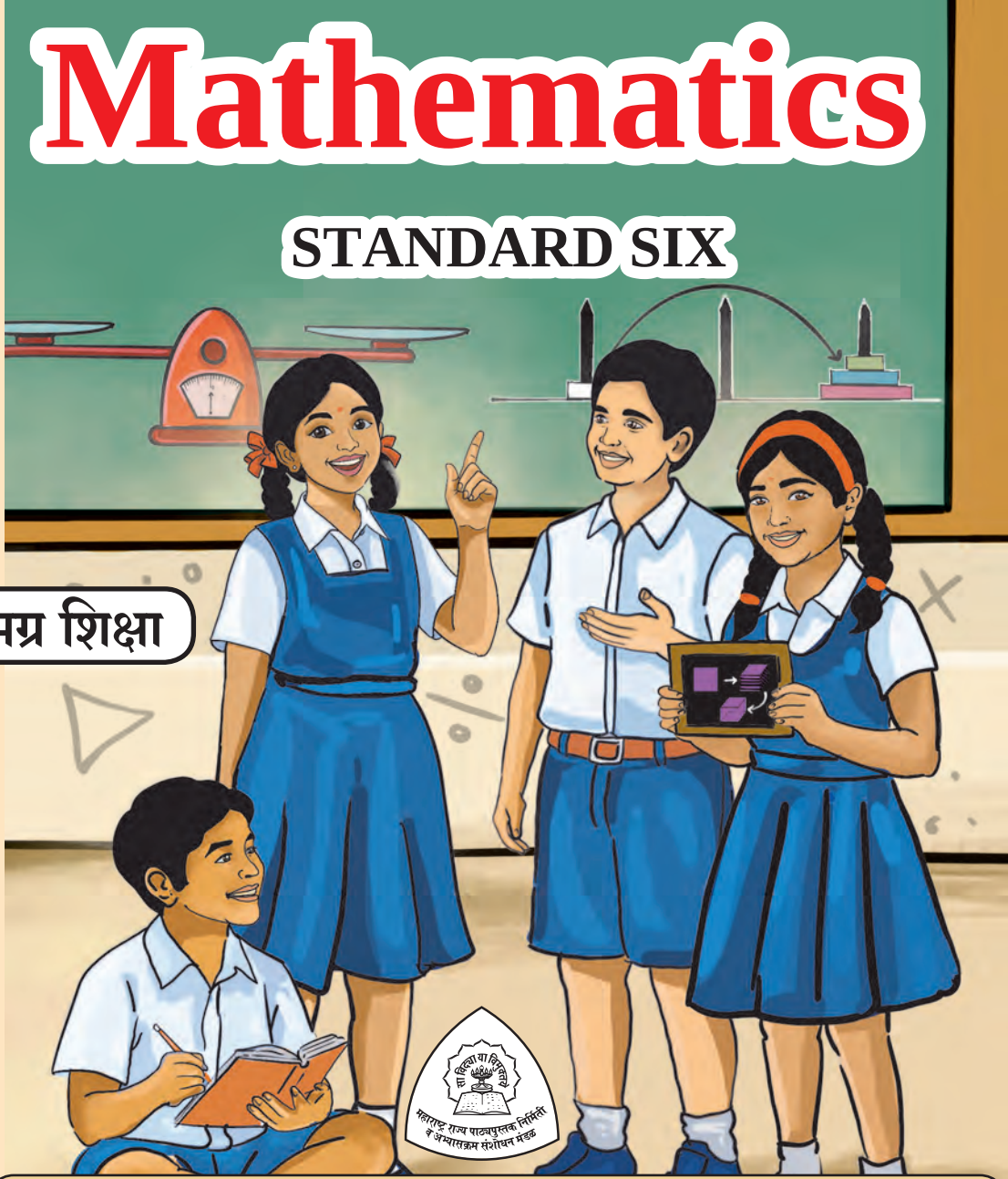


This textbook has been prepared as per the curriculum and syllabus prepared by the State Council of Educational Research and Training, Maharashtra, Pune and this textbook has been verified by them. The Co-ordination Committee constituted under Government Decision No. Abhyas-2116/(Pr.No.43/16) SD-4 dated 25.4.2016 has approved the implementation of this textbook from the academic year 2026-2027 in its meeting dated 19/05/2026.

# Mathematics

## STANDARD SIX

समग्र शिक्षा



Maharashtra State Bureau of Textbook Production and Curriculum Research, Pune 411 004.



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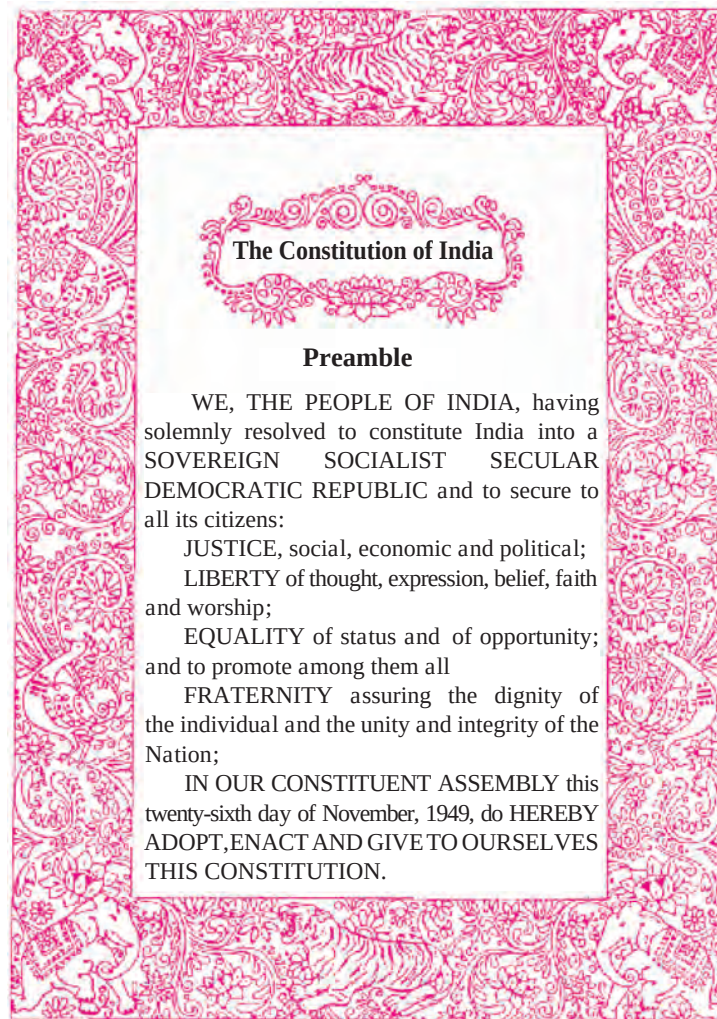
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## **The Constitution of India**

### **Part IV-A**

### **Fundamental Duties of Citizens**

#### **Article 51C**

**Fundamental Duties - It shall be the duty of every citizen of India -**

- (a) to abide by the Constitution and respect its ideals and institutions, the National Flag and the National Anthem.
- (b) to cherish and follow the noble ideals that inspired our freedom struggle.
- (c) to uphold and protect the sovereignty, unity and integrity of India.
- (d) to defend the country and render national service when called upon.
- (e) to promote harmony and the spirit of common brotherhood amongst all people, transcending religious, linguistic, regional or sectional diversities; to renounce practices derogatory to the dignity of women.
- (f) to value and preserve the rich heritage of our composite culture.
- (g) to protect and improve the natural environment including forests, lakes, rivers and wildlife, and to have compassion for living creatures.
- (h) to develop scientific temper, humanism and the spirit of inquiry and reform.
- (i) to safeguard public property and abjure violence.
- (j) to strive towards excellence in all spheres of individual and collective activity so that the nation constantly rises to higher levels of endeavour and achievement.
- (k) to provide opportunities for education to one's child or ward between the age of six and fourteen years.

## NATIONAL ANTHEM

Jana-gana-mana-adhināyaka jaya hē  
Bhārata-bhāgya-vidhātā,

Panjāba-Sindhu-Gujarāta-Marāthā  
Drāvida-Utkala-Banga

Vindhya-Himāchala-Yamunā-Gangā  
uchchala-jaladhi-taranga

Tava subha nāmē jāgē, tava subha āsisa māgē,  
gāhē tava jaya-gāthā,

Jana-gana-mangala-dāyaka jaya hē  
Bhārata-bhāgya-vidhātā,

Jaya hē, Jaya hē, Jaya hē,  
Jaya jaya jaya, jaya hē.

## PLEDGE

India is my country. All Indians  
are my brothers and sisters.

I love my country, and I am proud  
of its rich and varied heritage. I shall  
always strive to be worthy of it.

I shall give my parents, teachers  
and all elders respect, and treat  
everyone with courtesy.

To my country and my people,  
I pledge my devotion. In their  
well-being and prosperity alone lies  
my happiness.

## राज्यगीत

जय जय महाराष्ट्र माझा, गर्जा महाराष्ट्र माझा  
जय जय महाराष्ट्र माझा, गर्जा महाराष्ट्र माझा ॥१॥

भित्ती न आम्हा तुझी मुळी ही गडगडणाऱ्या नभा  
अस्मानाच्या सुलतानीला जबाब देती जिभा  
सह्याद्रीचा सिंह गर्जतो शिव शंभू राजा  
दरी दरीतून नाद गुंजला महाराष्ट्र माझा  
जय जय महाराष्ट्र माझा, गर्जा महाराष्ट्र माझा ॥१॥

काळ्या छातीवरी कोरली, अभिमानाची लेणी  
पोलादी मनगटे खेळती, खेळ जीवघेणी  
दारिद्र्याच्या उन्हात शिजला  
निढळाच्या घामाने भिजला  
देश गौरवासाठी झिजला  
दिल्लीचे ही तख्त राखितो, महाराष्ट्र माझा  
जय जय महाराष्ट्र माझा, गर्जा महाराष्ट्र माझा  
जय जय महाराष्ट्र माझा ॥२॥



## प्रस्तावना

Greetings to my students, teachers and parents,

It is with great pleasure that we present to you the mathematics textbook for Class VI prepared in accordance with the National Education Policy 2020. According to the State Curriculum Framework 2024, through Mathematics at the pre-secondary level (Class VI to VIII), students develop basic arithmetic skills along with logical thinking, creative problem solving, accurate oral and written communication. This textbook has been prepared with this idea in mind. It has tried to link mathematics with experiences from the environment as well as daily life.

It should be clear while teaching and learning what competencies students should acquire at the pre-secondary level. For this, the expected mathematical competencies and learning outcomes have been mentioned in the beginning of this textbook. The content of the textbook has been innovatively arranged in accordance with those competencies. To make it clear that mathematics is used in many places around us, examples from practice and daily life have been included in some places. To motivate students to think about the world around them and beyond, revision self-study activities based on previous knowledge, mathematical discussions between teachers and students, mathematical games for entertainment, magical mathematics and puzzles have been included in the lessons. Knowledge of mathematics at this level is the link for the secondary level, leading from concrete experiences to abstract experiences.

This textbook has been designed keeping in mind that the teaching-learning process should be student-centred, emphasis should be placed on self-study process and the learning process should be interesting and enjoyable. The concepts covered in the areas of geometry, arithmetic, number system, patterns, fractions, algebra, practical mathematics, information management are explained in simple language. At the end of each lesson, a summary of the lesson is given under the title 'At a Glance'. 'Historical references' related to the lesson are mentioned. Also, practice sets and self-study are given for practice. I. C. T. tools that will be useful for effective teaching-learning are suggested. Also, activities and projects are given according to the units of the lesson.

It is hoped that students, teachers and parents will welcome this book.



(Anuradha Oak)  
Director,

Maharashtra State Bureau  
of Textbook Production and  
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**Pune**

**Date:** 1 May 2026

**Indian Solar Date:** Vaishakh 11, Shaka 1948  
Maharashtra Din

## Curricular Goals, Competencies and Learning Outcomes

| Curriculum Objective CG 1: Understand numbers and sets of numbers (whole numbers, fractions, integers, rational numbers, and real numbers) and find patterns and understand relationships among them.                                                                           |                                                                                                                                                                                            |          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Competency                                                                                                                                                                                                                                                                      | Learning Outcome                                                                                                                                                                           | Page No. |
| C 1.1: Reads, writes, forms, compares, and estimates large whole numbers up to 20 digits. Develops understanding and ability to perform operations on these numbers. Writes the numbers in words and expresses them in scientific notation by writing them in exponential form. | Reads and writes large numbers according to the Indian numeral system.                                                                                                                     | 18,20    |
|                                                                                                                                                                                                                                                                                 | Explains the concept of counting numbers or natural numbers.                                                                                                                               | 18       |
|                                                                                                                                                                                                                                                                                 | Explains the concept and need for whole numbers.                                                                                                                                           | 18,19    |
|                                                                                                                                                                                                                                                                                 | Identifies and compares the smallness and largeness of two numbers.                                                                                                                        | 19,20    |
|                                                                                                                                                                                                                                                                                 | Compares (smallness-largeness) between two whole numbers.                                                                                                                                  | 19,20    |
|                                                                                                                                                                                                                                                                                 | Adds and subtracts the given whole numbers.                                                                                                                                                | 20       |
|                                                                                                                                                                                                                                                                                 | Creates verbal (word) examples.                                                                                                                                                            | 19       |
|                                                                                                                                                                                                                                                                                 | Explains through an example why the need for integers arose.                                                                                                                               | 20,21    |
|                                                                                                                                                                                                                                                                                 | Demonstrates where zero and negative numbers are used in everyday life.                                                                                                                    | 20,21    |
|                                                                                                                                                                                                                                                                                 | Reads and writes positive and negative numbers.                                                                                                                                            | 20       |
|                                                                                                                                                                                                                                                                                 | Classifies positive and negative numbers.                                                                                                                                                  | 21,22    |
|                                                                                                                                                                                                                                                                                 | Writes the opposite of a given number.                                                                                                                                                     | 21,22    |
|                                                                                                                                                                                                                                                                                 | Recognizes the smallness and largeness of integer numbers.                                                                                                                                 | 21       |
|                                                                                                                                                                                                                                                                                 | Adds and subtracts integer numbers.                                                                                                                                                        | 23,24    |
|                                                                                                                                                                                                                                                                                 | Finds that a fraction is part of a whole number.                                                                                                                                           | 42       |
|                                                                                                                                                                                                                                                                                 | Reads and writes the given fraction.                                                                                                                                                       | 43       |
|                                                                                                                                                                                                                                                                                 | Compares two given fractions.                                                                                                                                                              | 44,45    |
|                                                                                                                                                                                                                                                                                 | Recognizes unit fractions.                                                                                                                                                                 | 44       |
|                                                                                                                                                                                                                                                                                 | Identifies that mixed fractions have a whole number part and a fraction part.                                                                                                              | 48       |
|                                                                                                                                                                                                                                                                                 | Creates an equivalent fraction of a given fraction.                                                                                                                                        | 43,47    |
|                                                                                                                                                                                                                                                                                 | Gives a simplest form of a given fraction.                                                                                                                                                 | 43       |
|                                                                                                                                                                                                                                                                                 | Converts an improper fraction to a mixed fraction.                                                                                                                                         | 48       |
|                                                                                                                                                                                                                                                                                 | Adds and subtracts mixed fractions.                                                                                                                                                        | 48       |
|                                                                                                                                                                                                                                                                                 | Explains that fractions whose denominator is 10, 100, 1000,... are called decimal fractions.                                                                                               | 78       |
|                                                                                                                                                                                                                                                                                 | Converts a common fraction to a decimal fraction.                                                                                                                                          | 81       |
|                                                                                                                                                                                                                                                                                 | Converts a decimal fraction to a common fraction.                                                                                                                                          | 80       |
|                                                                                                                                                                                                                                                                                 | Adds and subtracts decimal fractions.                                                                                                                                                      | 81       |
| C 1.2: Discovers, identifies and extends number patterns. Describes rules in patterns (e.g., multiples of 7, powers of 3, prime numbers) and explains relations between different patterns.                                                                                     | Identifies even numbers, odd numbers, prime numbers, co-prime numbers, twin primes, triangular numbers, square numbers.                                                                    | 27,28    |
|                                                                                                                                                                                                                                                                                 | Finds patterns in number series (even, odd, prime numbers, coprime numbers, twin prime numbers, triangular numbers, square numbers). Expands, creates and arranges them in pictorial form. | 27,28    |
|                                                                                                                                                                                                                                                                                 | Writes the prime factorization of a number.                                                                                                                                                | 34       |
|                                                                                                                                                                                                                                                                                 | Explains the need for concepts GCD and LCM.                                                                                                                                                | 35,36,37 |
|                                                                                                                                                                                                                                                                                 | Calculates the GCD of given numbers by factorisation method.                                                                                                                               | 36       |
|                                                                                                                                                                                                                                                                                 | Calculates the GCD of given numbers by division method.                                                                                                                                    | 36       |
|                                                                                                                                                                                                                                                                                 | Calculates the LCM of given numbers by factorisation method.                                                                                                                               | 38       |
| C1.3 : Learns about the inclusion of zero and negative quantities as numbers, and the arithmetic operations on them, as given by Brahmagupta                                                                                                                                    | Applies GCD and LCM in particular situation.                                                                                                                                               | 37,39    |
|                                                                                                                                                                                                                                                                                 | Performs arithmetic operations on zero and negative numbers.                                                                                                                               | 21,22,23 |
|                                                                                                                                                                                                                                                                                 | Understands and uses Brahmagupta's formulation regarding zero and negative numbers.                                                                                                        | 23,24    |

|                                                                                                                                                                                                                               |                                                                                                                                                  |          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| C1.4 Explores and understands sets of numbers, such as whole numbers, fractions, integers, rational numbers, and real numbers, and their properties, and visualizes them on the number line.                                  | Shows whole numbers on a number line.                                                                                                            | 20,21    |
|                                                                                                                                                                                                                               | Analyses fractions.                                                                                                                              | 43       |
|                                                                                                                                                                                                                               | Shows fractional numbers on a number line.                                                                                                       | 45       |
|                                                                                                                                                                                                                               | Gives approximate fractions through demonstration (e.g. shows a container, pours water into it, and tells how full it is in fraction form.)      | 43       |
| C1.5 Explores the idea of percentage and applies it to solve problems.                                                                                                                                                        | Corelates percentage concept with daily transactions.                                                                                            | 90,91    |
|                                                                                                                                                                                                                               | Knows that percent is 100.                                                                                                                       | 90       |
|                                                                                                                                                                                                                               | Writes percent as a fraction.                                                                                                                    | 92       |
|                                                                                                                                                                                                                               | Write information in the form of a fraction as a percent.                                                                                        | 91       |
| C 1.6: Explores and applies fractions (both as ratios and in decimal form) in daily-life situations.                                                                                                                          | Finds where decimal fractions are used in practice.                                                                                              | 78       |
|                                                                                                                                                                                                                               | Converts a common fraction to a decimal fraction.                                                                                                | 81       |
|                                                                                                                                                                                                                               | Converts a decimal fraction to a common fraction.                                                                                                | 80       |
|                                                                                                                                                                                                                               | Writes the ratio as a fraction.                                                                                                                  | 51       |
|                                                                                                                                                                                                                               | Finds practical examples of proportion.                                                                                                          | 51       |
|                                                                                                                                                                                                                               | Writes the proportion of two numbers as a ratio.                                                                                                 | 52       |
|                                                                                                                                                                                                                               | Write a ratio in simplest form using equivalent fractions.                                                                                       | 51,52    |
|                                                                                                                                                                                                                               | When finding the ratio of two quantities of the same kind, expresses them in the same units.                                                     | 53       |
| <b>Curriculum Objective CG 2: Understands the concepts of variable, constant, coefficient, expression, and one variable equation and uses these concepts to solve meaningful daily-life problems with procedural fluency.</b> |                                                                                                                                                  |          |
| C 2.1: Understands equality between numerical expressions and learns to check arithmetical equations.                                                                                                                         | Explains the concept of equality.                                                                                                                | 67       |
|                                                                                                                                                                                                                               | Checks the properties of equality.                                                                                                               | 68       |
|                                                                                                                                                                                                                               | Creates and solves expressions using numbers and basic operations.                                                                               | 65,66    |
| C2.4: Poses and solves linear equations to find the value of an unknown, including to solve puzzles and word problems.                                                                                                        | Recognizes linear equations in one variable.                                                                                                     | 70,71    |
|                                                                                                                                                                                                                               | Formulates a linear equation in one variable.                                                                                                    | 72       |
|                                                                                                                                                                                                                               | Understand and explain that if the same operation is performed on both sides of an equation every time, the resulting equation remains balanced. | 71,72    |
|                                                                                                                                                                                                                               | Calculates the value of the variable by performing mathematical operations on both sides of the equation.                                        | 71,72,73 |
| C2.5: Develops own methods for solving puzzles and problems through algebraic thinking.                                                                                                                                       | Applies algebraic thinking to solve various problems.                                                                                            | 72,73    |
|                                                                                                                                                                                                                               | Enjoys discussing/arguing with/in pairs/groups about various methods and ways of solving puzzles/activities/problems                             | 146,147  |
| <b>Curriculum Objectives CG-3 Understands, formulates and applies properties and theorems regarding simple geometric shapes (2D and 3D).</b>                                                                                  |                                                                                                                                                  |          |
| C3.1 : Describes, classifies, and understands relationships among different types of two- and three-dimensional shapes using their defining properties/attributes                                                             | Identifies the concepts of Dimension, two-dimension and three-dimension.                                                                         | 1, 2, 3  |
|                                                                                                                                                                                                                               | Describes and classifies 2D and 3D shapes.                                                                                                       |          |
|                                                                                                                                                                                                                               | Understands the difference between prism and pyramid.                                                                                            | 7        |
|                                                                                                                                                                                                                               | Classifies prisms, identifies their names based on their types, and counts their components.                                                     | 4, 5     |
|                                                                                                                                                                                                                               | Classifies pyramids, identifies their names based on their types, and counts their components.                                                   | 5, 6     |
|                                                                                                                                                                                                                               | Identifies and describes the sphere as a 3D shape.                                                                                               | 6, 7     |



|                                                                                                                                                                                                                                                               |                                                                                                                                                                                         |            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| C 3.2: Outlines the properties of lines, angles, triangles, quadrilaterals, and polygons and applies them to solve related problems.                                                                                                                          | Identifies point, line, line segment, ray, and plane.                                                                                                                                   | 9, 11      |
|                                                                                                                                                                                                                                                               | States the concept of collinear and non-collinear points with examples.                                                                                                                 | 10         |
|                                                                                                                                                                                                                                                               | Identifies concepts of angles like zero angle, straight angle, complete angle, acute angle, obtuse angle, and right angle.                                                              | 14, 15     |
|                                                                                                                                                                                                                                                               | Reads triangle and its components.                                                                                                                                                      | 57         |
|                                                                                                                                                                                                                                                               | Identifies difference between triangle and triangular region.                                                                                                                           | 58         |
|                                                                                                                                                                                                                                                               | Identifies and state types of triangles based on sides and angles.                                                                                                                      | 59         |
|                                                                                                                                                                                                                                                               | Explains different components of a quadrilateral such as vertices, sides, angles, angles adjacent to diagonal, opposite angles, interior, exterior, rectangular region with activities. | 62, 63     |
| C 3.4: Draws and constructs geometric shapes, such as lines, parallel lines, perpendicular lines, angles, and simple triangles, with specified properties using a compass and straightedge.                                                                   | Identifies tools in a compass box. States their uses.                                                                                                                                   | 98         |
|                                                                                                                                                                                                                                                               | Measures and records length of given line segment.                                                                                                                                      | 98         |
|                                                                                                                                                                                                                                                               | Draws line segment of given length using a ruler.                                                                                                                                       | 98, 99     |
|                                                                                                                                                                                                                                                               | Draws angle of given measure using a protractor.                                                                                                                                        | 102        |
|                                                                                                                                                                                                                                                               | Measures and records given angles.                                                                                                                                                      | 107        |
|                                                                                                                                                                                                                                                               | Draws a circle using compass.                                                                                                                                                           | 98         |
|                                                                                                                                                                                                                                                               | Using compass and straight edge, draws angle of measure same as the measure of given angle.                                                                                             | 105        |
|                                                                                                                                                                                                                                                               | Using compass and straight edge, draws angle bisector.                                                                                                                                  | 105        |
|                                                                                                                                                                                                                                                               | Draws perpendicular line to a line through a point on it using a set-square.                                                                                                            | 101        |
|                                                                                                                                                                                                                                                               | Draws perpendicular lines to a line through a point on it using protractor.                                                                                                             | 102        |
|                                                                                                                                                                                                                                                               | Draws perpendicular lines to a line through a point on it using a compass.                                                                                                              | 102        |
|                                                                                                                                                                                                                                                               | Draws perpendicular lines to a line through a point outside it by folding paper.                                                                                                        | 103        |
|                                                                                                                                                                                                                                                               | Draws perpendicular lines to a line through a point outside it by using a set-square.                                                                                                   | 103        |
|                                                                                                                                                                                                                                                               | Draws perpendicular lines to a line through a point outside it by using compass and a straight edge.                                                                                    | 104        |
| <b>Curriculum Goal CG 4: Develops understanding of perimeter and area for 2D shapes and uses them to solve day-to-day life problems.</b>                                                                                                                      |                                                                                                                                                                                         |            |
| C 4.1: Discovers, understands, and uses formulae to determine the area of a square, triangle, parallelogram, and trapezium and develops strategies to find the areas of composite 2D shapes                                                                   | Finds rectangular and square shape things in the surrounding.                                                                                                                           | 109        |
|                                                                                                                                                                                                                                                               | Discovers formulas to find the area of rectangle and square.                                                                                                                            | 114, 115   |
|                                                                                                                                                                                                                                                               | Discovers formulas to find the perimeter of rectangle and square.                                                                                                                       | 108, 109   |
|                                                                                                                                                                                                                                                               | Calculates the perimeter of a triangle.                                                                                                                                                 | 110        |
|                                                                                                                                                                                                                                                               | Solves word problems based on formulas of area and perimeter of rectangle and square.                                                                                                   | 119        |
|                                                                                                                                                                                                                                                               | Derives formulas for area of right-angled triangle, acute-angled triangle, and obtuse-angled triangle from formula for area of rectangle.                                               | 115        |
|                                                                                                                                                                                                                                                               | Solves word problems using formulas for area.                                                                                                                                           | 117, 118   |
| Learns the Baudhayana-Pythagoras theorem on the lengths of the sides of a right-angled triangle, and discovers a geometric proof using areas of squares erected on the sides of the triangle, and other related geometric constructions from the Sulba-Sutras | Identifies the hypotenuse and the sides making the right angle in a right-angled triangle.                                                                                              | 122        |
|                                                                                                                                                                                                                                                               | States Baudhayana-Pythagoras theorem using lengths of sides of a right-angled triangle.                                                                                                 | 123        |
|                                                                                                                                                                                                                                                               | States the concept Pythagorean triplets based on natural numbers and obtains some Pythagorean triplets.                                                                                 | 125        |
|                                                                                                                                                                                                                                                               | Using the concept of Pythagorean triplets, decides whether a triangle is right-angled if the side lengths are given.                                                                    | 125, 126   |
| C 4.3: Constructs various designs (using tiling) on a plane surface using different 2D shapes and appreciates their appearances in art in India and around the world.                                                                                         | Makes designs of rangoli using dots.                                                                                                                                                    | 12, 112    |
|                                                                                                                                                                                                                                                               | Makes designs using two dimensional shapes.                                                                                                                                             | 12, 29, 30 |

|                                                                                                                                                                                                                                                                                                             |                                                                                                         |                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------|
| C 4.4: Develops familiarity with the notion of fractal and identifies and appreciates the appearances of fractals in nature and art in India and around the world.                                                                                                                                          | Describes the concept of fractals and identifies various Fractals occurring in the nature.              | 31             |
|                                                                                                                                                                                                                                                                                                             | Gives information about geometric shape fractals.                                                       | 29, 30         |
|                                                                                                                                                                                                                                                                                                             | Identifies fractals in 2D shapes.                                                                       | 29, 30         |
|                                                                                                                                                                                                                                                                                                             | Identifies and appreciates the appearances of fractals in nature and art in India and around the world. | 29, 30         |
|                                                                                                                                                                                                                                                                                                             | Finds various 2D shapes embedded in various fractals.                                                   | 29, 30         |
|                                                                                                                                                                                                                                                                                                             | Creates new fractals using 2D shapes.                                                                   | 29, 30         |
| <b>Curriculum Goal CG 5: Collects, organizes, represents (graphically and in tables), and interprets data/ information from daily-life experiences.</b>                                                                                                                                                     |                                                                                                         |                |
| C5.1 : Collects, organizes, and interprets the data using measures of central tendencies such as average/ mean, mode, and median.                                                                                                                                                                           | Collects statistical (Data) information.                                                                | 135            |
|                                                                                                                                                                                                                                                                                                             | Uses raw data, tally marks and frequency in statistical information.                                    | 135, 136       |
|                                                                                                                                                                                                                                                                                                             | Manages information in the form of statistical data.                                                    | 136            |
| C5.2 : Selects, creates, and uses appropriate graphical representations (e.g., pictographs, bar graphs, histograms, line graphs, and pie charts) of data to make interpretations                                                                                                                            | Represents data using pictograph.                                                                       | 138            |
|                                                                                                                                                                                                                                                                                                             | Interprets a pictograph.                                                                                | 138            |
|                                                                                                                                                                                                                                                                                                             | Observes a bar graph.                                                                                   | 138,139        |
|                                                                                                                                                                                                                                                                                                             | Represents data using bar graph.                                                                        | 139,140        |
|                                                                                                                                                                                                                                                                                                             | Interprets a bar graph.                                                                                 | 141            |
|                                                                                                                                                                                                                                                                                                             | Collects information about infographics.                                                                | 142            |
|                                                                                                                                                                                                                                                                                                             | Represents data using infographics.                                                                     | 142,143        |
|                                                                                                                                                                                                                                                                                                             | Experiences aesthetic and beauty of infographics.                                                       | 143            |
| <b>Curriculum Goal CG 6: Develops mathematical thinking and the ability to communicate mathematical ideas logically and precisely.</b>                                                                                                                                                                      |                                                                                                         |                |
| C6.1 : Applies both inductive and deductive logic to formulate definitions and conjectures, evaluate and produce convincing arguments or proofs to turn these definitions and conjectures into theorems or correct statements, particularly in the areas of algebra, elementary number theory, and geometry | Analyzes given information.                                                                             | 37,116,117,133 |
|                                                                                                                                                                                                                                                                                                             | Thinks logically about given data.                                                                      | 39, 120        |
|                                                                                                                                                                                                                                                                                                             | Puts thoughts about objects / information in own words precisely.                                       | 97             |
|                                                                                                                                                                                                                                                                                                             | Analyzes definitions in algebra, number theory and geometry.                                            | 3              |
|                                                                                                                                                                                                                                                                                                             | Puts collected information and theorem in own words accurately.                                         | 122–124        |
|                                                                                                                                                                                                                                                                                                             | Applies inductive and deductive methods to prove theorems and mathematical statements.                  | 108, 114       |
|                                                                                                                                                                                                                                                                                                             | Debates on Theorem, mathematical statements can be proved with the help of examples.                    | 125, 126       |
|                                                                                                                                                                                                                                                                                                             | Understands perspectives of other classmates by discussing with them.                                   | 88             |
| <b>Curriculum Goal CG 7: Engages with puzzles and mathematical problems and develops own creative methods and strategies to solve them.</b>                                                                                                                                                                 |                                                                                                         |                |
| C 7.1: Demonstrates creativity in discovering one's own solutions to puzzles and other problems, and appreciates the work of others in finding their own, possibly different, solutions.                                                                                                                    | Understands mathematical puzzles.                                                                       | 75             |
|                                                                                                                                                                                                                                                                                                             | Finds relationships between given elements to solve mathematical puzzles.                               |                |
|                                                                                                                                                                                                                                                                                                             | Thinks creatively to solve problems.                                                                    |                |
|                                                                                                                                                                                                                                                                                                             | Develops strategies for solving mathematical puzzles and problems.                                      | 148            |
|                                                                                                                                                                                                                                                                                                             | Encourages solutions found by others when solving mathematical puzzles.                                 | 146            |
| C7.2 : Engages in and appreciates the artistry and aesthetics of puzzle-making and puzzle-solving.                                                                                                                                                                                                          | Creates own mathematical puzzles.                                                                       | 77             |
|                                                                                                                                                                                                                                                                                                             | Solve puzzles in print and digital media.                                                               | 146            |
| <b>Course Objectives CG 8: Develops basic skills and capacities of computational thinking, namely, decomposition, pattern recognition, data representation, generalization, abstraction, and algorithms in order to solve problems where such techniques of computational thinking are effective.</b>       |                                                                                                         |                |

|                                                                                                                                                                                                                          |                                                                                                                                               |             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| C8.1 : Approaches problems using programmatic thinking techniques such as iteration, symbolic representation, and logical operations and reformulates problems into series of ordered steps (i.e., algorithmic thinking) | The student identifies and isolates the problem in the given information.                                                                     | 125,126     |
|                                                                                                                                                                                                                          | Identifies patterns.                                                                                                                          | 28          |
|                                                                                                                                                                                                                          | Represents the given information in various ways through diagrams, tables, etc.                                                               | 143         |
|                                                                                                                                                                                                                          | Explains the rules and logic in diagrams.                                                                                                     | 28,29       |
| <b>Course Objective CG 9: Knows and appreciates the development of mathematical ideas over a period and the contributions of past and modern mathematicians from India and across the world</b>                          |                                                                                                                                               |             |
| C 9.1: Recognizes how concepts (like counting numbers, whole numbers, negative numbers, rational numbers, zero, concepts of algebra, geometry) evolved over a period of time in different civilizations.                 | Study and identify counting methods from different periods in different civilizations.                                                        | 100         |
|                                                                                                                                                                                                                          | Study how concepts such as counting methods, whole numbers, negative numbers, rational numbers, zero, etc., developed.                        | 18, 21      |
|                                                                                                                                                                                                                          | Study how concepts algebra and geometry developed.                                                                                            | 10,14,15    |
| C 9.2: Knows and appreciates the contributions of specific Indian mathematicians, such as, Baudhayana, Pingala, Aryabhata, Brahmagupta, Virahanka, Bhaskara, Ramanujan and Mathematicians in states.                     | Getting introduced to Indian mathematician Baudhayana, learn about his contributions, and be proud of him.                                    | 123,124     |
|                                                                                                                                                                                                                          | Getting introduced to Indian mathematician Pingala, Aryabhata, Brahmagupta, Virahanka, Bhaskaracharya, Ramanujan, and be proud of them.       | 47, 146     |
|                                                                                                                                                                                                                          | Getting introduced to Dattatray Ramchandra Kaprekar, a mathematician from Maharashtra, and learn about his contributions and be proud of him. | 26          |
| <b>Curriculum Goal CG 10: Knows about and appreciates the interaction of Mathematics with each of their other school subjects.</b>                                                                                       |                                                                                                                                               |             |
| C10.1 : Recognizes interaction of Mathematics with multiple subjects across Science, Social Science, Visual Arts, Music, Vocational Education, and Sports                                                                | Explains the interaction of mathematics with science.                                                                                         | 11,21,67,71 |
|                                                                                                                                                                                                                          | Explains the interaction of mathematics with social science.                                                                                  | 9,12        |
|                                                                                                                                                                                                                          | Explains the interaction of mathematics with visual arts.                                                                                     | 12,15,30,31 |
|                                                                                                                                                                                                                          | Explains the interaction of mathematics with music.                                                                                           | 1           |
|                                                                                                                                                                                                                          | Explains the interaction of mathematics with into vocational education.                                                                       | 84,91,94    |
|                                                                                                                                                                                                                          | Explains the interaction of mathematics with sports.                                                                                          | 1,9,35,58   |
| <b>Curriculum Goal CG 11: To develop financial literacy to solve problems in everyday life.</b>                                                                                                                          |                                                                                                                                               |             |
| C 11.1: Understands the concepts of partnership, profit and loss and solves problems in daily life.                                                                                                                      | Identifies profit and loss in transactions.                                                                                                   | 85          |
|                                                                                                                                                                                                                          | Identifies factors that affect profit and loss.                                                                                               | 87          |
|                                                                                                                                                                                                                          | Calculates the profit and loss made in the transaction.                                                                                       | 86,87       |
| C 11.3: Solves daily life problems based on banking transactions, interest charges (simple interest, compound interest).                                                                                                 | Calculates simple interest when the principal, rate, and period are given.                                                                    | 133,134     |



### **When handling the textbook...**

- This textbook has been prepared in accordance with the National Education Policy (NEP-2020) and the State Curriculum Framework, therefore the following points should be considered while teaching or understanding the textbook.
- Use of materials from the environment and tools in the mathematical box available at the school to explain all mathematical concepts in the textbooks (units, activities, projects, demonstrations, mathematical puzzles).
- Use at your level by getting more references for new concepts.
- Headings with emojis (Revision, Understand, Think about, Mathematical Discussion, Practice Sets, Self-study, History References, At a Glance) will help to develop Conceptual Understanding, Core Elements and Values and Life Skills Development.
- The student should be encouraged to think critically about opportunities for application of acquired knowledge.
- Structured to achieve learning outcomes from each component. The instructions given in the textbook are guidelines, but flexibility is required in teaching.
- Students should develop critical thinking skills, logical abilities, and prediction skills through various mathematical dimensions (discussion, teaching, practice, games).
- Sign language and supplementary teaching methods should be used in the teaching-learning process for students with special needs. Such students should be given the opportunity to express themselves in whatever way possible.

# INDEX

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# 1 Three-Dimensional Shapes



## Revision

FOR FREE DISTRIBUTION

- \* Identify the shapes of the following objects and classify them as two-dimensional and three-dimensional shapes.



Drum



Surface of a table



Reed pipe/  
Mouthpiece



Carrom-board  
surface



Harmonium



Road



Playground



Spinning Top

| Sr.No. | Two dimensional shape | Three dimensional shape |
|--------|-----------------------|-------------------------|
| 1      | Surface of a table    |                         |
| 2      |                       |                         |
| 3      |                       |                         |
| 4      |                       |                         |

- \* Write down the names of some other shapes that you know and classify them.

## Dimensions



## Understand

- \* One Dimensional Shape (1D)

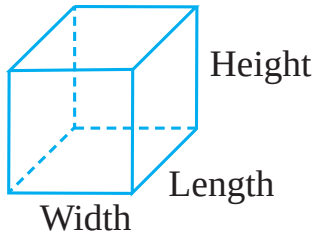


- \* Two Dimensional Shape (2D)

Two-dimensional shapes occupy space on a flat surface.



## \* Three Dimensional shape (3D)



Three-dimensional shapes occupy place in space.

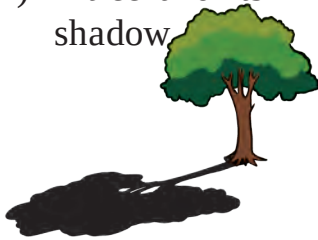
Observe two-dimensional and three-dimensional shapes through the Internet.



### Mathematical Discussion

Discuss the two-dimensional and three-dimensional shapes in the picture based on the following points.

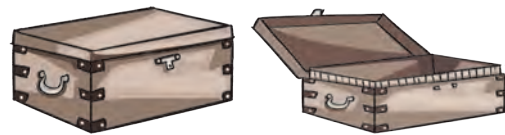
- 1) A tree and its shadow



- 2) Faces of cupboard and cupboard



- 3) Faces of trunk and trunk



### Do it.



I will fill a matchbox with wet soil and make brick out of it.

How can you create three-dimensional shapes?



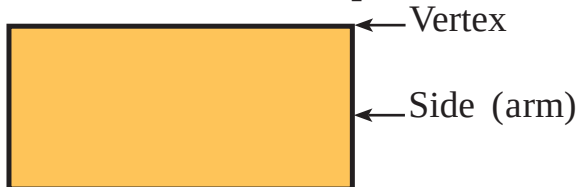
Correct.  
Is the resulting shape (Cuboid) (brick shape) three-dimensional?  
Check this

### Elements of shapes



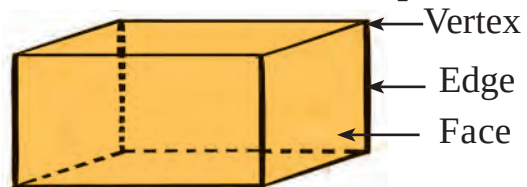
### Understand

#### Two dimensional shape



**Rectangle**

#### Three dimensional shape

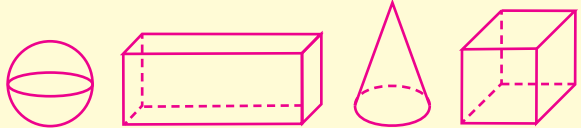


**Cuboid**

**Teacher Instructions :** Students should be given the opportunity to handle real objects to experience three-dimensional shapes.

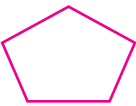
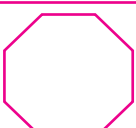




| Two dimensional shapes                                                                                                                                                                                                                                                 | Three dimensional shapes                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>* The common point of two sides is a vertex.</li> <li>* A line segment connecting two adjacent vertices of a two-dimensional shape is a side of that shape.</li> <li>* A two-dimensional shape has only one surface.</li> </ul> | <ul style="list-style-type: none"> <li>* A common point of three or more edges is a vertex.</li> <li>* In three-dimensional shape, the common side of two surfaces is called an edge.</li> <li>* Three-dimensional shape is formed by the combination of many surfaces.</li> </ul> |
|                                                                                                                                                                                       |                                                                                                                                                                                                  |

## Two Dimensional Shapes

- \* **Polygon:** A closed figure with three or more sides is a polygon.
- \* **Complete the table below.**

| Name          | Figure                                                                              | Number of Vertices | Number of Sides |
|---------------|-------------------------------------------------------------------------------------|--------------------|-----------------|
| Triangle      |    | 3                  | 3               |
| Quadrilateral |  |                    |                 |
| Pentagon      |  |                    |                 |
| Hexagon       |  |                    |                 |
| Heptagon      |  |                    |                 |
| .....         |  |                    |                 |



**Think about**

If we keep on increasing the number of sides of a polygon, what closed figure will be formed?



**Do it.**

Create various polygons using straws/sticks.



## Three Dimensional Shapes

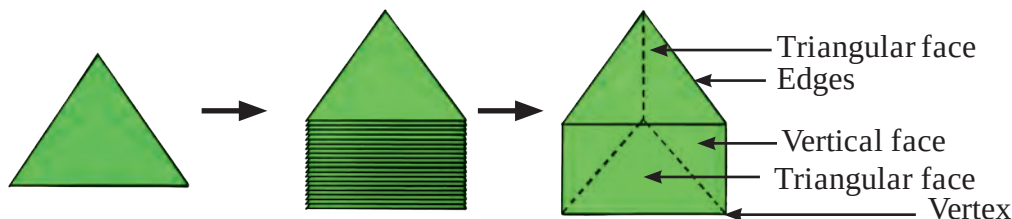
### Prism



Do it.

#### (1) Triangular Prism

- Activity 1 :** (1) Cut 10 to 15 triangles of equal size from thick cardboard.  
(2) Place the cut triangles on top of each other.  
(3) Write the elements of the three-dimensional shape formed.

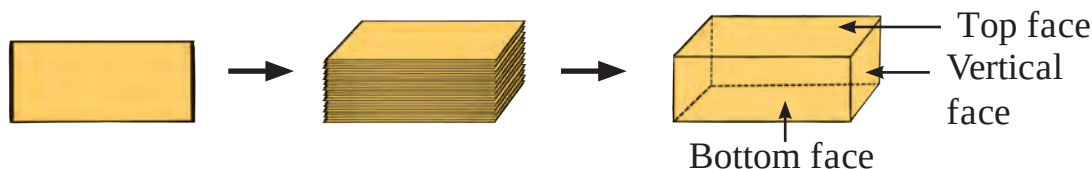


The three-dimensional figure formed is a triangular prism.

Triangular prism has  vertices,  triangular faces,  rectangular faces,  edges.

#### (2) Cuboid Or Rectangular Prism

- Activity 2 :** (1) Cut 10 to 15 rectangles of equal size from thick cardboard.  
(2) Place the cut rectangles on top of each other.  
(3) Write the elements of the three-dimensional shape formed.



The three-dimensional figure formed is a Cuboid.

Cuboid has  vertices,  rectangular faces,  edges.

#### Demonstration / Practical

As in activity No. 1 and 2 above, make a pile of different shapes like squares, pentagons, hexagons and stack them on top of each other. Record their names, vertices, faces and edges.

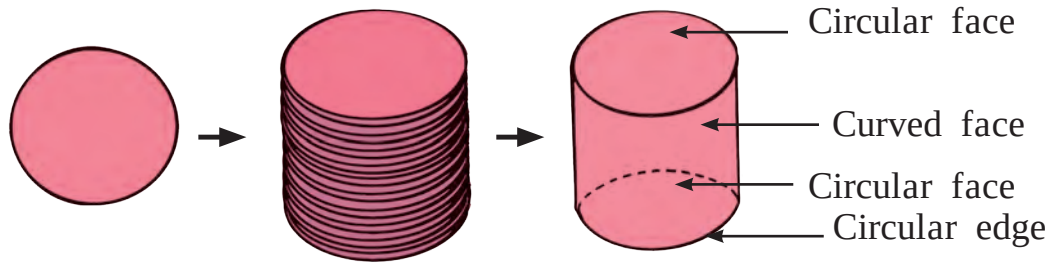
#### • Cylinder (Circular Prism)

- Activity 3:** (1) Cut 10 to 15 circles of equal radius from thick cardboard.  
(2) Place the cut circles on top of each other.

**Teacher instructions :** Have a discussion with students about how the name of the Prism is determined.



(3) Write the elements of the three-dimensional shape formed.



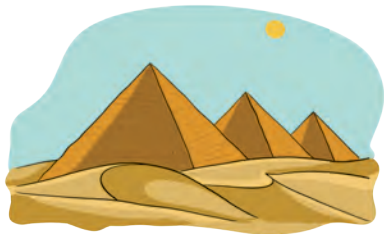
The three-dimensional figure formed is a cylinder. A cylinder has  circular faces,  curved face,  circular edges.



### Understand

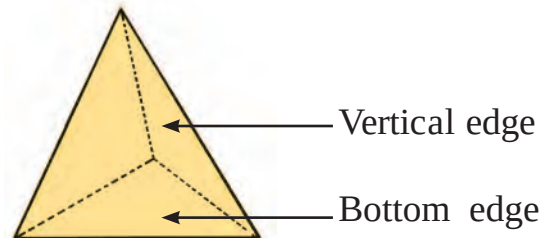
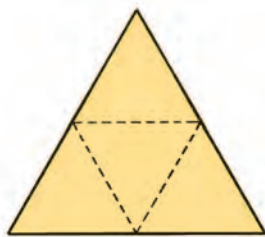
## Pyramid

Observe the pyramid and the tent shown in the figure below. This figure has a structure whose vertical faces are triangular and meet at the top. Such shapes are Pyramids.



### (1) Triangular Pyramid

The figure below shows a triangle with equal sides. The midpoints of each side are connected. The three vertices are joined by folding on the dotted line. The three-dimensional shape formed is a triangular prism.

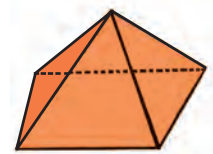
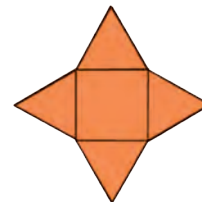


Triangular pyramid has  4 vertices,  4 triangular faces,  6 edges.

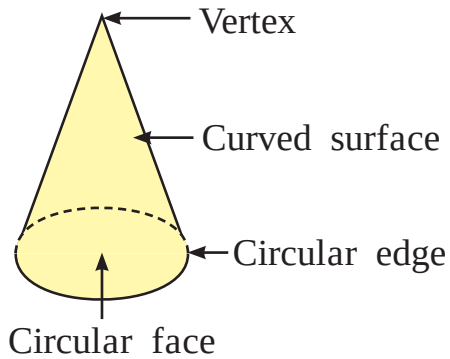
### (2) Square Pyramid

The adjacent three-dimensional shape is a square pyramid.

It has  vertices,  triangular faces,  edges,  square face



## • Cone



## Activity :

- (1) Draw a circle with a radius of 6 cm on a piece of paper.
- (2) Cut the circle with scissors. Cut out two semicircles from it.
- (3) Glue the two radii of a semicircle together.
- (4) Draw another circle with radius 3 cm on the paper. Cut it out and stick it on the circumference of the shape you have made.
- (5) The three-dimensional shape formed is a cone.
- (6) Write the elements of a cone.



Vertex



Circular edge

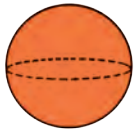


Circular face



Curved face

## • Sphere



We see shapes like ball and laddoo in our daily life. The shape of a ball is spherical. This three-dimensional figure does not have edges and vertices, but has a curved surface.



## Discuss.

- (1) How can the height of prism, pyramid, cylinder and cone be shown?
- (2) Can the height of a three-dimensional shape sphere be shown?



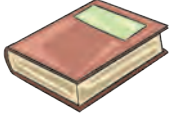
1) Complete the table below.

| Pyramid / Prism         | Triangular |            | Square |         | Pentagonal |         |
|-------------------------|------------|------------|--------|---------|------------|---------|
|                         | Prism      | Pyramid    | Prism  | Pyramid | Prism      | Pyramid |
| Picture                 |            |            |        |         |            |         |
| Bottom face             | Triangular | Triangular |        |         |            |         |
| No. of vertices         | 6          | 4          |        |         |            |         |
| No. of edges            | 9          | 6          |        |         |            |         |
| No. of vertical faces   | 3          | 3          |        |         |            |         |
| No. of bottom/top faces | 2          | 1          |        |         |            |         |





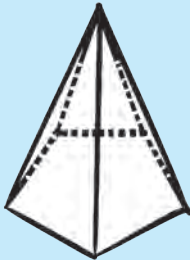
- 2) Complete the table by writing the names of the shapes that appear in the surrounding area.

|                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                     |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |  |  |
| Cone                                                                              |                                                                                   |                                                                                   |                                                                                    |                                                                                     |

## Difference between Prism and Pyramid



### Understand

|          | Prism                                                                                                            | Pyramid                                                                                         |
|----------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Faces    | The bottom face and the top face are triangular, squares shapes or polygonal.<br>Vertical faces are rectangular. | The bottom face is triangular, square shape or polygonal.<br>The vertical faces are triangular. |
| Vertices | They are on the bottom and top surfaces.                                                                         | One vertex at the top and other vertices at the bottom.                                         |
| Edges    | Three times the number of sides of top (or bottom) surface.                                                      | Twice the number of sides of the bottom surface.                                                |
| Diagram  |                               |            |



### Think about

- 1) Is cube a Prism or a Pyramid ?
- 2) In which shape are all the faces triangular in shape?
- 3) If we keep on increasing the number of sides of base polygon then which 3D shape will be formed ?





## At a glance

- A **polygon** is a closed figure with many sides.
- Dimensions: One-dimensional, two-dimensional and three-dimensional shapes.



## Exercise 1

(1) Select the correct option.

(i) What shape will be formed if carrom pieces are placed one on top of the other?

- A) Cuboid      B) Cylinder      C) Cone      D) Cube

(ii) By which 2 similar shapes is hourglass  formed ?

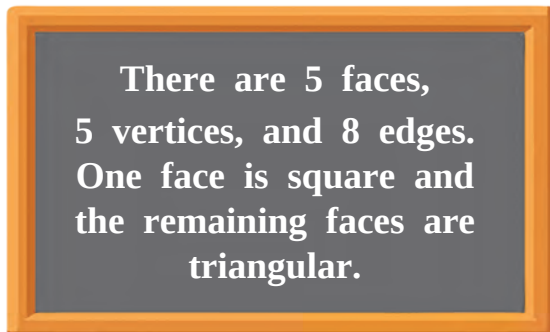
- A) Cube      B) Triangular prism      C) Sphere      D) Cone

(iii) Which of the following shapes is not three-dimensional?

- A) square      B) Sphere      C) Cylinder      D) Cone

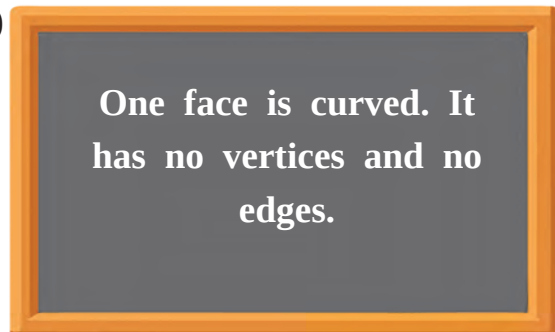
(2) Draw three-dimensional figures from the numbers given below and write the names..

(i)



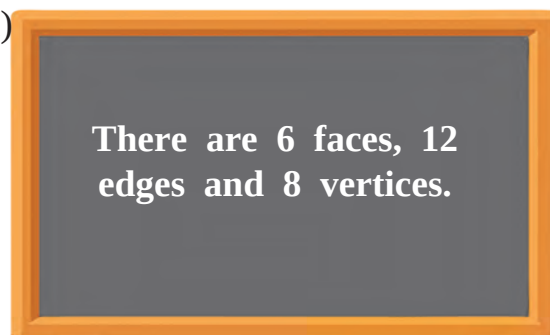
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(ii)



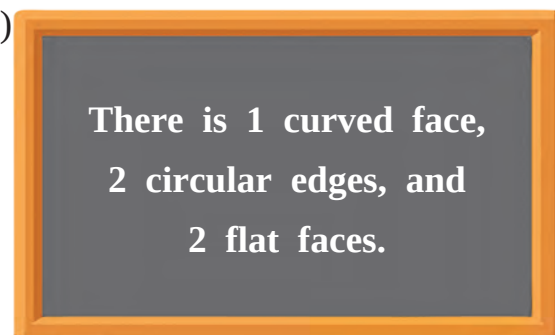
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(iii)



-----

(iv)



-----

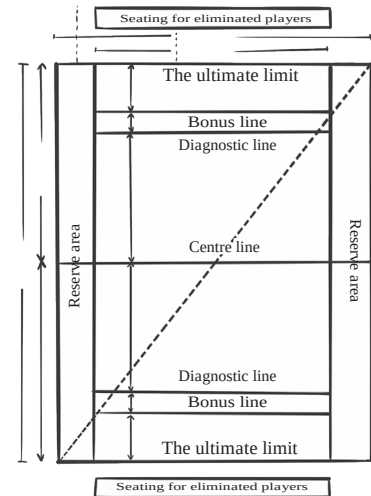




### Revision

Geometric shapes and basic concepts are used in designing any playground. The adjacent picture shows a kabaddi field being designed. Some basic geometric concepts have been used in designing it, such as point, line and plane.

Now we will study the basic concepts: a point, a line, and a plane. These elements will help us understand the design and analysis of various shapes in geometry.

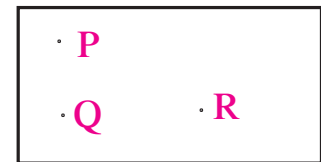


### Point



### Understand

Place a sharp pencil on the paper and make a mark. This mark is a point. A point has no length, width or thickness. Capital letters in English are used to denote Points. For example, point P, point Q, point R.



### Do it

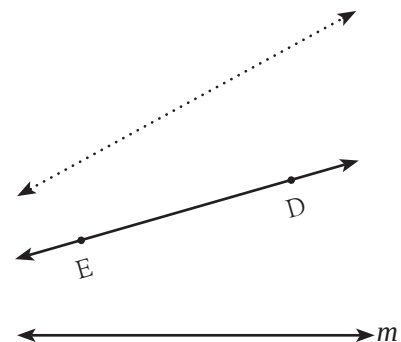
Indicate the following cities with points on the neighbouring outline map of Maharashtra.

- |             |           |
|-------------|-----------|
| 1) Pune     | 2) Mumbai |
| 3) Nashik   | 4) Nagpur |
| 5) Kolhapur | 6) Satara |



### Line

The figure below is a line. Line is infinite on both sides. To show this, arrow marks are drawn at both ends. Line is one-dimensional and there are an infinite number of points on a line. The name of a line is written by using any two points on the line. For example, the name of the line in the adjacent figure is line DE or line ED. Sometimes, the name of the line is also written using a single letter in the lowercase English script, such as line *m*, line *n*, line *a*.



## Collinear and Non-collinear points

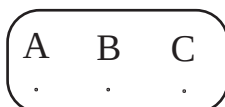


### Mathematical Discussion

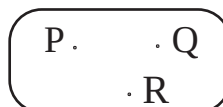
- (1) Take a point on a paper. Draw lines passing through it. How many such lines can be drawn?
- (2) Now take two points on a paper. Draw lines passing through both points. How many such lines can be drawn?
- (3) Take three points on a paper. How many lines can be drawn through all three points?

Sarita and Afroz drew those three points as follows.

**Sarita**

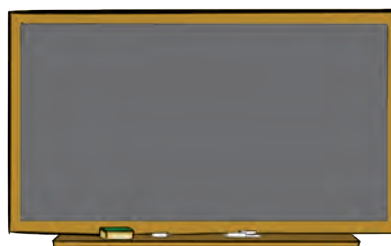
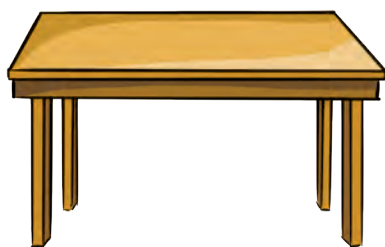


**Afroz**

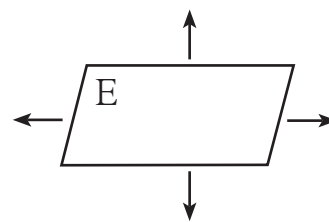


|                                                                                                                                                |                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Only one line can be drawn through all three points drawn by Sarita.</li> </ul>                         | <ul style="list-style-type: none"> <li>No line can be drawn that passes through all three points drawn by Afroz.</li> </ul>                         |
| <ul style="list-style-type: none"> <li>If a single line passes through three or more such points, they are called collinear points.</li> </ul> | <ul style="list-style-type: none"> <li>If a line cannot be drawn through three or more such points, they are called non-collinear points</li> </ul> |
| <b>Birds seated on a cable</b>                                                                                                                 |                                                                                                                                                     |
|                                                                                                                                                |                                                                                                                                                     |

### Plane



The surfaces of a table and board in the picture are flat. These flat surfaces are parts of a surface that is infinite on all sides. A flat surface that is infinite on all sides is a plane. This boundless is shown by arrows. The name of a plane is indicated by a capital letter in English, such as plane E.



**Point, line and plane are basic concepts in geometry.**



## Segment



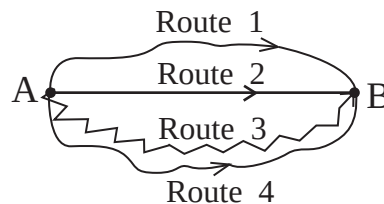
Do it.

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Draw line  $x$  on a paper. Take points  $P$  and  $Q$  on the line  $x$ . The part of the line from point  $P$  to point  $Q$  is a line segment (part of line or segment of line). Points  $P$  and  $Q$  are the endpoints of the line segment. This line segment is called line segment  $PQ$  or line segment  $QP$  and is read and written as  $\text{seg } PQ$  or  $\text{seg } QP$ .



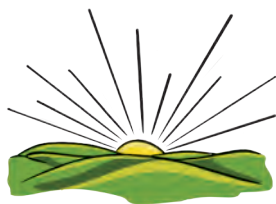
Take two points  $A$  and  $B$  on a paper. Connect them in different ways. There are many ways to get from  $A$  to  $B$ , the shortest of which (path 2) is the line segment  $AB$ .



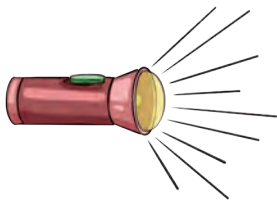
## Ray



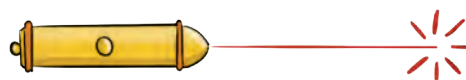
Understand



Sun rays



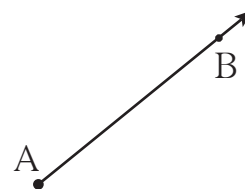
Torch light



Light coming from a laser gun

A ray is a part of a line that starts from a point and extends indefinitely in one direction. The point from which the ray starts is called the starting point of the ray.

In the adjacent diagram, point  $A$  is the starting point of ray  $AB$  and  $B$  is a point on the ray other than point  $A$ . The ray is read and written from its starting point, as ray  $AB$  and not ray  $BA$ .



(1) Look at the adjacent diagram and write the names of the terms given below.

(i) Points

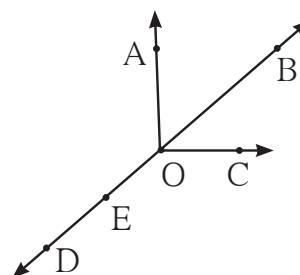
(ii) Line segments

(iii) Rays

(iv) Collinear points

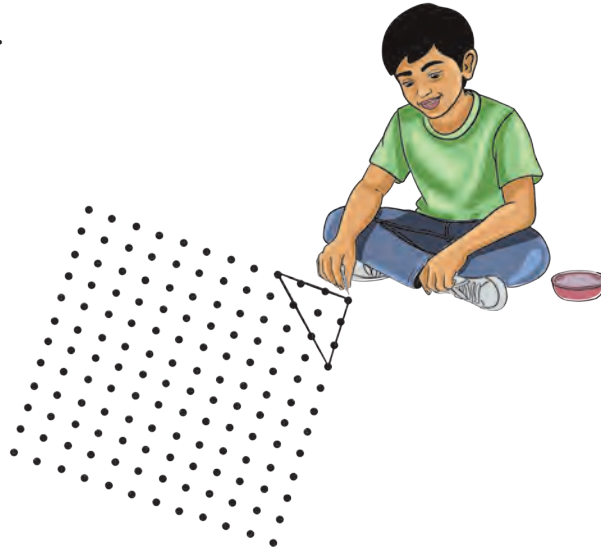
(v) Line

(vi) Non-collinear points





- (2) Draw the following figures.  
 (i) Line BT    (ii) Ray DE    (iii) Line MP    (iv) Line y    (v) Point M
- (3) Draw diagrams that illustrate the following points.  
 (i) Point P lies on line AB.  
 (ii) Points E and F are on line m. But point D is not on line m.  
 (iii) Draw ray OP and ray OQ.
- (4) Show the points, line segments and planes from the rangoli in the accompanying picture.



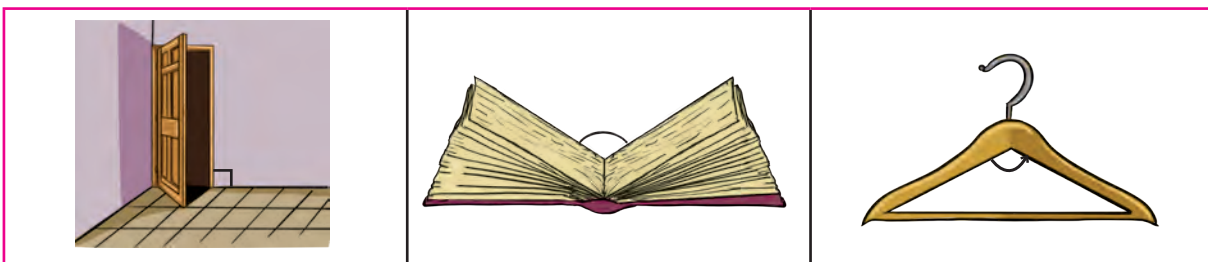
- (5) Four important places are marked with points on the map of India. Draw as many line segments as possible connecting these places.



## Angle



### Revision



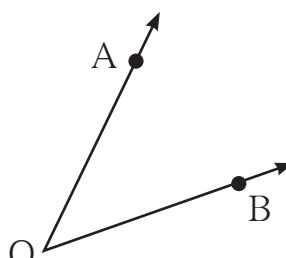
Which shape is formed at the corner in the above pictures.





## Understand

An angle is formed by two line segments that have a common point or rays that have a common starting point. These line segments or rays that form the angle are called the 'arms of the angle' or 'sides of the angle'. The common point is called the 'vertex of the angle'. In the figure, ray OA and ray OB are the sides of the angle and the point O is the vertex of the angle.



The name of the adjacent angle is read as angle AOB or angle BOA or angle O. Symbol ( $\angle$ ) is used to write the same as  $\angle AOB$  or  $\angle BOA$  or  $\angle O$ .

\* Complete the following table.

|                   | (i)                          | (ii) | (iii) | (iv) |
|-------------------|------------------------------|------|-------|------|
| Figure            |                              |      |       |      |
| Name of the angle | $\angle KYZ$ or $\angle ZYK$ |      |       |      |
| Vertex            | Y                            |      |       |      |
| Sides/ arms       | ray YK<br>ray YZ             |      |       |      |

In the adjacent figure can we write name of the angle as  $\angle B$ ?

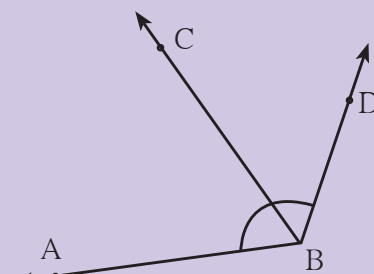
How do we know exactly which angle is  $\angle B$ ?

Because there are three rays with initial point B.

How many angles are there? Write the names of those angles.

As  $\angle ABC$ ,  $\angle$   and  $\angle$

To represent a particular angle three letters are used to write an angle. Among them, the name of the vertex is in the middle.



Which angle is larger among  $\angle ABC$  and  $\angle ABD$ ?

Which angle is larger among  $\angle CBD$  and  $\angle ABD$ ?

How will you compare these angles ?





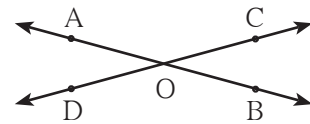
### Do it

- (1) Draw several rays from a point. Write the number of rays and the number of angles formed. Prepare a table up to 6 rays.
- (2) Measure the angles given below. Observe the angles. Record your observations considering the sides of the angles and the measures of the angles.



Does increasing or decreasing the sides of an angle change the measure of the angle?

- (3) In the adjacent figure, line AB and line DC intersect each other at point O. Write the names of all the angles formed in the figure. In the figure, which figure is formed by ray OA and ray OB? Write its name.



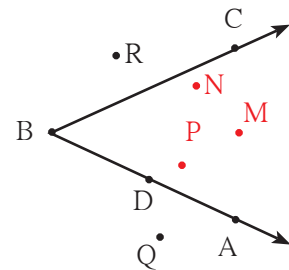
## Interior and Exterior of Angle



### Understand

In the adjacent figure, point P, point M, point N are in the interior of the angle.

Point Q and point R are points in the exterior of the angle and point A, point B, point C and point D are points on the angle.



## Types of angles



### Do it

Draw the hour hand and the minute hand in the wall clock showing time in the corresponding digital clock. Write the type of the angle formed.



1. -----



2. -----



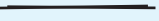
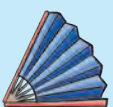
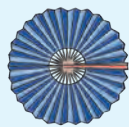
3. -----



4. -----



- What kind of angle will the hands of a clock make at 6 o'clock?  
Now to understand all types angles, let us make use of paper fan. By keeping one edge of the fan fixed, the position of the other edge, forms different types of angles.

| Picture                                                                                               | Type of angle                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Zero angle       | <ul style="list-style-type: none"> <li>When a paper fan is closed, a zero angle is formed between the two edges of the fan. The measure of a zero angle is written as <math>0^\circ</math>.</li> </ul>                                                                              |
| <br>Acute angle      | <ul style="list-style-type: none"> <li>By keeping one edge of the fan fixed and rotating the other edge, ..... angle is formed there.</li> <li>Its measure is greater than ..... and less than....</li> </ul>                                                                       |
| <br>Right angle      | <ul style="list-style-type: none"> <li>Here.... angle is formed.</li> <li>The measure of a right angle is .....</li> </ul>                                                                                                                                                          |
| <br>Obtuse angle     | <ul style="list-style-type: none"> <li>Here .....angle is formed.</li> <li>Its measure is greater than .....and less than.....</li> </ul>                                                                                                                                           |
| <br>Straight angle  | <ul style="list-style-type: none"> <li>This is a straight line formed by two edges. This is called a straight angle.</li> <li>The measure of straight angle is <math>180^\circ</math>.</li> </ul>                                                                                   |
| <br>Reflex angle   | <ul style="list-style-type: none"> <li>The angle formed in this picture is reflex angle.</li> <li>The angle formed here is larger than a straight angle.</li> <li>The measure of this angle is greater than <math>180^\circ</math> and less than <math>360^\circ</math>.</li> </ul> |
| <br>Complete angle | <ul style="list-style-type: none"> <li>In this picture, a complete fan has been formed and both the edges of the fan meet exactly. This angle is called a complete angle. The measure of a complete angle is <math>360^\circ</math>.</li> </ul>                                     |



### Mathematical Discussion

- What angle is formed between the hour hand and the minute hand of a clock when they are pointed towards with two adjacent numbers?



### Do it

- Identify the types of angles in the picture and draw a similar picture using different types of angles.
- During the drill, observe the angles formed by moving your arms up and down and write down the names of the types of angles.



## Mathematical puzzle

### I am an acute angle.

- If you double my measure, you get an acute angle.
- If you triple my measure, again you will get an acute angle.
- If you quadruple (four times) my measure, you will get an acute angle again.
- But if you multiply my measurement by 5, you will get an obtuse angle.
- What could be my measure ?



- (1) Find the given letters of English alphabets given below whose lines form right angles.  
A, E, N, K, L, H
- (2) Write the type of each angle based on the measures of the angle given below.  
(i)  $40^\circ$  (ii)  $0^\circ$  (iii)  $90^\circ$  (iv)  $155^\circ$  (v)  $180^\circ$  (vi)  $245^\circ$
- (3) Draw the reflex angles of different measures.
- (4) Answer the following questions.
  - (i) How many right angles are included in a straight angle?
  - (ii) What are the type of angles formed between two adjacent fingers of hand?
  - (iii) What kind of angle is formed between the hour hand and the minute hand of a clock when it is 12 o'clock?



### At a glance

- Point, line, and plane are basic concepts in geometry.
- A point defines a location in a plane and is represented by a capital letter in English.
- A line is infinite on both sides. The name of a line is indicated by any two points on the line, such as line AB or line BA, or by a single letter, such as line m.
- A line segment is a part of a line.
- A ray starts from a starting point and extends indefinitely in one direction.
- A plane is a surface that is infinite in all directions.
- An angle is formed by two line segments or rays that have the same starting point. These rays that form the angle are called the sides or arms of the angle and the common starting point is called the 'vertex of the angle'.
- Types of angles: zero angle ( $0^\circ$ ), acute angle (greater than  $0^\circ$  but less than  $90^\circ$ ), right angle ( $90^\circ$ ), obtuse angle (greater than  $90^\circ$  but less than  $180^\circ$ ), straight angle ( $180^\circ$ ), reflex angle (greater than  $180^\circ$  and less than  $360^\circ$ ), complete angle ( $360^\circ$ ).



### Exercise 2

- (1) Choose the correct option
  - i) A complete angle contains ..... right angles.

(a) 1

(b) 2

(c) 3

(d) 4





- ii) If a straight angle is divided into three equal parts, each resulting angle will be ....  
 (a) zero angle (b) acute angle (c) obtuse angle (d) right angle
- iii) The starting point of ray OP and ray OQ is .....  
 (a) O (b) P (c) Q (d) P and Q
- iv) ..... is infinite on both sides.  
 (a) Line (b) Line segment (c) Ray (d) Plane

(2) Match the pairs.

Group A

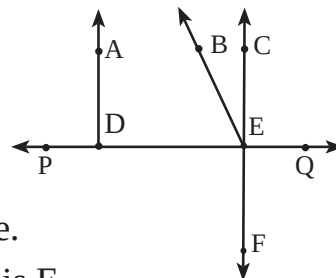
- i)  $91^\circ$   
 ii)  $180^\circ$   
 iii)  $360^\circ$   
 iv)  $215^\circ$   
 v)  $90^\circ$

Group B

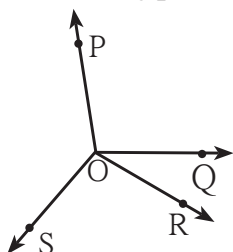
- a) Straight angle  
 b) Complete angle  
 c) Reflex angle  
 d) Right angle  
 e) Obtuse angle

(3) Write the answers by looking at the figure.

- i) Write the names of all the rays in the figure.  
 ii) On which lines the point E is located?  
 iii) Write the names of the line segments in the figure.  
 iv) Write the names of the rays whose starting point is E.  
 v) What type of angle is BEQ? Write the names of its sides.



(4)



- Write the names and types of the angles in the diagram.

(5) Find the number of acute angles in each of the following figures.



(a)



(b)



(c)

(d)

From this, draw the correct figure that comes in place (d) in the above series. How many acute angles are formed? What pattern is found in the number of acute angles?

(6) Draw a freehand picture with acute angle, right angle and obtuse angle and fill it with colours.

### Activity

List the angles in objects found in the surrounding. Classify them according to their types.

**Use of ICT:** Practice drawing figures such as points, lines, line segments, planes, rays and angles using the software GEOGEBRA. Experience the never-ending line.





## Revision

3 0 5 9 7

- Obtain the largest and smallest five-digit numbers using each digit only once.
- Write the place value of the digit in the hundred's place in each of the numbers formed.

## Natural Numbers and Whole Numbers



## Understand

- We use the numbers 1, 2, 3, ... to count objects in a group. These numbers are counting numbers. Counting numbers are also called as natural numbers.
- Natural number group : 1, 2, 3, ..., 951, ..., 29761, ...  
Natural numbers are infinite. We use large natural numbers in many places in our daily lives.

**For example** 1) According to the 2011 census, the population of a city was 1,24,42,373.

2) The number of books in a book exhibition was 23,74,333.



## Think about

1, 2, 3, ... are natural numbers.  
 $7 + 5 = 12$  is a natural number.  
 $7 - 5 = 2$  is also a natural number.  
 Is  $5 - 5 = 0$  a natural number?

- The smallest natural number is 1.
- Can you tell the largest natural number ?

## I'm zero talking ...

Dear student, do you know me? I am the zero in your daily acquaintance. I am used to show nothing. '0' is my symbol. I born in India. If I had not been discovered, there would have been a lot of confusion in the world. When there were only nine digits from one to nine, where did the word **deca** come from in the name of the decimal system? Where is this tenth digit? This tenth digit is me. Outside India, I am known by the names Void, Cipher, Zephyro Zipra. Zephyro has been transformed into Zero in English. My importance in mathematics is extremely unique. Once I am after a number, the value of that number increases tentimes.

When zero is included to the set of natural numbers it is called the set of whole numbers. The set of whole numbers is 0, 1, 2, 3, 4, .....,951, 952,.....



### Understand

- The following shows gives altitude (height) of each of five the peaks in the world in meters above sea level.  
Observe the table and write the answers to the given questions.

| Names of peaks in the world | Altitude (in meters) |
|-----------------------------|----------------------|
| K-2                         | 8611                 |
| Lao Tse                     | 8516                 |
| Mount Everest (Sagarmatha)  | 8849                 |
| Makalu                      | 8485                 |
| Kanchanjunga                | 8586                 |

- How many meters higher is the Mount Everest than the Makalu?
- Write the names of the peaks in ascending order of their height.
- Write the names of the peaks in descending order of their height.



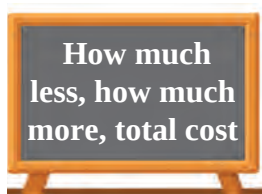
### Revision

- The smallest whole number is 0.
- Can you tell the largest whole number?



### Mathematical Discussion

- Asawari wants to buy a laptop and a mobile phone. Write down the approximate amount she will need to buy those items. Frame and solve word problems of addition and subtraction using the words written on the board.



## Practice Set 3.1

- (1) Write the following numbers in words.
  - (i) 49,38,67,521      (ii) 28,41,57,649      (iii) 2,37,50,145      (iv) 6,79,81,049
- (2) Write the appropriate symbols in the box  $<$ ,  $=$ ,  $>$ .
  - (i) 59,21,492  64,79,426      (ii) 7,579  0
  - (iii) 48,579  48,579      (iv) 0  700
- (3) Solve.
  - (i)  $95,754 - 38,254 = ?$       (ii)  $85,39,652 + 53,27,541 = ?$
- (4) The expenditure on school nutrition in the month of June was ₹ 87,000. If it became ₹ 1,69,000 in the month of July, what was the total expenditure for the months of June and July? How much more was the expenditure in the month of July than in June?
- (5) In the financial year, an organization deposited ₹ 9,25,498 in income tax in the month of February and ₹ 11,49,505 in income tax in the month of March with the Income Tax Department. How much total income tax was deposited in the two months?



### Understand

## Integers



I read in the newspaper that the temperature in the Siachen area fell to  $-40$  degrees Celsius due to snowfall. Is zero degree celsius?

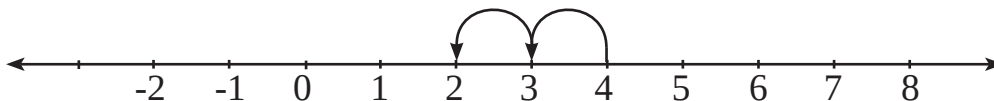
Yesterday when we went to uncle's place, uncle's car parking was two floors below the ground floor. To go there, we had to press the button written  $(-2)$  in the elevator. What do  $(-40)$  and  $(-2)$  mean in the above examples? Let's try to understand.



- i)  $4 - 2 = ?$       ii)  $2 - 4 = ?$

You know that while subtracting a number from a given number, on the number line, given number moves to the left of it by the number to be subtracted.

- i)  $4 - 2 = 2$  is a whole number.

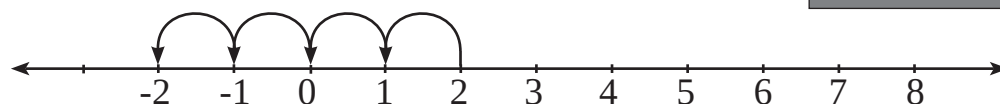


The numbers 1, 2, 3, 4.... at equal distances to the right of the number 0 on the number line. This distance remains constant. Numbers written the left of zero.  $-1$ ,  $-2$ ,  $-3$ ... are negative numbers. There is no need to write a  $(+)$  sign when writing positive numbers. But there is a need to write a  $(-)$  sign for negative numbers.



ii)  $2 - 4 = ?$

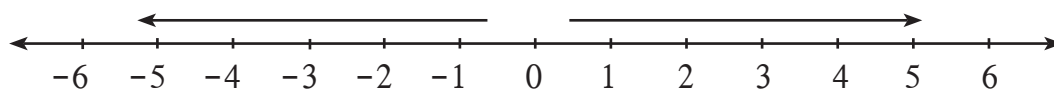
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$2 - 4 = -2$

The set of numbers formed by zero, all positive numbers and all negative numbers is called the set of integers.

Negative numbers to the left of zero    Positive number to the right of zero

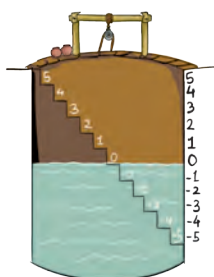


$(-3) < (-2) < (-1) < 0$

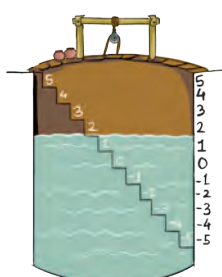
$0 < 1 < 2 < 3$

On the number line a number in the right is always greater than a number in the left.

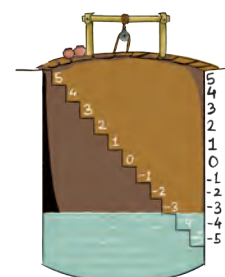
## Comparison of Integers



General condition  
water level (0)



Monsoon season  
water level (+2)



Summer seasons  
water level (-3)

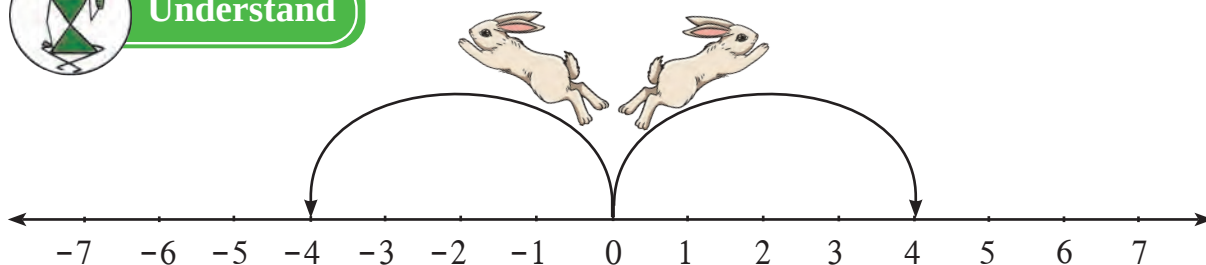
The above picture shows a well with steps and generally six steps of that well are always submerged in water. We have considered this level as zero. During the monsoon, the water level rises by two steps. The rise in water means (+2), during the summer the water level decreases and the water goes down by three steps, which means (-3).

From this example, you will notice that  $(+2) > (-3)$ ,  $0 > (-1)$ ,  $(-1) > (-2)$



Understand

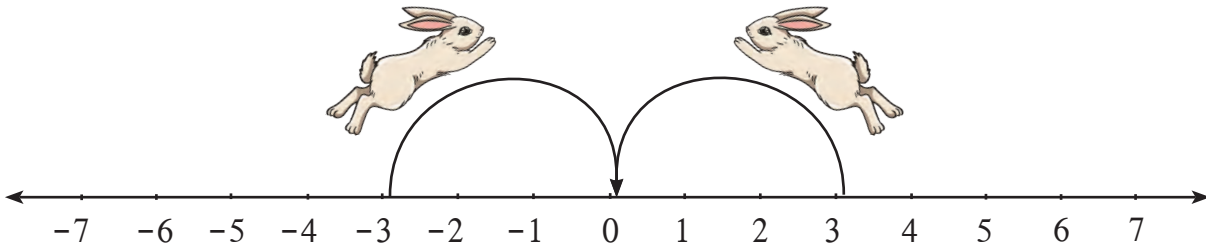
## Opposite Numbers



When the rabbit jumps 4 units to the right from 0, it goes to the number +4 and when it jumps 4 units to the left from 0, it goes to the number -4. Both the distances from 0 are the same, but the directions in which the jumps are made are opposite to each other. Here +4 and -4 are opposite numbers.







Two rabbits on a number line jumped 3 units. The rabbit standing on -3 jumped to the right and the rabbit standing on +3 jumped to the left. Where will they reach?

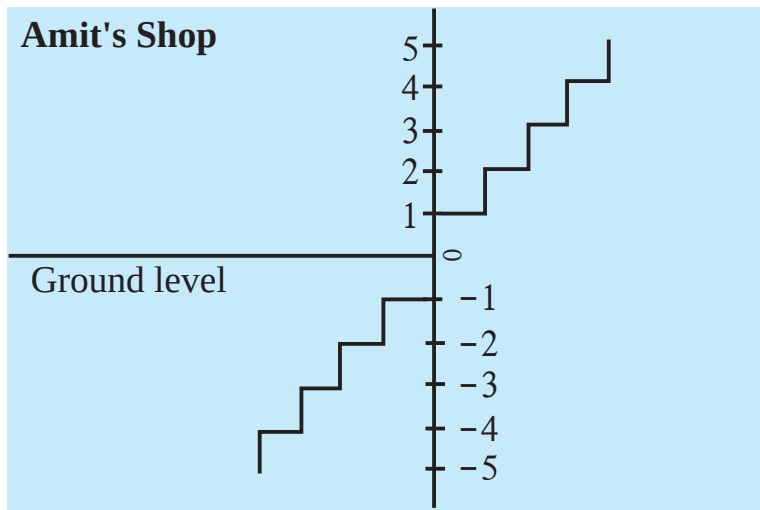
$$(-3) + (+3) = 0 \text{ or } (+3) + (-3) = 0$$

**On a number line, opposite numbers are at the same distance from 0 but in opposite directions.**

**The sum of opposite numbers is zero. Zero does not have opposite number.**



### Mathematical Discussion



In the above diagram, Amit's shop has stairs leading from the ground level to the upper floor and the basement. The stairs leading to the upper floor are shown as positive integers 1, 2, 3, ... respectively and the stairs leading to the basement are shown as negative integers -1, -2, -3, .... respectively and the ground level is shown as zero. Write the number on the step, Amit will reach in each movement, as an integer.

- 1) Amit went up three steps from ground level.
- 2) Amit came down two steps from ground level.
- 3) Amit went up three steps from ground level and then went back up two steps.
- 4) Amit went down two steps from ground level and came back down two steps.
- 5) Amit went up two steps from ground level and came down two steps again.





## Understand

### Sum of whole numbers

⊕ This disk is (+1), ⊖ this disk is (-1).

The sum of two opposite numbers is zero, means value of the pair ⊕ and ⊖ is zero.  $\oplus + \ominus = 0$  is zero pair.

|       |    |
|-------|----|
| ⊕ ⊕ ⊕ | +3 |
| ⊖ ⊖ ⊖ | -3 |

|                   |                                          |      |
|-------------------|------------------------------------------|------|
| $(+1) + (-1) = 0$ | $\cancel{\oplus} + \cancel{\ominus} = 0$ | zero |
|-------------------|------------------------------------------|------|

### Sum of integers with the same sign

1)  $(+4) + (+3) =$

|                                 |                    |
|---------------------------------|--------------------|
| ⊕ ⊕ ⊕ ⊕ + ⊕ ⊕ ⊕ = ⊕ ⊕ ⊕ ⊕ ⊕ ⊕ ⊕ | $(+4) + (+3) = +7$ |
|---------------------------------|--------------------|

2)  $(-3) + (-2) = ?$

|                         |                    |
|-------------------------|--------------------|
| ⊖ ⊖ ⊖ + ⊖ ⊖ = ⊖ ⊖ ⊖ ⊖ ⊖ | $(-3) + (-2) = -5$ |
|-------------------------|--------------------|

Rule: While adding integers with the same signs, the numbers should be added regardless of the sign and the resulting sum should have the same sign.

### Sum of integers with different signs

1)  $(+4) + (-2) = ?$

|                                                                                        |                    |
|----------------------------------------------------------------------------------------|--------------------|
| ⊕ ⊕ $\cancel{\oplus}$ $\cancel{\ominus}$ + $\cancel{\ominus}$ $\cancel{\ominus}$ = ⊕ ⊕ | $(+4) + (-2) = +2$ |
|----------------------------------------------------------------------------------------|--------------------|

2)  $(+3) + (-5) = ?$

|                                                                                                                                            |                    |
|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| $\cancel{\oplus}$ $\cancel{\oplus}$ $\cancel{\oplus}$ + $\cancel{\ominus}$ $\cancel{\ominus}$ $\cancel{\ominus}$ $\ominus$ $\ominus$ = ⊖ ⊖ | $(+3) + (-5) = -2$ |
|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------|

Rule: While adding integers with different signs, subtract the smaller number from the larger number, regardless of the signs, and give sign of larger number to the answer.

### Subtraction of integers

#### Subtraction of integers with the same sign

1)  $(+5) - (+3) = ?$

|                                                                   |                    |
|-------------------------------------------------------------------|--------------------|
| Here for this subtraction from five ⊕ disks reduce three ⊕ disks. |                    |
| ⊕ ⊕ $\cancel{\oplus}$ $\cancel{\oplus}$ $\cancel{\oplus}$ = ⊕ ⊕   | $(+5) - (+3) = +2$ |



2)  $(-5) - (-3) = ?$

Here for this subtraction from five  disks reduce three  disks

 = 

$(-5) - (-3) = -2$

From this it can be seen that subtracting the second number from the first means adding the opposite of the second number to the first number

### Subtraction of integers with different signs

1)  $(+4) - (-2)$

Subtracting  $(-2)$  from 4 means adding the opposite of  $(-2)$  to 4.

$(+4) + (+2)$

 = 

$(+4) - (-2) = 6$

2)  $(-4) - (+2)$  means  $(-4) + (-2)$

 = 

$(-4) - (+2) = -6$



(1) Write the correct symbol  $>$ ,  $<$  in the box below.

(i)  $(-10)$

(ii)  $(-19)$    $(-5)$

(iii)  $(-3)$    $0$

(iv)  $8$    $12$

(v)  $5$    $0$

(vi)  $(-3)$    $(-9)$

(2) Solve.

(i)  $4 + 7$

(ii)  $(-4) + (-5)$

(iii)  $(-7) + 2$

(iv)  $10 + (-2)$

(v)  $(-63) + (-32)$

(vi)  $(-71) + (+71)$

(3) Solve.

(i)  $78 - (-45)$

(ii)  $(-19) - (-27)$

(iii)  $148 - (-37)$

(iv)  $(-120) - 200$

(v)  $(-57) - (-48)$

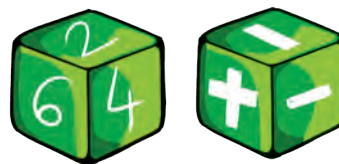
(vi)  $(-15) - (-24)$

### References in history

In ancient India, the book 'Arthashastra' written by Kautilya records negative numbers in the accounts while recording the expenditure. Even in the Bakhshali manuscripts, negative numbers were shown with a special symbol. In the book 'Brahmasphuta Siddhanta' written by Brahmagupta, the basic mathematical operations regarding positive numbers, negative numbers and zero were explained for the first time.



## Mathematical Game

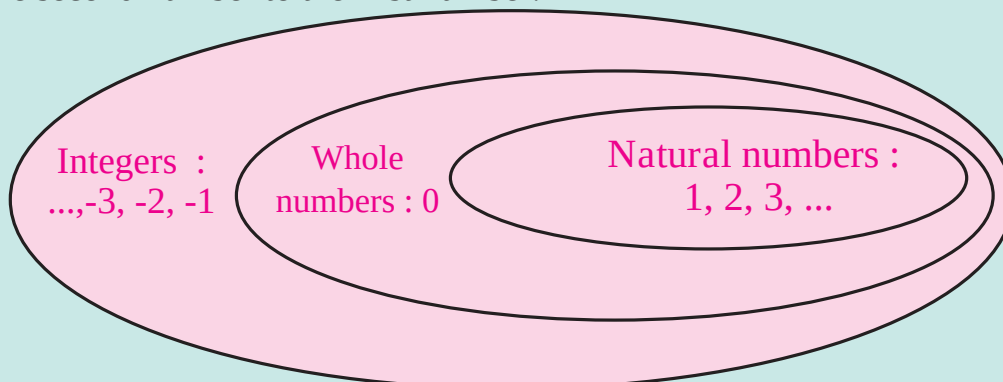


- Take a number strip (as shown) with numbers from -25 to +25.
- On one die, write numbers from 1 to 6 on its faces. On the second die, put a plus sign (+) on 3 faces and minus sign (-) on the other 3 faces.
- Players place different coloured markers (buttons, plastic pieces) on the zero position of the number track. Roll both dice at the same time and see what symbol and number appear on the faces that come up.
- Suppose if the first die shows (-) sign and the second die shows a 3, then the marker should be moved 3 units to the left of zero, assuming that the marker have a chance of winning (-3).
- If the first die shows (+) sign and the second die shows a 5, then the marker should be considered to have been awarded (+5) and the marker should be moved 5 units to the right from the zero position of the number strip.
- The player whose score goes to (-25) is eliminated and the player whose score goes to (+25) first is the winner.



### At a glance

- The set formed after adding zero to the set of natural numbers is called the set of whole numbers. 0, 1, 2, 3, ...
- The set of numbers formed by zero, all positive numbers and all negative numbers is called the set of integers.
- While adding integers with the same signs, the numbers should be added without considering the signs and the resulting sum should have the same sign.
- While adding integers with different signs, subtract the smaller number from the larger number, regardless of the signs, and give the resulting subtraction the sign of the larger number.
- The sum of two opposite numbers is always zero.
- Subtracting a number from another number means adding the opposite of the second number to the first number.





### Exercise 3

(1) Choose the correct option.

(i) -5 is .....

a) a whole number b) a natural number c) an integer d) a fraction

(ii) -5    0

a) > b) < c) = d) None of the above

(2) Write whether the following statements are true or false.

(i) Zero is the smallest natural number.

(ii) All natural numbers are whole numbers.

(iii) All whole numbers are natural numbers.

(iv) The sum of opposite numbers is zero.

(3) Write the following four numbers in ascending order.

87,29,35,104

87,39,27,401

85,93,21,793

89,51,23,307

(4) Obtain five-digit numbers using the digits 1 to 9 only once. Read them and write them in words and write the largest and smallest numbers.

(5) (i) Draw a number line on your notebook and show the numbers -5, 3, -3 and 7 on it.

(ii) What are the numbers in the right of zero called?

(iii) What are the numbers on the left of zero called?

(6) A district spent ₹ 69,27,762 in the financial year 2023-24 and ₹ 88,36,584 in the financial year 2024-25 for environmental protection. What is the total expenditure for environmental protection in these two financial years? How much more money was spent in 2024-25 compared to 2023-24?

(7) Write the numbers 1 to 9 in the given box in such a way that each number appears only once and their vertical, horizontal and diagonal sums are equal.

|   |   |   |
|---|---|---|
| 8 | 1 | 6 |
| 3 | 5 |   |
|   |   | 2 |

### Activity

**Kaprekar Constant:** Shri. Dattatreya Ramchandra Kaprekar (17/01/1905-04/07/1986) was a famous mathematician from Maharashtra. He did in valuable work for 'Digits and Numbers'. He introduced important concepts like the Kaprekar Constant. The '**Kaprekar Constant**' for four-digit numbers is '6174'. Do you know its characteristics? Find out the information about 'Demlo Number' and 'Harshad Number' discovered by Kaprekar and share it with your class.

Find out about the concepts and contribution made by Brahmagupta regarding 'Zero and Negative Numbers' using the internet or a library. Present your findings to the class.







## Revision

**\* Complete the table below.**

|                   |                                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------------|
| Even Numbers      | 2, 4, <input type="text"/> , <input type="text"/> , <input type="text"/> , <input type="text"/> .       |
| Odd Numbers       | 1, 3, 5, <input type="text"/> , <input type="text"/> , <input type="text"/> , <input type="text"/> .    |
| Prime Numbers     | 2, 3, 5, 7, <input type="text"/> , <input type="text"/> , <input type="text"/> , <input type="text"/> . |
| Composite Numbers | 4, 6, 8, 9, <input type="text"/> , <input type="text"/> , <input type="text"/> , <input type="text"/> . |

- Two prime numbers are called twin primes if difference between them is two.
- There are 8 pairs of twin prime numbers between 1 and 100
- Coprime numbers are numbers whose common divisor is 1 only.



(1) Fill in the blanks.

- ..... number is neither prime nor composite.
- ..... is the smallest prime number.
- ..... is the only even prime number.
- ..... is the smallest odd prime number.

(2) Write five pairs of coprime numbers between 1 and 50.

(3) Complete the pattern of prime numbers.

2, , 5, 7, , 13, 17, , 23, 29, , 37, , , 47.

(4) Identify the 'twin prime' numbers from the following and write a reason why they are twin prime numbers.

- |               |              |                |
|---------------|--------------|----------------|
| i) 3 and 5    | ii) 7 and 9  | iii) 11 and 13 |
| vi) 23 and 29 | v) 29 and 31 | vi) 41 and 47  |



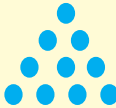
## Understand

## Pattern of Numbers

- Even numbers: 2, 4, 6, 8, 10, ...
- Odd numbers: 1, 3, 5, 7, 9, ...
- Virahanka Numbers 0, 1, 1, 2, 3, 5, 8, ...

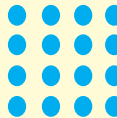
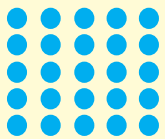


#### 4) Triangular numbers

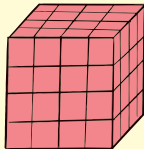
|          |                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                     |
|----------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Figure   |  |  |  |  |  |
| Number   | 1                                                                                 | 3                                                                                 | 6                                                                                 | 10                                                                                 | 15                                                                                  |
| Sum form | 1                                                                                 | $1 + 2$                                                                           | $1+2+3$                                                                           | $1+2+3+4$                                                                          | $1+2+3+4+5$                                                                         |

The sum of two consecutive triangular numbers gives the square  
(a number obtained by multiplying a number by itself) number.

#### 5) Square numbers

|                     |                                                                                   |                                                                                   |                                                                                   |                                                                                     |                                                                                     |
|---------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Figure              |  |  |  |  |  |
| Number              | 1                                                                                 | 4                                                                                 | 9                                                                                 | 16                                                                                  | 25                                                                                  |
| Multiplication form | $1 \times 1$                                                                      | $2 \times 2$                                                                      | $3 \times 3$                                                                      | $4 \times 4$                                                                        | $5 \times 5$                                                                        |

#### 6) Cube numbers

|                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |
|---------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Figure              |  |  |  |  |
| Number              | 1                                                                                   | 8                                                                                   | 27                                                                                  | 64                                                                                   |
| Multiplication form | $1 \times 1 \times 1$                                                               | $2 \times 2 \times 2$                                                               | $3 \times 3 \times 3$                                                               | $4 \times 4 \times 4$                                                                |



Do it

- With the help of dots show that the number 36 is a triangular and square number.

#### Mathematical fun

|                                             |                  |                      |       |
|---------------------------------------------|------------------|----------------------|-------|
| 1                                           | $1 \times 1$     | 1 square $(1)^2$     | 1     |
| $1 + 2 + 1$                                 | $2 \times 2$     | 2 square $(2)^2$     | 4     |
| $1 + 2 + 3 + 2 + 1$                         | $3 \times 3$     | 3 square $(3)^2$     | 9     |
| $1 + 2 + 3 + 4 + 3 + 2 + 1$                 | $4 \times 4$     | 4 square $(4)^2$     | 16    |
| $1 + 2 + 3 + 4 + 5 + 4 + 3 + 2 + 1$         | $5 \times 5$     | 5 square $(5)^2$     | 25    |
| $1 + 2 + 3 + 4 + 5 + 6 + 5 + 4 + 3 + 2 + 1$ | $6 \times 6$     | 6 square $(6)^2$     | 36    |
| $1+2+3+....+99+100+99+....+3+2+1$           | $100 \times 100$ | 100 square $(100)^2$ | 10000 |



## Practice Set 4.2

Identify the pattern in each of the number series given below and write the next three numbers.

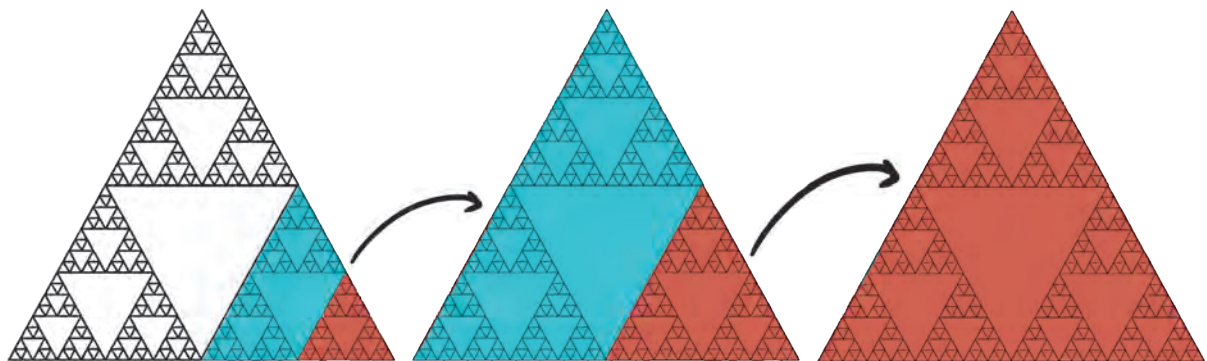
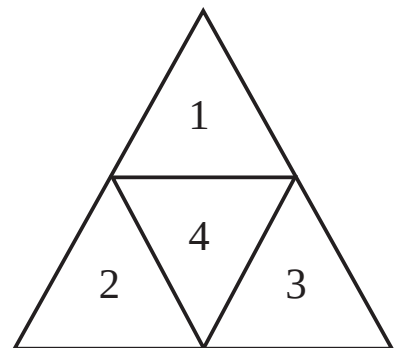
|     |                                            |
|-----|--------------------------------------------|
| 1)  | 1, 1, 1, 1, 1, 1, ..., ..., ...            |
| 2)  | 1, 2, 3, 4, 5, 6, 7, ..., ..., ...         |
| 3)  | 1, 3, 5, 7, 9, ..., ..., ...               |
| 4)  | 2, 4, 6, 8, 10, 12, ..., ..., ...          |
| 5)  | 1, 3, 6, 10, 15, 21, ..., ..., ...         |
| 6)  | 1, 4, 9, 16, 25, 36, ..., ..., ...         |
| 7)  | 1, 8, 27, 64, 125, ..., ..., ...           |
| 8)  | 0, 1, 1, 2, 3, 5, 8, 13, 21, ..., ..., ... |
| 9)  | 1, 2, 4, 8, 16, 32, 64, ..., ..., ...      |
| 10) | 1, 3, 9, 27, 81, ..., ..., ...             |



**Do it**

### Fractals

- 1) Draw a triangle with all three sides of equal length.
  - 2) Connect the midpoints of each side.
  - 3) Four equilateral triangles will be formed.
  - 4) Repeat all the steps from step 2 for the remaining triangles except triangle number 4.
- What will you see?

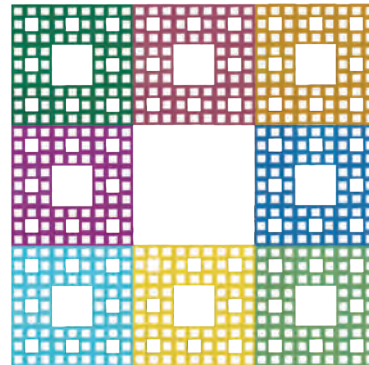


- A small part of a complete figure, no matter how small, is the same as the complete figure (self-similar).
- No matter how much you zoom in, the shape looks like the complete figure.



## Activity

- What should be the procedure for a square as shown in the adjacent figure so that when a small part of the square is zoomed, it looks like a complete square?



## Understand

### What exactly are fractals ?



Yesterday I saw a strange shape of plant besides the road.

Try putting the photo in Google Lens. Otherwise, we'll ask the teacher.

Children, this spiral structure is a fractal.



## Think about

We can draw and study regular (sides of equal length and angles of equal measure) shapes easily; but to draw irregular (non-regular) shapes in nature like; rough surfaces of planets, clouds in the sky, jagged seashores, winding rivers (meanders), mountains, forests, branches of trees, broccoli, pineapples, a new geometry needs to be studied. This geometry is called Fractals. This is natural geometry.



The Latin word "fractals" means "to break or to break into pieces". It is from this meaning that mathematician Benoit Mandelbrot coined the term "fractals" for this new geometry in 1975.



Fractals generally exist in the following forms: Branching, Geometric, Spiral, and Algebraic.



### At a glance

- Various number series such as counting numbers, even numbers, odd numbers and square numbers can be arranged in patterns.
- Sometimes numbers in a number series can be connected to each other in a specific way.
- The different shapes found in nature can be understood through fractal geometry.

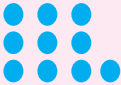
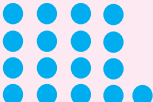




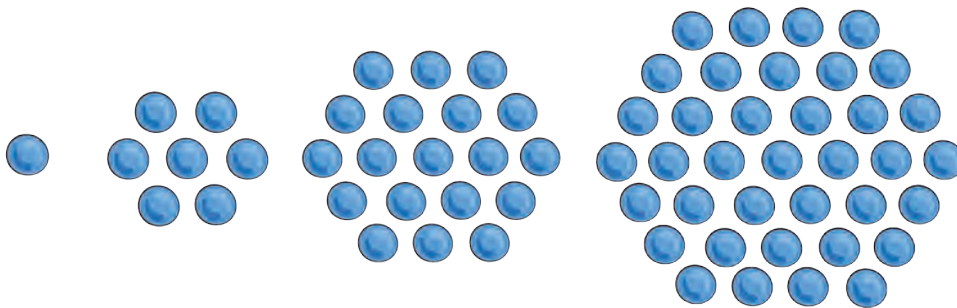


## Exercise 4

(1) Complete the table below.

|                                                                                   |                                                                                   |                                                                                   |                                                                                   |  |  |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--|--|
|  |  |  |  |  |  |
| 2                                                                                 | 5                                                                                 | 10                                                                                | 17                                                                                |  |  |

(2) The figures showing hexagonal numbers are drawn below. Draw them in your notebook. What is the next number in this figure? Show it with the help of the figure.



(3) Observe the number pattern and write the next three numbers.

2, 9, 28, 65, ..., ..., ...

(4) Write the first five triangular numbers. What is the tenth triangular number?

(5) What number series is formed if the counting numbers are added in ascending order?

(6) Write the sequence of numbers formed by adding the odd numbers starting from 1.

(Ex. Square numbers:  $1, 1+3, 1+3+5, \dots$ )

(7) What is the sum of  $1+2+3 + \dots + 50 + 49 + \dots + 3+2+1$ ?

## Activity

(1) Collect samples of various fractals found in the nature, display and discuss them in class.

(2) Find various fractals using mobile phone. Create them using different shapes and show them to the class and ask your friends to observe.

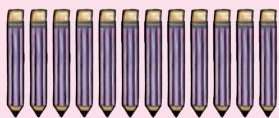




Do it.

## Divisor and Dividend

- Take 12 pencils of the same size. Divide them into equal groups.

|                                              |                                                                                     |               |
|----------------------------------------------|-------------------------------------------------------------------------------------|---------------|
| A group of 12                                |    | $12 \times 1$ |
| Division of 12 into two equal groups         |    | $6 \times 2$  |
| Division of 12 into three equal groups       |    | $4 \times 3$  |
| Division of 12 into four equal groups        |    | $3 \times 4$  |
| 12 cannot be divided into five equal groups. | cannot be divided.                                                                  | -             |
| Division of 12 into six equal groups         |  | $2 \times 6$  |
| 7 equal groups of 12                         | cannot be divided.                                                                  | -             |
| 8 equal groups of 12                         | cannot be divided.                                                                  | -             |
| 9 equal groups of 12                         | cannot be divided.                                                                  | -             |
| 10 equal groups of 12                        | cannot be divided.                                                                  | -             |
| 11 equal groups of 12                        | cannot be divided.                                                                  | -             |
| 12 equal groups of 12                        |  | $1 \times 12$ |

By observing the above table, we can divide 12 pencils equally into groups of 1, 2, 3, 4, 6 and 12. Therefore, the divisors of the number 12 are 1, 2, 3, 4, 6 and 12. The number 12 is divisible by numbers 1, 2, 3, 4, 6 and 12. **A number** that divides another number and leaves no remainder is called a divisor (or a factor). **A number that leaves no remainder when divided by another number is a multiple.**

We could not divide 12 pencils equally into groups of 5, 7, 8, 9, 10 and 11. Therefore 5, 7, 8, 9, 10 and 11 are not divisors of the number 12.



## Understand

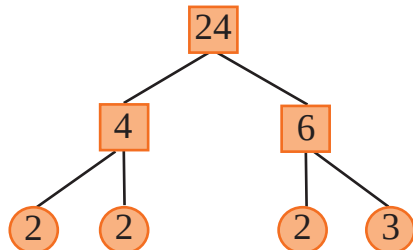
Numbers that are multiples of a number are divisible by that number.

For example, the number 12 is multiple of 2.

∴ Number 12 is divisible by number 2 and number 2 is a divisor (or factor) of number 12. Divisors of 12 are : 1, 2, 3, 4, 6, 12

## Methods of finding prime factors:

(1) Find prime factors of number 24.



### Horizontal arrangement

$$24 = 2 \times 12$$

$$= 2 \times 2 \times 6$$

$$= 2 \times 2 \times 2 \times 3$$

### Vertical arrangement

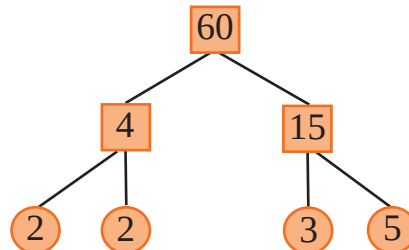
|   |    |
|---|----|
| 2 | 24 |
| 2 | 12 |
| 2 | 6  |
| 3 | 3  |
|   | 1  |

### • Prime factors of number 24 :

2 and 3.

- Number 24 is divisible by 2 and 3.
- Numbers 2 and 3 are divisors (or factors) of the number 24.

(2) Find prime factors of number 60.



### Horizontal arrangement

$$60 = 2 \times 30$$

$$= 2 \times 2 \times 15$$

$$= 2 \times 2 \times \boxed{\phantom{00}} \times \boxed{\phantom{00}}$$

### Vertical arrangement

|                                                                         |                                                                         |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------|
| 2                                                                       | 60                                                                      |
| <div style="border: 1px solid black; width: 40px; height: 20px;"></div> | 30                                                                      |
| 3                                                                       | 15                                                                      |
| 5                                                                       | <div style="border: 1px solid black; width: 40px; height: 20px;"></div> |
|                                                                         | 1                                                                       |

### • Prime factors of number 60 :

, ,

- Number 60 is divisible by 2, 3, and 5.
- Numbers 2, 3, and 5 are divisors (or factors) of number 60.



## Practice Set 5.1

(1) Write each number as product of prime factors.

(i) 69                      (ii) 27                      (iii) 78                      (iv) 126

(2) Write all divisors of each of the following numbers.

(i) 18                      (ii) 56                      (iii) 120                      (iv) 256                      (v) 67



## GCD : Greatest Common Divisor

There are 12 students in group 'A' and 16 students in group 'B'. For the exercise, separate rows are to be made with equal number of students from both the groups. What is the maximum number of students that can be lined up in each row such that no student should be left out?

Group 'A'  Group 'B' 

| Sr. No. | Group 'A'<br>(Number of students-12)                                                                    | Group 'B'<br>(Number of students-16)                                                                    | The same number of students in each row from both groups |
|---------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| 1       | <br>$2 \times 6$       | <br>$2 \times 8$       | ✓                                                        |
| 2       | <br>$3 \times 4$       | <br>$3 \times 5 + 1$   | ×                                                        |
| 3       | <br>$4 \times 3$      | <br>$4 \times 4$      | ✓                                                        |
| 4       | <br>$5 \times 2 + 2$ | <br>$5 \times 3 + 1$ | ×                                                        |
| 5       | <br>$6 \times 2$     | <br>$6 \times 2 + 4$ | ×                                                        |

By observing the above table, we were able to divide the students into groups of two and four. The largest group was formed of four students therefore, 4 is the greatest common divisor G.C.D (or H.C.F.) of 12 and 16.

### Divisor method

**Illustrative example:** Find the GCD of numbers 12 and 16.

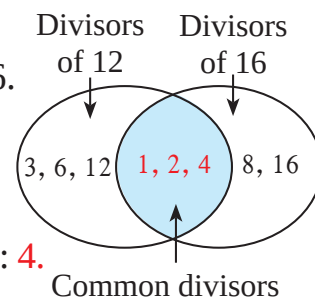
→ The divisors of number 12 : 1, 2, 3, 4, 6, 12.

Divisors of number 16 : 1, 2, 4, 8, 16

Common divisors of two numbers 12 and 16 : 1, 2, 4

The greatest common divisor of two numbers 12 and 16 : 4.

∴ The GCD of numbers 12 and 16 is 4.



## Methods of finding GCD

### Illustrative Example:

Find the GCD of numbers 24 and 36.

#### (1) Divisor Method:

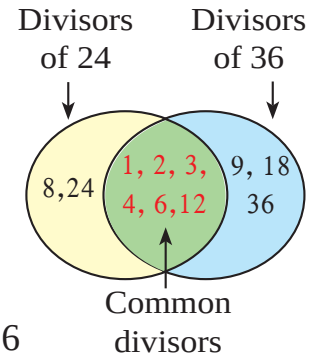
The divisors of number 1, 2, 3, 4, 6, 8, 12, 24

The divisors of the number 36: 1, 2, 3, 4, 6, 9, 12, 18, 36

The common divisors of the numbers 24 and 36 : 1, 2, 3, 4, 6, 12

The greatest common divisor is 12.

∴ The GCD of numbers 24 and 36 is 12.



#### (2) Prime factorisation Method :

$$24 = 2 \times 12$$

$$= 2 \times 2 \times 6$$

$$= 2 \times 2 \times 2 \times 3$$

$$36 = 2 \times 18$$

$$= 2 \times 2 \times 9$$

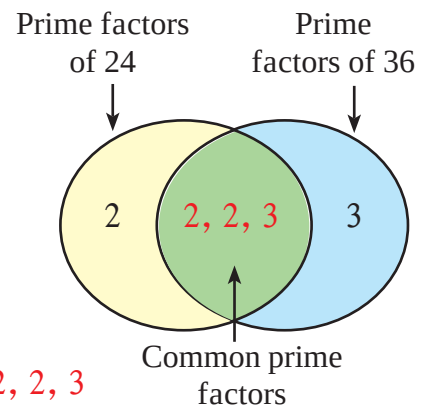
$$= 2 \times 2 \times 3 \times 3$$

Common prime factors of the numbers 24 and 36 = 2, 2, 3

G.C. D. of the numbers 24 and 36 = Product of their common factors

$$= 2 \times 2 \times 3 = 12$$

∴ The GCD of the numbers 24 and 36 is 12.



#### (3) Division method:

- While finding the GCD of two numbers, divide the larger number by the smaller number.
- If there is a remainder in the division, divide the divisor by that remainder.
- This process should continue until the remainder is zero (0).
- When the remainder is zero, the last divisor is the GCD

$$\begin{array}{r} 1 \\ 24 \overline{) 36} \\ - 24 \\ \hline 12 \end{array}$$

$$\begin{array}{r} 2 \\ 12 \overline{) 24} \\ - 24 \\ \hline 00 \end{array}$$

- Here the remainder is 0 and the divisor is 12, so the GCD of the numbers 24 and 36 is 12.





## Practice Set 5.2

- (1) Find the GCD of the following numbers.  
i) 36, 48      ii) 31, 2      iii) 65, 13      iv) 108, 132 v) 30, 24, 48
- (2) The length and breadth of a room are 15 meters and 9 meters respectively. What is the maximum length of each tile to be laid in square tiles in this room?
- (3) A shopkeeper has two rolls of cloth of length 90 m and 120 m. He wants to cut both of rolls into pieces of equal maximum length. What can be the maximum length of each piece ?
- (4) A garland seller has 48 red and 60 white roses. He wants to make separate garlands of both types of flowers. If the number of flowers in each garland is the same, then what is the maximum number of flowers in each garland?

### LCM : Least Common Multiple



#### Mathematical Discussion



**Grandfather:** Oh, Shaurya, I want to go to the doctor in the taluka for my eyes and knee pain.

**Shaurya:** Grandpa, when do you want to go?

**Grandfather:** The eye specialist doctor comes to the taluka on the 6th of every month and then every sixth day thereafter.

**Shaurya:** When does the knee specialist come?

**Grandfather:** The knee specialist comes on the 8th of every month and then every eighth day thereafter.

**Shaurya:** Do we have to go to the taluka twice?

**Grandfather:** No. Let's do both the tests on the same day.

**Shaurya:** Okay, Grandpa! On which date of this month will both the doctors be available at the earliest? Now we will see.

Write the answers by looking at the calendar provided.

- 1) On what dates of the month does the eye specialist visit?
- 2) On what dates of the month does the knee specialist visit?
- 3) If both checkups are to be done on the same day, on what date of the month will both doctors be available at the earliest?

| दिनदर्शिका |   |    |    |    |    |
|------------|---|----|----|----|----|
| र          |   | 7  | 14 | 21 | 28 |
| सो         | 1 | 8  | 15 | 22 | 29 |
| मं         | 2 | 9  | 16 | 23 | 30 |
| बु         | 3 | 10 | 17 | 24 | 31 |
| गु         | 4 | 11 | 18 | 25 |    |
| शु         | 5 | 12 | 19 | 26 |    |
| श          | 6 | 13 | 20 | 27 |    |

Here we will find the multiples of the numbers 6 and 8.

The multiples of the number 6 : 6, 12, 18, 24, 30, 36, 42, 48, 54, 60, 66, 72, 78, 84, 96...

The multiples of the number 8 : 8, 16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96...

Common multiples of the numbers 6 and 8 : 24, 48, 72, 96

The smallest (least) common multiple of the numbers 6 and 8 : 24

∴ The least common multiple (LCM) of the numbers 6 and 8 is 24.

## Methods for finding LCM

**Illustrative Example:** Find the LCM of the numbers 8, 12 and 16.

### (1) Listing method

The multiples of 8 are 8, 16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96,...

The multiples of 12 are 12, 24, 36, 48, 60, 72, 84, 96, ...

The multiples of 16 are 16, 32, 48, 64, 80, 96, ...

Common multiples of 8, 12 and 16 are 48, 96,.... etc

The smallest common multiple of the above numbers is 48

∴ The LCM of the numbers 8, 12 and 16 is 48

### (2) Prime Factorisation method:

Let's write the prime factors of numbers.

$$8 = 2 \times 4$$

$$= 2 \times 2 \times 2$$

$$12 = 2 \times 6$$

$$= 2 \times 2 \times 3$$

$$16 = 2 \times 8$$

$$= 2 \times 2 \times 4$$

$$= 2 \times 2 \times 2 \times 2$$

Common prime factors of the numbers 8, 12 and 16 : 2, 2

Common prime factor of numbers 8 and 16 : 2

Uncommon factors : 2, 3

LCM = product of common and uncommon prime factors :  $2 \times 2 \times 2 \times 2 \times 3 = 48$

∴ LCM of the numbers 8, 12 and 16 is 48

### (3) Vertical layout method

|   |   |    |    |
|---|---|----|----|
| 2 | 8 | 12 | 16 |
| 2 | 4 | 6  | 8  |
| 2 | 2 | 3  | 4  |
| 2 | 1 | 3  | 2  |
| 3 | 1 | 3  | 1  |
|   | 1 | 1  | 1  |

$$\text{GCD: } 2 \times 2 = 4$$

$$\text{LCM: } 2 \times 2 \times 2 \times 2 \times 3 = 48$$

|   |    |    |    |
|---|----|----|----|
| 3 | 15 | 25 | 45 |
| 3 | 5  | 25 | 15 |
| 5 | 5  | 25 | 5  |
| 5 | 1  | 5  | 1  |
|   | 1  | 1  | 1  |

$$\text{GCD: } 5$$

$$\text{LCM: } 5 \times 3 \times 3 \times 5 = 225$$

The LCM of two or more numbers is the product of their common and uncommon prime factors. The LCM is the smallest number that is exactly divisible by all the given numbers.



The product of two given numbers is equal to the product of their GCD and LCM.

| Find the GCD of numbers<br>18 and 30.                                  | Find the LCM of the numbers<br>18 and 30                                        |
|------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| The divisors of the number 18 are<br><b>1, 2, 3, 6</b> , 9, 18         | Multiples of 18 : 18, 36, 54, 72, <b>90</b> ,<br>108, 126, 144, 162, <b>180</b> |
| The divisors of the number 30 are<br><b>1, 2, 3, 5, 6</b> , 10, 15, 30 | Multiples of 30:<br>30, 60, <b>90</b> , 120, 150, <b>180</b> , 210              |
| Common divisors : <b>1, 2, 3, 6</b>                                    | Common multiples : <b>90, 180</b> , ...                                         |
| Greatest common divisor is <b>6</b>                                    | Least common multiple : <b>90</b>                                               |
| Therefore the GCD of 18 and 30 is 6                                    | The LCM of the 18 and 30 is 90.                                                 |

$$18 \times 30 = 540 \text{ (multiplication of both numbers)}$$

$$6 \times 90 = 540 \text{ (multiplication of GCD and LCM)}$$

$$18 \times 30 = 6 \times 90$$

**Product of given numbers = Product of GCD and LCM.**



(1) Find the LCM of the following numbers.

- i) 16, 24      ii) 30, 40      iii) 72, 96      iv) 9, 15, 20      v) 30, 45, 60

(2) Solve the following examples.

- Find the smallest three-digit number that is divisible by 5, 10 and 15.
- The product of two numbers is 360 and their GCD is 6, then what is their LCM?
- The LCM of two numbers is 200 and the GCD is 10. If one of the given numbers is 40, what is the other number?
- A farmer has 250 kg of wheat and 150 kg of jowar. Both the grains are to be filled in sacks so that their weight will be same. What is the maximum weight of grain that can be filled in each sack ?
- Three friends were practicing running on a circular track. It took them 2, 3 and 4 minutes respectively to complete one lap. If they start running from the same place at the same time, then after how many minutes will they meet again at the same place for the first time?
- If the children in the class are lined up 6 in each row or 8 in each row, then no child is left, what is the minimum possible number of children in the class?
- Three buses going to different places leave from the same station every 10, 15 and 20 minutes. If all the three buses first leave at 8 am, then at what earliest will they leave at the same time again?



## Mathematical puzzle

A box with numbers from 1 to 100 is provided.

- (1) Neha and Sharvari have blue and green triangular counters respectively. Neha placed her blue counters in those cells whose numbers are divisors of number 16. Sharvari placed her green counters in the cells whose numbers are divisors of number 24.

\* In which cell did the two counters come together?

\* Which of them is the largest number?

The divisors of the number 16 are 1,2,4,8,16 and the divisors of the number 24 are : 1,2,3,4,6,8,12,24. The common divisors of both the numbers are 1, 2, 4, 8. The largest common divisor is 8. This means that the greatest common divisor (GCD) of 16 and 24 is 8.

|    |    |    |    |    |    |    |    |    |     |
|----|----|----|----|----|----|----|----|----|-----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10  |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20  |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30  |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40  |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50  |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60  |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70  |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80  |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90  |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

- (2) Aditya and Shubhankar have red and yellow counters. Aditya arranged his red counters in the cells whose numbers are multiples of 16. Shubhankar arranged his yellow cells whose numbers are multiples of 24. (The multiples of 16 and 24 are infinite. But here we have considered numbers up to 100.)

\* In which cell did both counters first come together?

The multiples of 16 are 16, 32, 48, 64, 80, 96, ...

The multiples of 24 are 24, 48, 72, 96, ...

The common multiples of both numbers are 48,96,...

∴ LCM of the numbers 16 and 24 is 48.

Both counters first come together in the cell numbered 48.

**Teacher's Note:** Play the game of finding GCD (and LCM) of two different numbers by placing counters in the cells whose numbers are divisor (and multiples) of given numbers.





### At a glance

- Divisors: Numbers which divides given number exactly are called divisors of the given number. Divisor is also called as a factor of the given number.
- Prime factors: Factors among the divisors of a number which are prime numbers are called prime factors.
- Divisible: Numbers that are multiples of a number are divisible by that number.
- GCD : The product of all the common factors of given numbers is called their GCD.
- LCM: The LCM of given numbers is the product of their common and uncommon prime factors. LCM is the smallest number that is divisible by all the given numbers.
- The product of two numbers is always equal to the product of their GCD and LCM.



### Exercise 5

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(1) Choose the correct option.

- What is the smallest prime factor of the number 75?  
a) 1      b) 3      c) 5      d) 7
- What is the least common divisor of the numbers 17 and 18?  
a) 1      b) 2      c) 3      d) 4

(2) State True or False.

- Every prime number has only two factors.
- The product of two prime numbers is always a prime number.
- GCD is a divisor of LCM.
- LCM is greater than the given numbers or equal to the smallest number between those numbers.

(3) Express the number as a product of prime factors.

- 91      ii) 160      iii) 89

(4) Find GCD of the following numbers.

- 21, 42      ii) 17, 9      iii) 120, 180, 240

(5) Find LCM of the following numbers.

- 11, 16      ii) 7, 49      iii) 16, 48, 6

(6) Solve.

- The product of two numbers is 120. If their LCM is 60, then what is their GCD?





- ii) The GCD of a number and 16 is 4 and their LCM is 48. What is that number?
- iii) A carpenter has two wooden strips. The length of one strip is 54 inches and the length of the other strip is 72 inches. He wants to cut both the strips into equal lengths so that no wooden strip is left over. What is the maximum length of the strips that he can cut?
- iv) Samip has 84 mangoes and 105 apples. He wants to place these fruits in boxes, so that each box has the same number of fruits and each box contains only one type of fruit.
- 1) What is the maximum number of fruits he should put in a box?
  - 2) How many boxes will he need to store the fruits?

### Mathematical puzzle

#### Puzzle

|    |    |     |     |
|----|----|-----|-----|
|    |    |     | 30  |
|    |    |     | 105 |
|    |    |     | 70  |
| 12 | 75 | 245 | *   |



#### The answer to the Puzzle

|    |    |     |     |
|----|----|-----|-----|
| 2  | 3  | 5   | 30  |
| 3  | 5  | 7   | 105 |
| 2  | 5  | 7   | 70  |
| 12 | 75 | 245 | *   |

What rules could there be to solve this puzzle? Think about it.

- Solve the puzzle given below like this.

i)

|   |    |    |    |
|---|----|----|----|
|   |    |    | 8  |
|   |    |    | 18 |
|   |    |    | 30 |
| 8 | 18 | 30 | *  |

ii)

|    |    |     |    |
|----|----|-----|----|
|    |    |     | 42 |
|    |    |     | 45 |
|    |    |     | 66 |
| 12 | 27 | 385 | *  |



### Think about

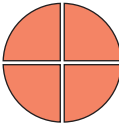
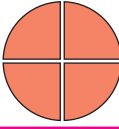
- If the GCD increases, does the LCM increase?
- How do you decide whether to calculate the GCD or the LCM when solving a verbal example?





## Revision

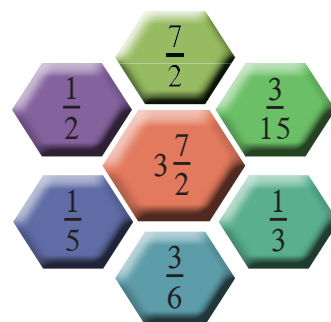
The circular nut-brittle (chikki) was cut into four equal parts. It is distributed between two boys as follows.

|                                                                                   | Ramesh                                                                                          | Suresh                                                                                            |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
|  |  $\frac{1}{4}$ |  $\frac{3}{4}$ |
|  |  $\frac{2}{4}$ |  $\frac{2}{4}$ |

Such a distribution can be expressed as a fraction (vulgar fraction or common fraction). For example,  $\frac{1}{4}$  has numerator 1 and a denominator 4. This is called a vulgar fraction.

- Classify in the fractions in the adjacent picture.

| Like fractions | Equivalent fractions | Improper fraction | Reduced form fraction (simplest form) | Mixed Fraction |
|----------------|----------------------|-------------------|---------------------------------------|----------------|
|                |                      |                   |                                       |                |



## Demonstration



Try the following experiment

**Materials:** Bucket, mug and water.

- It takes   mug water to fill the bucket completely with water.
- Garden plants were poured   mug water from the bucket full of water.

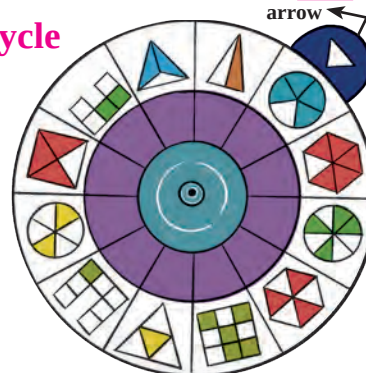
Write the remaining water in the bucket as a fraction  .



## Do it.

## Mathematical game: Fraction cycle

- Place a fraction wheel on a flat surface.
- Call students one by one to spin the fraction wheel, the wheel will stop on any one figure.
- When wheel stops, identify the fraction that represents the shaded part of the figure in front of the arrow.
- If the answer is correct, the student should be given 1 point.



5. This game should be continued by giving every student a chance.
6. In the end the student with the highest score will be declared the 'Fraction Champion'



## Understand

## Unit Fraction

- When a rectangular piece of paper is divided into four equal parts, each part  $\frac{1}{4}$  represents a fraction. In the picture, the coloured part =  $\frac{1}{4}$ , while the uncoloured part =  $\frac{3}{4}$



- When a rectangular piece of paper is divided into three equal parts, each part becomes  $\frac{1}{3}$ . In the picture, the coloured part =  $\frac{1}{3}$ . The uncoloured part =



- One of the two equal parts of a circle =

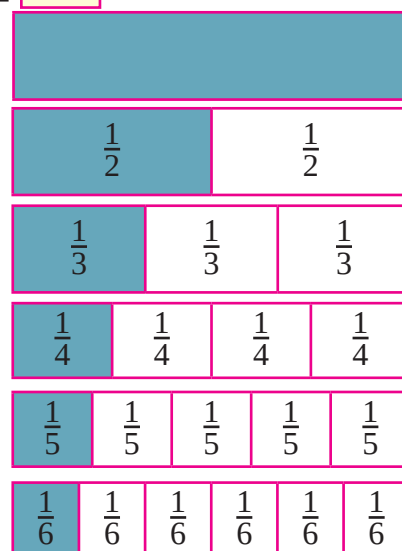


The coloured part =  The uncoloured part =



## Do it.

- Prepare 6 strips of paper of equal sizes as shown in the accompanying diagram.
- Divide the 5 strips into 2, 3, 4, 5, 6 equal parts respectively.
- Colour 1 part of each strip. Write the fraction of the coloured part.
- Observe the fractional parts shown in the figure.
- Compare fractions of coloured parts. Which is bigger/smaller



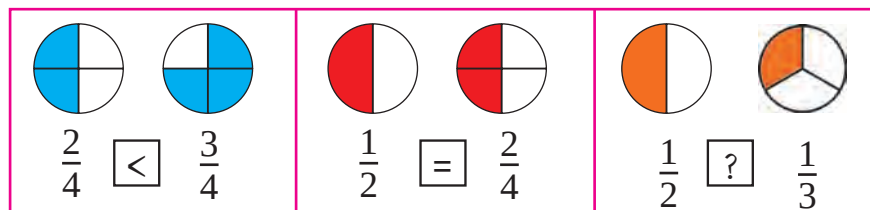
From the above activity, it is observed that,

When a whole is divided into several equal parts, each individual part is a **unit fraction**  $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}$  are all unit fractions and  $\frac{1}{2} > \frac{1}{3} > \frac{1}{4} > \frac{1}{5} > \frac{1}{6}$

## Comparison of Fractions



## Understand

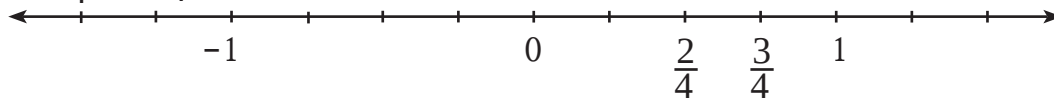


It is easy to compare from a diagram. But let's see how to compare without diagram.



Fractions can be compared on a number line.

As,  $\frac{2}{4}$  and  $\frac{3}{4}$



On number line, the number on the right side is always larger than the number of the left side, so  $\frac{2}{4} < \frac{3}{4}$

To compare fractions with different denominators on a number line, you first need to make their denominators equal.

### Illustrative Examples :

1) Which of the fractions  $\frac{1}{2}$  and  $\frac{1}{3}$  is larger?

→ To make their denominators equal, we find the LCM of denominators.

Let's Make the denominators of both fractions equal

$$\frac{1}{2} = \frac{1 \times 3}{2 \times 3} = \frac{3}{6} \text{ and } \frac{1}{3} = \frac{1 \times 2}{3 \times 2} = \frac{2}{6} \text{ But } \frac{3}{6} > \frac{2}{6} \therefore \frac{1}{2} > \frac{1}{3}$$

$\frac{1}{2}$  and  $\frac{1}{3}$  by making denominators equal, They can be shown on a numberline and compared.

2) Which of fractions  $\frac{4}{5}$  and  $\frac{7}{9}$  is larger?

→ The LCM of 5 and 9 is 45. Make the denominators of the given fractions 45.

$$\frac{4}{5} = \frac{4 \times 9}{5 \times 9} = \frac{36}{45} \text{ and } \frac{7}{9} = \frac{7 \times 5}{9 \times 5} = \frac{35}{45} \text{ But } \frac{36}{45} > \frac{35}{45} \therefore \frac{4}{5} > \frac{7}{9}$$

3) Let's compare fractions  $\frac{7}{9}$  and  $\frac{17}{21}$ .

$$\rightarrow \frac{7}{9} = \frac{7 \times \square}{9 \times \square} = \frac{49}{\square} \text{ and } \frac{17}{21} = \frac{17 \times \square}{21 \times \square} = \frac{51}{\square} \text{ But } \frac{49}{63} < \frac{51}{63} \therefore \frac{7}{9} < \frac{17}{21}$$

### Ascending and descending order of fractions

#### Illustrative Examples:

1) Arrange fractions  $\frac{5}{7}$ ,  $\frac{2}{7}$ ,  $\frac{8}{7}$  in ascending and descending order.

→ All the above fractions are like fractions.

Therefore, while determining their order, the order should be determined by comparing the numbers in the numerator ?

$$\text{Ascending order : } \frac{2}{7}, \frac{5}{7}, \frac{8}{7} \quad \text{Descending order : } \frac{8}{7}, \frac{5}{7}, \frac{2}{7}$$

2) Arrange the fractions  $\frac{5}{2}$ ,  $\frac{5}{3}$ ,  $\frac{5}{4}$  in ascending and descending order

→ All the above fractions are unlike fractions. Find the LCM of 2, 3, 4. The LCM of these denominators is 12. Let's make denominators of these fractions 12.

$$\frac{5}{2} = \frac{5 \times 6}{2 \times 6} = \frac{30}{12}, \quad \frac{5}{3} = \frac{5 \times 4}{3 \times 4} = \frac{20}{12}, \quad \frac{5}{4} = \frac{5 \times 3}{4 \times 3} = \frac{15}{12}$$



Ascending order :  $\frac{15}{12}, \frac{20}{12}, \frac{30}{12}$  Descending order :  $\frac{30}{12}, \frac{20}{12}, \frac{15}{12}$

$\therefore$  Ascending order :  $\frac{5}{4}, \frac{5}{3}, \frac{5}{2}$  Descending order :  $\frac{5}{2}, \frac{5}{3}, \frac{5}{4}$

3) Arrange the fractions  $\frac{2}{3}, \frac{1}{6}, \frac{5}{9}$  in ascending and descending order.

→ All the above fractions are unlike fractions. Find the LCM of 3, 6, 9. The LCM of these denominators is 18.

$$\frac{2}{3} = \frac{2 \times 6}{3 \times 6} = \frac{12}{18}, \quad \frac{1}{6} = \frac{1 \times 3}{6 \times 3} = \frac{3}{18}, \quad \frac{5}{9} = \frac{5 \times 2}{9 \times 2} = \frac{10}{18}$$

Ascending order :  $\frac{3}{18}, \frac{10}{18}, \frac{12}{18}$  Descending order :  $\frac{12}{18}, \frac{10}{18}, \frac{3}{18}$

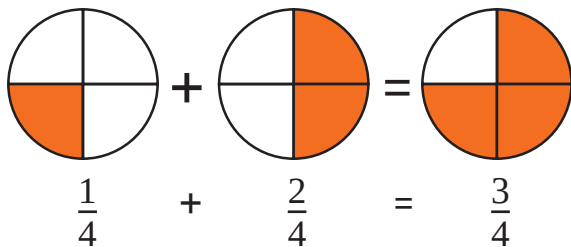
$\therefore$  Ascending order :  $\frac{1}{6}, \frac{5}{9}, \frac{2}{3}$  Descending order :  $\frac{2}{3}, \frac{5}{9}, \frac{1}{6}$

## Addition and Subtraction of fractions

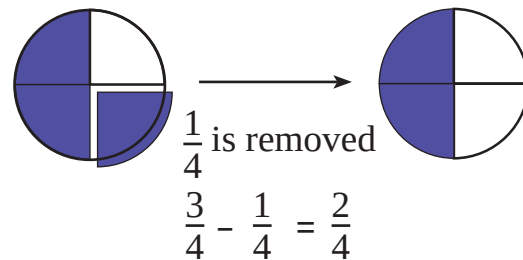


### Understand

#### Addition of like fractions

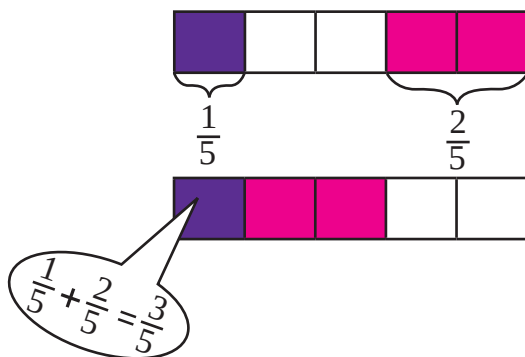


#### Subtraction of like fractions



#### Illustrative Examples

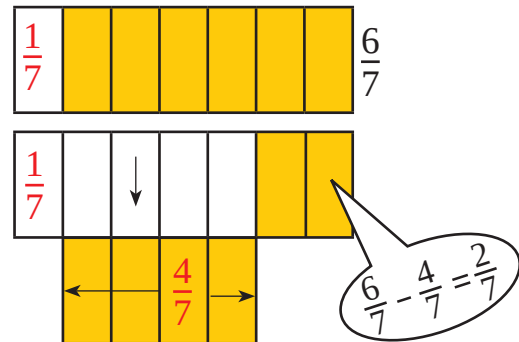
(1)  $\frac{1}{5} + \frac{2}{5} = \frac{1+2}{5} = \frac{3}{5}$



(2)  $\frac{4}{7} + \frac{6}{7} = \frac{4+6}{7} = \frac{10}{7}$

#### Illustrative Examples

(1)  $\frac{6}{7} - \frac{4}{7} = \frac{6-4}{7} = \frac{2}{7}$



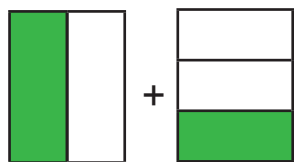
(2)  $\frac{3}{5} - \frac{1}{5} = \frac{3-1}{5} = \frac{2}{5}$



## Addition of unlike fractions

### Illustrative Examples

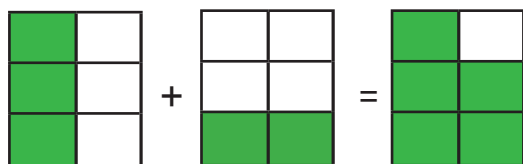
1)  $\frac{1}{2} + \frac{1}{3} = ?$



$$\frac{1}{2} + \frac{1}{3}$$

The LCM of 2 and 3 is 6.

To make denominators of fractions 6 both diagrams are divided into 6 equal parts.



$$\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$$

$$\therefore \frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$$

## Subtraction of unlike fractions

### Illustrative Examples

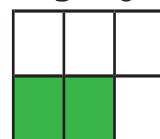
1)  $\frac{1}{2} - \frac{1}{3} = ?$

The LCM of 2 and 3 is 6.

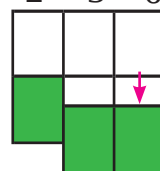
To make denominators of fractions 6 both diagrams are divided into 6 equal parts.

$$\frac{1}{2} = \frac{3}{6}$$

$$\frac{1}{3} = \frac{2}{6}$$



$$\frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$



$$\therefore \frac{1}{2} - \frac{1}{3} = \frac{3}{6} - \frac{2}{6} = \frac{1}{6}$$

2)  $\frac{1}{4} + \frac{2}{3} = ?$

The LCM of 4 and 3 is 12.

Make denominators of both fractions 12

$$\frac{1 \times 3}{4 \times 3} + \frac{2 \times 4}{3 \times 4} = \frac{3}{12} + \frac{8}{12} = \frac{3+8}{12} = \frac{11}{12}$$

2)  $\frac{3}{4} - \frac{2}{3} = ?$

The LCM of 4 and 3 is 12. Make denominators of both fractions 12

$$\frac{1 \times 3}{4 \times 3} - \frac{2 \times 4}{3 \times 4} = \frac{9}{12} - \frac{8}{12} = \frac{9-8}{12} = \frac{1}{12}$$

## References from History

### Brahmagupta's method for adding and subtracting fractions.

While adding or subtracting fractions -

- \* Find the LCM of the denominators of the fractions.
- \* Make denominators of fractions same as LCM and write the equivalent fraction of each fraction.
- \* Add or subtract the equivalent fractions obtained. To do this, keep the denominator same and simply add or subtract the numerators.



## Addition of mixed fractions

### Illustrative Examples :

$$\begin{aligned} 1) \quad & 3\frac{3}{4} + 2\frac{3}{5} \\ &= \frac{3 \times 4 + 3}{4} + \frac{2 \times 5 + 3}{5} \\ &= \frac{12 + 3}{4} + \frac{10 + 3}{5} = \frac{15}{4} + \frac{13}{5} \\ & \text{(The LCM of 4 and 5 is 20.)} \\ &= \frac{15 \times 5}{4 \times 5} + \frac{13 \times 4}{5 \times 4} = \frac{75}{20} + \frac{52}{20} \\ &= \frac{75 + 52}{20} = \frac{127}{20} = 6\frac{7}{20} \end{aligned}$$

## Subtraction of mixed fractions

### Illustrative Examples :

$$\begin{aligned} 1) \quad & 3\frac{3}{4} - 2\frac{3}{5} \\ &= \frac{3 \times 4 + 3}{4} - \frac{2 \times 5 + 3}{5} \\ &= \frac{12 + 3}{4} - \frac{10 + 3}{5} = \frac{15}{4} - \frac{13}{5} \\ & \text{(The LCM of 4 and 5 is 20)} \\ &= \frac{15 \times 5}{4 \times 5} - \frac{13 \times 4}{5 \times 4} = \frac{75}{20} - \frac{52}{20} \\ &= \frac{75 - 52}{20} = \frac{23}{20} = 1\frac{3}{20} \end{aligned}$$



1) Compare the following fractions.

i)  $\frac{8}{3}$  ,  $\frac{5}{2}$       ii)  $\frac{4}{9}$  ,  $\frac{3}{7}$       iii)  $\frac{7}{10}$  ,  $\frac{9}{14}$

2) Write the following fractions in ascending and descending order.

i)  $\frac{11}{8}$  ,  $\frac{2}{4}$  ,  $\frac{7}{2}$       ii)  $\frac{16}{9}$  ,  $\frac{5}{3}$  ,  $\frac{7}{4}$

3) Solve.

i)  $\frac{7}{10} + \frac{3}{10}$       ii)  $\frac{3}{4} + \frac{2}{6}$       iii)  $\frac{1}{3} + \frac{7}{5}$

iv)  $\frac{3}{10} + \frac{4}{5}$       v)  $\frac{5}{3} - \frac{2}{3}$       vi)  $\frac{3}{5} - \frac{3}{7}$

4) A house is built on  $\frac{2}{5}$  part of a plot and a garden is made on the remaining part. What part of the plot is made for the garden?



### At a glance

- While comparing fractions, their numerators are compared after making their denominators same.
- While adding or subtracting fractions
  - Find the LCM of the denominators of all the fractions.
  - Make denominators of fractions same as LCM and write the equivalent fraction of each fraction.
  - Add or subtract the equivalent fractions obtained.





## Exercise 6

(1) Write the correct number in the blank box.

i)  $\frac{5}{7} + \frac{2}{7} = \square$

ii)  $\frac{9}{8} - \square = 1$

iii)  $\square + \frac{6}{7} = \frac{12}{7}$

iv)  $\frac{11}{7} - \square = \frac{17}{14}$

v)  $\frac{16}{5} - \square = \frac{16}{5}$

vi)  $\frac{10}{5} - 1 = \square$

(2) Solve.

i)  $\frac{1}{9} + \frac{2}{9}$

ii)  $\frac{6}{10} + \frac{3}{5}$

iii)  $2\frac{2}{3} + 3\frac{5}{7}$

iv)  $\frac{13}{4} - \frac{9}{4}$

v)  $\frac{5}{4} - \frac{7}{8}$

vi)  $4\frac{1}{2} - 2\frac{1}{2}$

- (3) Out of the 12th pass students of a junior college,  $\frac{1}{5}$  students joined the Indian Army,  $\frac{2}{5}$  students took admission for professional education and the remaining students took admission for higher education elsewhere. How many students took admission for higher education?
- (4) Out of the open space in front of the school,  $\frac{3}{4}$  space was allocated for a football field and  $\frac{1}{5}$  space for indoor games. The remaining space was kept for the school garden. How much space was kept for the school garden?
- (5) Mother gave one guava each to Ramesh, Seema and their three friends. They divided each guava into four equal parts. If each of them gave one part of their guava to mother, What part of guava did mother get?
- (6)  $\frac{6}{7}$  is less than  $\frac{9}{11}$  because 6 is less than 9. Is this correct?
- (7) Write any two fractions whose sum is 1.





## 7 Ratio and Proportion



### Mathematical Discussion

In this computer age, we are all very familiar with digital photographs. We often resize these photographs as per need. Below are four photographs of a deer.



A



B



C



D

**Tai:** Observe all the pictures above. All the pictures are of different sizes. Which of these pictures look the same? Which pictures look different?

**Soham:** Tai, although pictures A and C have different shapes, they look the same.

**Sarika:** Pictures B and D look different. The deer in picture B is long and in picture D it is narrow and thick.

**Tai:** Very nice observation. Now let's actually measure the length and width of this picture frame.

**Sarika:** Pictures A and C look similar. Their length and width are 60 mm, 40 mm and 30 mm, 20 mm respectively. That is, the length and width of the picture have changed by the same factor

**Soham:** Tai, pictures B and D do not look the same because their length and width have not changed by the same factor.

| Picture | length (mm) | Width (mm) |
|---------|-------------|------------|
| A       | 60          | 40         |
| B       | 60          | 20         |
| C       | 30          | 20         |
| D       | 30          | 40         |

**Tai:** To show such a proportional relationship in mathematics, we will



understand the concept of ratio. In figure A, we can say that for a length of 60 mm, there is a width of 40 mm.

In picture A the ratio of length to width = 60:40.

In picture C the ratio of length to width = 30:20.

**Soham:** Tai, pictures A and C have been changed by the same amount. Will their ratios be same? How to check this?

**Tai:** Exactly right. Now let us check if the ratios are equal. Multiply both the terms by  $\frac{1}{2}$ ,  $60 \times \frac{1}{2} : 40 \times \frac{1}{2} = 30 : 20$

i.e. 30 : 20 is the ratio of the length and width of both the pictures A and C. This ratio can be written in simplest form as 3:2.

**Sarika:** This means, Tai, to compare whether the ratios are in proportion or not, we have to reduce them in their simplest form!



### Understand

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## Ratio and Proportion

- 1) To make concrete for construction, 3 parts of sand are mixed with one part of cement. The proportion of cement to sand in concrete is 1:3.

1:3 is read as 'one is to three'. It is also written as the ratio of cement to sand in concrete is  $\frac{1}{3}$

- 2) Sanjeev made laddus by adding 2 cups of jaggery to 5 cups of peanut butter. That is, the proportion of peanut butter to jaggery is 5:2 (five as to two). This proportion 5:2 is also written as a ratio  $\frac{5}{2}$

**A ratio is a number that shows how many times one quantity is compared to another.**

### Illustrative Examples:

- 1) Kundan's current age is 12 years and his father's age is 48 years. Kundan has obtained the ratio of his age with his father's age as follows.

→ The ratio was found as follows.

$$\frac{(\text{Kundan's age})}{(\text{Father's age})} = \frac{12}{48} = \frac{12 \div 12}{48 \div 12} = \frac{1}{4}$$

$\frac{1}{4}$  is the simplest form of  $\frac{12}{48}$



**Ratios are always written in simplest form.**



2) To make a garland of marigold flowers if a florist uses 20 yellow flowers and 12 orange flowers then

i) Find the ratio of yellow flowers to orange flowers.

$$\rightarrow \frac{(\text{yellow flowers})}{(\text{orange flowers})} = \frac{20}{12} = \frac{(20 \div 4)}{(12 \div 4)} = \frac{5}{3}$$

$\therefore$  The ratio of yellow flowers to orange flowers is  $\frac{5}{3}$ .

ii) Find the ratio of orange flowers to yellow flowers

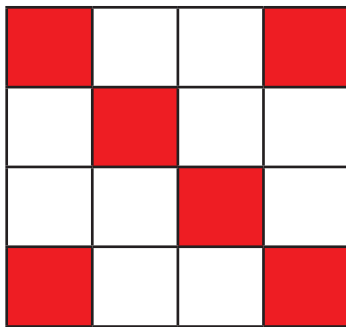
$$\rightarrow \frac{(\text{orange flowers})}{(\text{yellow flowers})} = \frac{12}{20} = \frac{(12 \div 4)}{(20 \div 4)} = \frac{3}{5}$$

$\therefore$  The ratio of orange flowers to yellow flowers is  $\frac{3}{5}$



**Do it.**

**Observe the diagram below and find the answers.**



- Count and write total number of cells in the figure
- Count and write the number of shaded cells
- Count and write the number of unshaded cells.
- Find the ratio of shaded cells to the unshaded cells.
- Find the ratio of shaded cells to total number of cells.
- Find the ratio of unshaded cells to the total number of cells.

• If a and b are two numbers or quantities, then the proportion of a to b is a:b and this proportion is written as a ratio  $\frac{a}{b}$ . Here 'a' is called the first term and 'b' is called the second term.



**Understand**

Anil and Sandeep built protective walls around their houses. Anil built a 50 feet long wall and Sandeep built a 25 feet long wall. For this, Anil used 10 bags of cement and Sandeep used 5 bags. Anil thought his wall would be stronger than the wall built by Sandeep because he used more cement than him. Give your opinion.



For this, we will calculate the ratio of the length of the wall (in feet) built by them to the number of bags of cement.

The ratio of the length of the wall built by Anil to the length of the wall built by Sandeep =  $50:25 = 2:1$

The ratio of number of cement bags used by Anil to number of cement bags used by Sandeep =  $10:5 = 2:1$

Since both ratios are the same, the length of the wall and the amount of cement used are in proportion.



### Mathematical Discussion

Suresh walks 1700 meters every morning, while his friend Ramesh walks 2 kilometers every day. Find the ratio of the distance covered by Suresh to the distance covered by Ramesh.

| Student 'A' solves as follows                                                                                                          | Student 'B' solves as follows.                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\text{Ratio} = \frac{\text{Distance covered by Suresh}}{\text{Distance covered by Ramesh}}$ $= \frac{1700 \text{ m.}}{2 \text{ km.}}$ | $\text{Ratio} = \frac{\text{Distance covered by Suresh(in meter)}}{\text{Distance covered by Ramesh(in meter)}}$ $= \frac{1700 \text{ m.}}{2000 \text{ m.}} = \frac{17}{20}$ |

**Which of the above answers is correct? Discuss.**

- To find the ratio of two quantities, their units must be made equal.
- A ratio does not have any unit.

### Illustrative Examples:

1) Find the ratio of 5 rupees to 80 paise.

→ 1 rupee = 100 paise

∴ 5 rupees =  $5 \times 100$  paise = 500 paise

The ratio of 5 rupees to 80 paise is the ratio of 500 paise to 80 paise.

$$\therefore \frac{500 \text{ paise}}{80 \text{ paise}} = \frac{500}{80} = \frac{500 \div 20}{80 \div 20} = \frac{25}{4}$$





## Practice Set 7.1

- (1) Write the ratio of the first number to the second number in it's simplest form.
  - i) 36, 88      ii) 72, 18      iii) 91, 65      iv) 85, 102
- (2) Write the ratio of the second number to the first number in it's simplest form.
  - i) 28, 84      ii) 120, 32      iii) 75, 135      iv) 220, 100
- (3) Find the ratio of the first quantity to the second quantity in it's simplest form.
  - i) 5 years, 10 months      ii) 750 grams, 1.5 kg      iii) 5 l, 500 ml
  - iv) 135 minutes, 2 hours      v) 2.1 m, 70 cm
- (4) 320 workers are working in a factory. Out of them, 200 are male workers. Solve the following questions based on this.
  - a) Find the ratio of the number of male workers to the number of female workers.
  - b) Find the ratio of the number of male workers to the total number of workers.
  - c) Find the ratio of the number of female workers to the total number of workers.



### Understand

## Unitary Method

### Illustrative Examples:



- 1) Suyash wants to honour 10 teachers in his school by giving them Tulsi saplings on the occasion of Guru Purnima. The nursery owner quoted the price of a dozen Tulsi saplings as Rs. 144. So how much will Suyash have to pay for 10 saplings?

Cost of 12 saplings = Rs. 144.

From this, first calculate the cost of one sapling.

Price of 1 sapling = Rs.  $144 \div 12 = 12$ .

Cost of 10 saplings =  $10 \times 12 = \text{Rs. } 120$ .

**Suyash will have to pay Rs. 120 for 10 saplings.**

- 2) Sanchita wants to distribute 1 pencil each to 75 friends on her birthday. If a box of 10 pencils costs Rs. 70, how much will it cost her in total?

Here, the cost of 10 pencils = Rs 70.

First we calculate the cost of one pencil.

The price of a pencil is  $70 \div 10 = 7$  rupees.



From this, the cost of 75 pencils =  $75 \times 7 = 525$  rupees.

**For this Sanchita's total expenditure will be Rs. 525.**

- While solving such problems, find the price of one item from the price of many items using division and from the price of that one item find the price of many items using multiplication, is called the **Unitary Method**.



- (1) A charitable organization wants to gift compass boxes to 67 needy students of a school, one each. The cost of a pack containing 10 compass boxes is ₹850. How much will the organization spend on purchasing compass boxes?
- (2) A goods train covers a distance of 240 km at a constant speed in 4 hours, then how much distance will that goods train cover in 7 hours?



### At a glance

- If 'a' and 'b' are two numbers or quantities, then the proportion of a to b is written as a:b and ratio as  $\frac{a}{b}$ . Here a is called the first term and b is called the second term.
- While calculating the ratio of two quantities, their units must be made the equal.
- A ratio does not have any unit.
- Finding the price of one item from the price of many items by division and from the price of that one item finding the price of many items by multiplication. This method is called the unitary method.



### Exercise 7

- (1) Choose the correct option

i) What is the ratio of the proportion 5 : 7 ?

- a)  $\frac{5}{7}$                       b)  $\frac{12}{24}$                       c)  $\frac{7}{5}$                       d)  $\frac{10}{21}$

ii) What is the simplest form of  $\frac{60}{36}$  ?

- a)  $\frac{10}{6}$                       b)  $\frac{20}{12}$                       c)  $\frac{30}{18}$                       d)  $\frac{5}{3}$



iii) What is the ratio of 2 kg to 800 grams?

a)  $\frac{2}{800}$

b)  $\frac{200}{8}$

c)  $\frac{5}{2}$

d)  $\frac{2}{5}$

(2) In a class of 45 students, 15 students play hockey, 20 students play kabaddi and the remaining students play kho-kho. If each student plays only one game, solve the following questions.

i) Find the ratio of number of students playing hockey to number of students playing kho-kho.

ii) Find the ratio of number of students playing Kabaddi to the total number of students.

iii) Find the ratio of number of students playing hockey to the total number of students.

(3) Colour any three squares in the adjacent figure with the colour of your choice and write the answers to the following questions.

i) The ratio of the number of squares you have coloured to the total number of squares = ....

ii) Ratio of number of uncoloured squares to total number of squares = ...

iii) Ratio of number of coloured squares to number of uncoloured squares = .....



(4) The length of a rectangular sheet is 1.20 m and the width is 48 cm. Find the ratio of the width to the length of the sheet.

(5) Michael's monthly income of ₹52,000 out of which he saves ₹3,000 monthly. Then find the ratio of his monthly income to his expenditure.

(6) A school wanted to purchase 50 saplings for planting trees on the occasion of Environment Day. The school purchased 35 saplings from 'Yashoda' nursery for ₹ 455 and the remaining saplings from 'Amol' nursery for ₹240. From which nursery will the school get the saplings purchased at low price?

(7) A cook prepares 10 kilograms of rice every day for 100 students in a school as part of mid-day meal. But one day, due to heavy rain, only 80 students were present in the school. How many kilograms of rice should be cooked on that day?



**Do it.**

- Measure the length of your full arm with your hand-span and calculate the ratio of the length of your arm to the length of your hand-span.

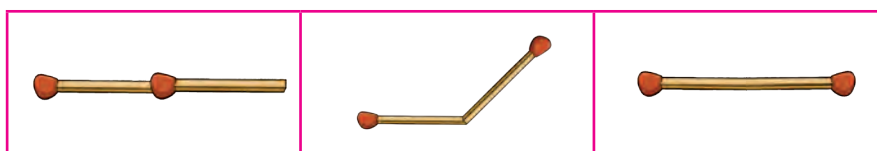




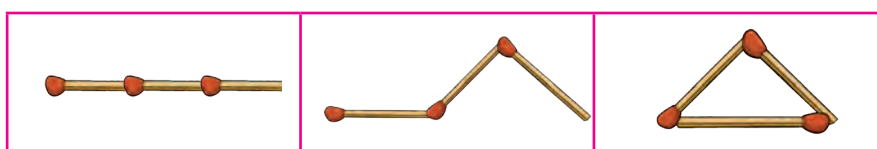
### Revision

**Make possible shapes by joining the match sticks**

**2 sticks**



**3 sticks**



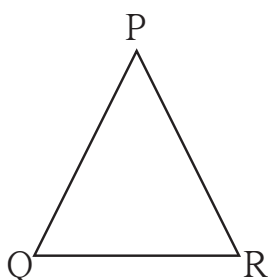
- 1) Can a closed shape be formed using 2 match sticks?
- 2) What is minimum number of match sticks to make a closed shape?

## Triangle



### Understand

- Action:**
- (1) Take three non-collinear points P, Q and R on a paper.
  - (2) Join all three points with line segments.



The resulting shape is a triangle PQR. It is written as  $\triangle PQR$ . Symbol  $\triangle$  is used to represent a triangle

In  $\triangle PQR$

**Vertices:** Point P, Point Q, Point R

**Sides:** Side PQ, Side QR, Side PR

**Angles:**  $\angle R$  i.e  $\angle QRP$  or  $\angle PRQ$

$\angle P$  i.e  $\angle QPR$  or  $\angle RPQ$

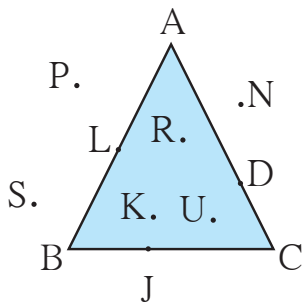
$\angle Q$  i.e  $\angle PQR$  or  $\angle RQP$

**Triangle:** A triangle is a closed shape formed by joining three non-collinear points with line segments. A triangle has three vertices, three sides (arms), and three angles.

## Interior and Exterior of Triangle and Triangular Region



### Understand



In  $\triangle ABC$ , the points on the triangle are A, B, C, L, J, D.

The shaded area in the figure is the interior of the triangle. Points R, K, U are the interior points of the triangle.

The area outside the triangle is called the exterior of the triangle. Points P, S, N are the exterior points of the triangle.

A triangle and its interior form a triangular region.

### Games :

- (1) Draw a large triangle on the ground and name its vertices as N, P, G.
- (2) Ask students stand on the vertices and sides of the triangles.
- (3) Play this game: Jump inside the triangle when you are in the pond, and outside the triangle when you are in the garden.

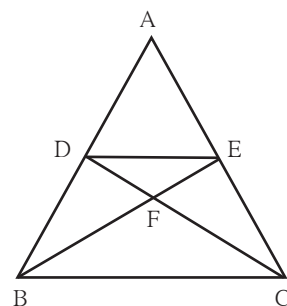


### Practice Set 8.1

- (1) Complete the table below by observing the diagram.

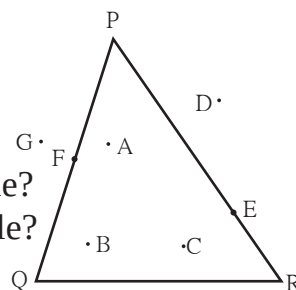
| Sr. No. | Figure | Vertices | Sides | Angles |
|---------|--------|----------|-------|--------|
| 1       |        |          |       |        |
| 2       |        |          |       |        |
| 3       |        |          |       |        |

- (2) Observe the adjacent figure and write the names of all the triangles.



- (3) Observe the adjacent figure and write the answers to the following questions.

- Write the name of the triangle.
- Write the names of the points on the triangle.
- Which are the points in the interior of the triangle?
- Which are the points in the exterior of the triangle?
- Write the names of the points in the triangular region.



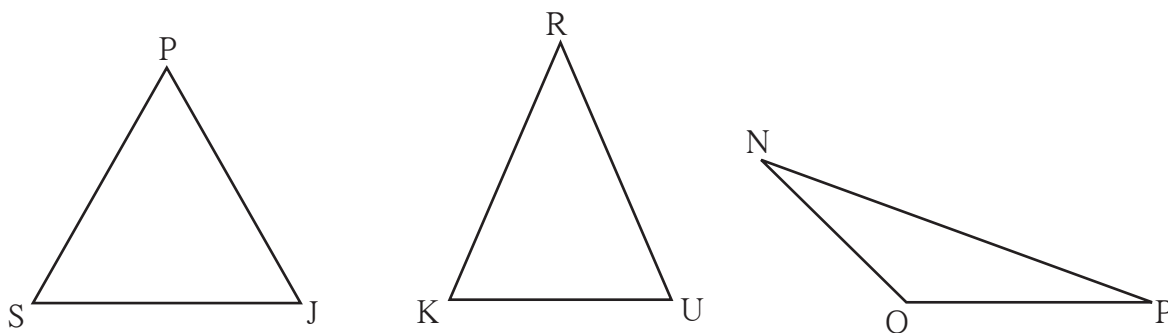
## Types of Triangles



### Understand

There are six types of triangles, three types based on sides and three types based on angles.

#### 1) Types of triangles based on sides:



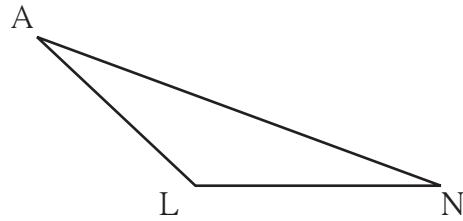
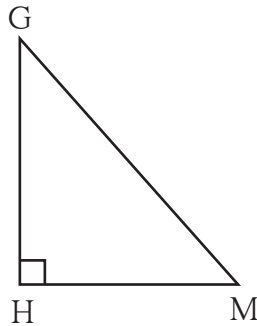
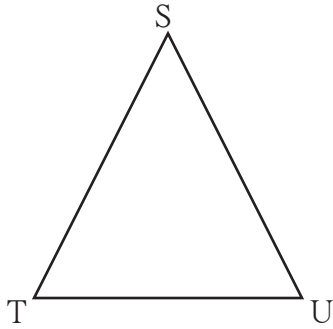
Measure the sides of the triangles and complete the table.

| $\triangle PSJ$                                                                     | $\triangle RKU$                                                                     | $\triangle NOP$                                                                     |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| $l(PS) = \text{----- cm}$<br>$l(SJ) = \text{----- cm}$<br>$l(PJ) = \text{----- cm}$ | $l(RK) = \text{----- cm}$<br>$l(KU) = \text{----- cm}$<br>$l(UR) = \text{----- cm}$ | $l(NO) = \text{----- cm}$<br>$l(NP) = \text{----- cm}$<br>$l(OP) = \text{----- cm}$ |
| All three sides are of equal length.                                                | The length of two sides are of equal                                                | The length of all three sides are different                                         |
| This is an equilateral triangle.                                                    | This is an isosceles triangle.                                                      | This is a scalene triangle.                                                         |





## 2) Types of triangles from angles:



Find the measures of all three angles of each triangle and complete the table given below.

| $\triangle STU$                                                                                                                                                                             | $\triangle GHM$                                                                                                                                                                                                      | $\triangle ALN$                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Measure of $\angle S = m\angle S = \dots^\circ$<br>Measure of $\angle T = m\angle T = \dots^\circ$<br>Measure of $\angle U = m\angle U = \dots^\circ$<br>All three angles are acute angles. | Measure of $\angle G = m\angle G = \dots^\circ$<br>Measure of $\angle H = m\angle H = \dots^\circ$<br>Measure of $\angle M = m\angle M = \dots^\circ$<br>One angle is a right angle and two angles are acute angles. | Measure of $\angle A = m\angle A = \dots^\circ$<br>Measure of $\angle L = m\angle L = \dots^\circ$<br>Measure of $\angle N = m\angle N = \dots^\circ$<br>One angle is obtuse and two angles are acute. |
| This is an acute angled triangle.                                                                                                                                                           | This is a right angled triangle                                                                                                                                                                                      | This is an obtuse angled triangle.                                                                                                                                                                     |



### Think about

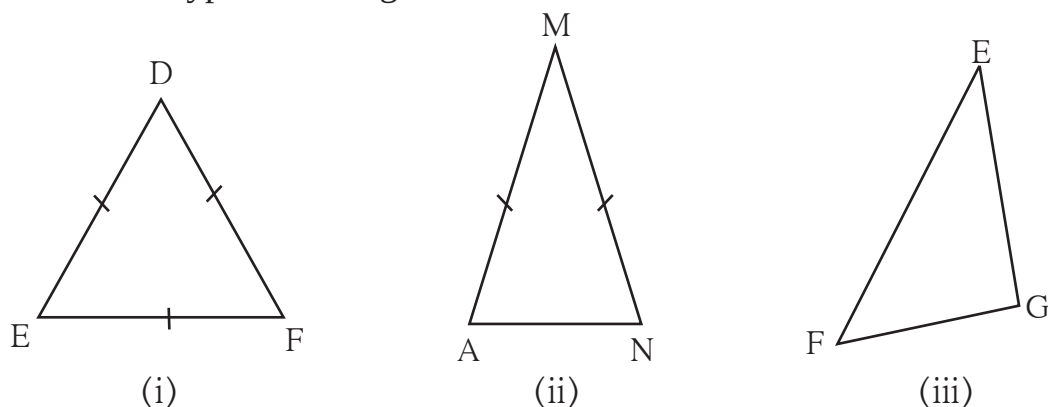
- Can an obtuse angled triangle be an acute angled triangle?
- Can a triangle with all three sides of equal length be an acute angled triangle?
- Can an isosceles triangle be a right angled triangle?

### Mathematical Game

- 1) First form a group of 12 students.
- 2) Take a 12 meter long rope. Tie 11 knots, one knot for every meter.
- 3) Have 11 students stand holding 11 knots, one on each knot.
- 4) A total of 12 students will stand, one holding both ends of the rope.
- 5) Ask the students to make all three types of triangles based on sides.
- 6) Ask the students to make all three types of triangles based on angles.



- (1) In the figure below, sides of equal are shown by the same marking. Write the type of triangle based on it.



- (2) Write the type of triangle from the information given below

- (i) In  $\triangle ABC$ ,  $m\angle A = 63^\circ$ ,  $m\angle B = 52^\circ$ ,  $m\angle C = 65^\circ$
- (ii) In  $\triangle RST$ , seg RS and seg ST are of equal length.
- (iii) In  $\triangle LMN$ ,  $\angle LMN$  is a right angle.
- (iv) In  $\triangle SDO$ ,  $m\angle OSD = 135^\circ$ .
- (v) All the sides of  $\triangle RTM$  are equal.
- (vi) The sides of  $\triangle DEF$  are 5.3 cm, 4.2 cm and 6.6 cm respectively

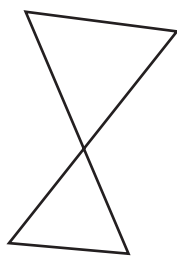
## Quadrilateral



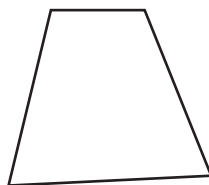
Do it.

Take four non-collinear points A, B, C, D on a paper. Join all these points in such a way that if any two points are joined, the remaining two points will be on the same side of the line joining them. This closed figure is a quadrilateral.

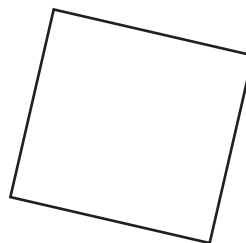
Which of the following figures are quadrilaterals, why?



(1)



(2)



(3)

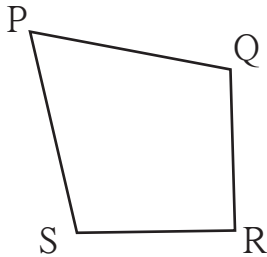


## Reading and writing of Quadrilaterals



### Understand

Reading and writing a quadrilaterals can be done starting from any vertex of the quadrilaterals. However, they are read or written in a **clockwise** or **counterclockwise** (anti clockwise) direction.



• Complete the table below :

| Reading            | Writing        | Reading            | Writing        |
|--------------------|----------------|--------------------|----------------|
| Clockwise          |                | Counter clockwise  |                |
| Quadrilateral PQRS | $\square$ PQRS | Quadrilateral PSRQ | $\square$ PSRQ |
|                    |                | Quadrilateral SRQP | $\square$ SRQP |
| Quadrilateral RSPQ | $\square$ RSPQ |                    |                |
|                    |                |                    |                |



### Think about

How many maximum ways can quadrilateral be named?

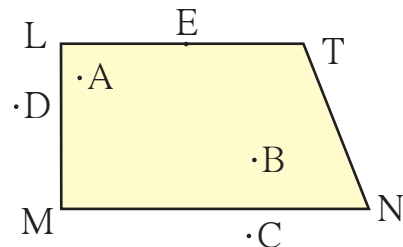
## Interior and Exterior of a Quadrilateral and Region of a Quadrilaterals

Points inside the quadrilateral: Point A, Point B

Points outside the quadrilateral: Point D, Point C

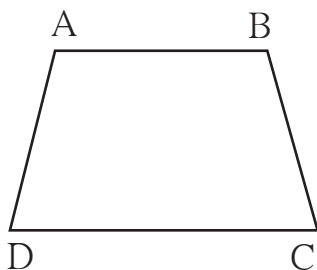
Points on the quadrilateral: Point L, Point E, Point M, Point N, Point T

A quadrilateral and its interior together form the region of quadrilateral.



## Elements of Quadrilaterals

Observe it  $\square$ ABCD.



**1) Opposite sides :** The sides that do not have common vertex are called opposite sides. In  $\square$ ABCD, side AB and side DC do not have any common vertex. Therefore, they are opposite sides of the quadrilateral. There are two pairs of opposite sides.

The opposite side of side AD is side BC.



2) **Adjacent sides:** If two sides of a quadrilateral have a common vertex, then those two sides are adjacent sides. In  $\square ABCD$ , side AB and side BC have common vertex B. Therefore, side AB and side BC is a pair of adjacent sides of the quadrilateral. There are four pairs of adjacent sides. Find the remaining pairs.

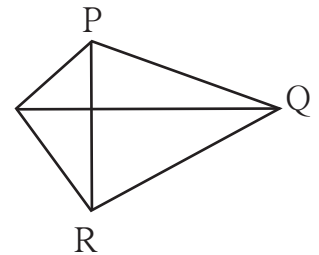
3) **Adjacent angles:** If two angles of a quadrilateral have one common side, then those angles are adjacent angles of the quadrilateral.

In  $\square ABCD$ , the side BC is common to both angles  $\angle ABC$  and  $\angle BCD$ . Therefore,  $\angle ABC$  and  $\angle BCD$  are adjacent angles. There are four pairs of adjacent angles in  $\square ABCD$ .

4) **Opposite angles:** If two angles of a quadrilateral have no common side, then those angles are opposite to each other.

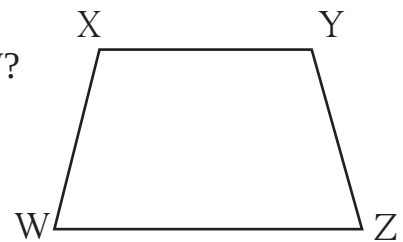
In  $\square ABCD$ ,  $\angle ABC$  and  $\angle ADC$  do not have a common side. Therefore,  $\angle ABC$  and  $\angle ADC$  are opposite angles. There are two pairs of opposite angles in ABCD.

5) **Diagonal:** The line segment joining the vertices of the opposite angles of a quadrilateral is a diagonal of the quadrilateral. The line segments joining the vertices of the opposite angles  $\angle P$  and  $\angle R$  as well as  $\angle S$  and  $\angle Q$  of  $\square PQRS$  are the two diagonals of the quadrilateral.



(1) Observe  $\square XYZW$  and answer the following questions

- Which side is opposite to side XY?
- Which are the adjacent sides of side XW?
- Which is the opposite angle of W?
- Which are the adjacent angles of  $\angle Z$ ?
- Which are the diagonals of  $\square XYZW$ ?



(2) Write the opposite angles in  $\square PQRS$ .



### At a glance

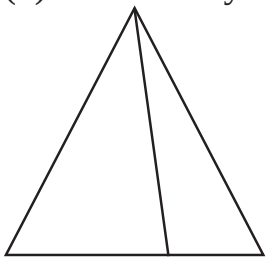
- A triangle has three sides (arms), three vertices and three angles.
- Types of triangles based on sides: equilateral triangle, isosceles triangle and scalene triangle.
- Types of triangles based on angles: acute angled triangle, right angled triangle, obtuse angled triangle.

- A quadrilateral has four vertices, four sides, four angles and two diagonals.
- A quadrilateral has two pairs of opposite sides, four pairs of adjacent sides, four pairs of adjacent angles and two pairs of opposite angles

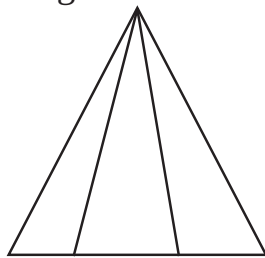


### Exercise 8

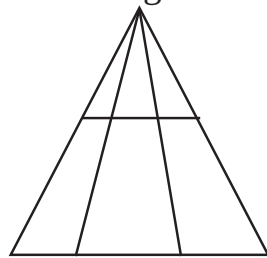
- (1) Fill in the blanks.
  - i) A triangle in which all three sides have the same length is called ..... triangle.
  - ii) A triangle in which only two sides have equal lengths is called ..... triangle.
  - iii) A triangle whose one angle is ..... degrees is called a right angled triangle.
  - iv) The line segment joining the vertices of the opposite angles of a quadrilateral is called a ..... of the quadrilateral.
  - v) In  $\square LMNO$ , side LM and side..... are adjacent sides.
- (2) State whether true or false.
  - (i) A triangle can have two right angles.
  - (ii) A right angled triangle can be an isosceles triangle.
  - (iii) An acute-angled triangle cannot be an equilateral triangle.
- (3) Draw  $\square UVWX$  and write the answers to the following questions from it.
  - i) Write the names of the quadrilateral in different ways.
  - ii) Write the pairs of opposite sides.
  - iii) Write the pairs of adjacent sides.
  - iv) Write the names of the diagonals.
  - v) Write the pairs of opposite angles.
  - vi) Write the pairs of adjacent angles.
- (4) How many triangles are there in the figure below?



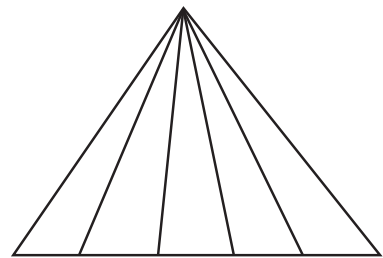
(i)



(ii)

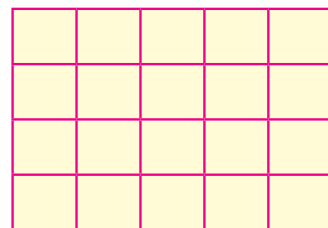


(iii)



(iv)

- 5) Count and write the total number of quadrilaterals in the adjacent figure.





## Revision

Prakash gave 8 honey jars out of 23 jars to Sunita. That is,  $23 - 8 = \square$ . Of the remaining jars 6 were given to Deepali, that is,  $\square - 6 = \square$ . Let's see how we can write this arrangement!



## Understand

## Expressions

We have studied how to perform basic operations on numbers such as addition, subtraction, multiplication and division. Salim, Arya and Rajat have performed some mathematical operations.

$$14 + 5 - 3$$



$$23 - 8 + 2$$



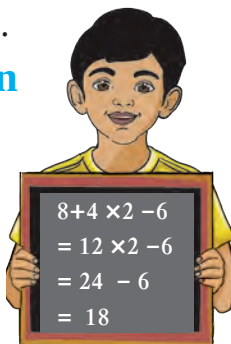
$$18 \div 3 + 4 \times 2$$



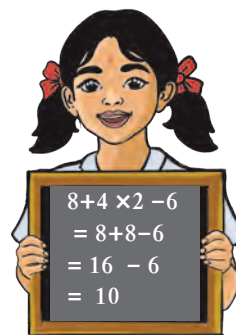
An arrangement of mathematical operations involving two or more numbers is called an '**expression**'.

- Aman initially had 8 pencils. Rajat gave him 2 more packets containing 4 pencils each. Aman distributed 6 of them to his friends. How many pencils did Aman have left at the end? Rohan and Gracie solved this example by formulating the following expressions.

Rohan



Gracie



Rohan's answer is 18.

Gracie's answer is 10.



## Think about

The answers given by both Rohan and Gracie are different. Why this happened?

- There are rules for performing mathematical operations while solving expression.

- The operation given in brackets (parentheses) should be performed first.
- Then, if there is more than one operation in the expression, the multiplication and division operations should be performed in the order in which they occur from left to right.
- Then, the addition and subtraction operations should be performed in the order in which they occur from left to right.



- 4) If there is more than one operations in brackets, follow the above two rules and do them first.
- 5) Sometimes, more than one brackets is used to make the sequence of actions clear. In such cases, along with the round brackets ' ( ) ', square brackets ' [ ] ' and curly braces ' { } ' are used. If more than one type of brackets is used, the actions in the innermost brackets should be performed first.

| Order of symbols evaluating expressions           |                                |                    |                          |                    |                       |
|---------------------------------------------------|--------------------------------|--------------------|--------------------------|--------------------|-----------------------|
| B<br>( )<br>Evaluate<br>operations<br>in brackets | O<br>×<br>of<br>Multiplication | D<br>÷<br>Division | M<br>×<br>Multiplication | A<br>+<br>Addition | S<br>-<br>Subtraction |

**\* Use these rules to determine the correct answer to the expression.**

$$\begin{aligned}
 &8 + 4 \times 2 - 6 \\
 &= \underline{8 + 8} - 6 \quad \dots \text{multiplication first} \\
 &= \underline{16} - 6 \quad \dots \text{addition} \\
 &= 10 \quad \dots \text{subtraction}
 \end{aligned}$$

∴ Of Rohan and Gracie,  's answer is correct.

### Illustrative Examples:

- 1)  $81 \div (5 + 4) - 2$   
 $= 81 \div 9 - 2$  ... By solving the brackets  
 $= 9 - 2$  ... By dividing  
 $= 7$
- 2)  $[18 + 5(7-3)] + 64$   
 $= [18 + 5 \times 4] + 64$  ... Solving the simple brackets  
 $= [18 + 20] + 64$  ... multiplying  
 $= 38 + 64$  ... solving the square  
 $= 102$
- 3) There are 24 marigold flowers in a basket. If Shefali and Shreejay each add 5 more flowers to that basket, how many flowers are there in the basket now?  
 $24 + 2 \times 5$   
 $= 24 + 10$  ... by multiplying  
 $= 34$  ... by adding



(1) Complete the following steps.

(i)  $32 \times 4 \div 8 - 6$

$$\begin{aligned}
 &= \boxed{\phantom{00}} \div 8 - \boxed{\phantom{00}} \\
 &= \boxed{\phantom{00}} - 6 = \boxed{\phantom{00}}
 \end{aligned}$$

(ii)  $65 \div 5 \times 2 - 20$

$$\begin{aligned}
 &= \boxed{\phantom{00}} \times 2 - 20 \\
 &= \boxed{\phantom{00}} - 20 = \boxed{\phantom{00}}
 \end{aligned}$$



$$(iii) 80 \times (15+8-3)+5$$

$$= 80 \times (\square - 3) + 5$$

$$= 80 \times \square + 5$$

$$= \square + 5$$

$$= \square$$

$$(iv) 100 \div (18+5-3)+7$$

$$= 100 \div (\square - 3) + 7$$

$$= 100 \div \square + 7$$

$$= \square + 7$$

$$= \square$$

$$(v) 4 + \{100 \div [3(7-5) + 4]\}$$

$$= 4 + \{100 \div [3 \times \square + 4]\}$$

$$= 4 + \{100 \div [\square + 4]\}$$

$$= 4 + \{100 \div \square\}$$

$$= 4 + \square$$

$$= \square$$

$$(vi) 210 \div \{125 - [5 + 3(14-9)]\}$$

$$= 210 \div \{125 - [5 + 3 \times \square]\}$$

$$= 210 \div \{125 - [5 + \square]\}$$

$$= 210 \div \{125 - \square\}$$

$$= 210 \div \square$$

$$= \square$$

(2) Find the values of the following expressions.

$$(i) 28 + 12 - 25$$

$$(ii) 8 \times 4 - 3 \times 6$$

$$(iii) 112 - 18 \times 2 \div 9 + 2$$

$$(iv) 13 - [4 + (32 \div 4 - 5)] \quad (v) \{100 \div [3(18 - 12) + 7]\} + 6$$

(3) Sharvari made some expressions by using numbers from 1 to 4 which have value of 21. Help her to make more such expressions.

$$(i) 3 \times 4 + 2 \times 4 + 1 \quad (ii) 4(3 + 2) + 1$$

(4) Obtain a verbal example from the expressions given below.

$$(i) (5 + 4) - 8$$

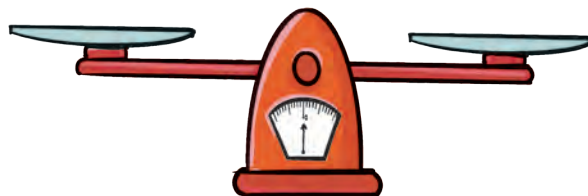
$$(ii) (6 \times 12) \div 4$$

$$(iii) 2 \times 6 + 3 \times 4 + 1$$



Understand

## Equality



$$2 \times 7 = \square$$

$$3 + 4 = \square$$

$$20 - 6 = \square$$

$$14 \div 2 = \square$$

What do you see if you complete the actions in the examples above ?

In the above example, the values of the expressions  $2 \times 7$  and  $20 - 6$  are equal. Similarly, the values of the expressions  $3 + 4$  and  $14 \div 2$  are equal. The arrangement  $2 \times 7 = 20 - 6$  and  $3 + 4 = 14 \div 2$  is equality. From this, the values in the left and right sides of the '=' sign are equal.



## Illustrative Examples:

Determining whether the given expressions are equal or not:

(i)  $6 \times 5$  ;  $14 + 16$

First number  $6 \times 5 = 30$

Second number  $14 + 16 = 30$

$\therefore 6 \times 5 = 14 + 16$

These expressions are equal.

(ii)  $30 \div 15$  ;  $18 - 8$

First number  $30 \div 15 = 2$

Second number  $18 - 8 = 10$

$\therefore 30 \div 15 \neq 18 - 8$  These expressions are not equal.

(The  $\neq$  symbol is used to indicate that there is no equality.)



- (1) In each of the following examples, find the values of the expressions given on the left and right sides of the box. From that, write the appropriate symbol '=' or ' $\neq$ ' in the box.

(i)  $5 + 9$    $2 \times 7$

(ii)  $18 \div 3$    $9 \times 3$

(iii)  $6 \div 2 + 4$    $13 - 6$

- (2) In the example below, write the correct number in the blank space so that the values on the left and right sides are equal.

(i)  $16 + 4 =$    $+ 8$

(ii)  $26 \div$    $= 5 + 8$

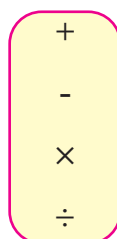
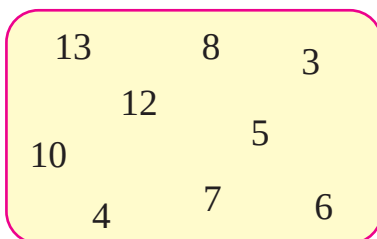
(iii)  $38 -$    $= 21 \div 3$

- (3) Create your own examples of equality using the numbers and symbols for operations on numbers given below.

Numbers

Symbols for operations

Examples of Equality



1)  $13 + 7 = 4 \times 5$

2) .....

3) .....

4) .....

## Properties of Equality:

Example, let's check the equality property for  $(6 + 4) = (5 \times 2)$

| Properties of Equality | Left side                             | Right side                                                                      | Property                                                         |
|------------------------|---------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------|
| Addition               | $(6 + 4) + 5$<br>$= 10 + 5$<br>$= 15$ | $(5 \times 2) + 5$<br>$= 10 + 5$<br>$= 15$                                      | Adding the same number to both sides results in equality.        |
| Subtraction            | $(6 + 4) - 2$<br>$= 10 - 2$<br>$= 8$  | $(5 \times 2) - 2$<br>$= 10 -$ <input type="text"/><br>$=$ <input type="text"/> | Subtracting the same number from both sides results in equality. |



| Properties of Equality | Left side                                               | Right side                                                     | Property                                                       |
|------------------------|---------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------|
| Multiplication         | $(6+4) \times 3$<br>$= \square \times 3$<br>$= \square$ | $(5 \times 2) \times 3$<br>$= \square \times 3$<br>$= \square$ | Multiplying both sides by the same number results in equality. |
| Division               | $(6+4) \div 2$<br>$= \square \div 2$<br>$= \square$     | $(5 \times 2) \div 2$<br>$= \square \div 2$<br>$= \square$     | Dividing both sides by the same number results in equality.    |

### Arithmetic Expression

The statement 'the sum of numbers five and eight' is written as '5+8' using symbols. Use of symbols makes writing manageable and easy to understand. Use of letters for unknown numbers also makes solving examples easier.

- There are two ways to use letters for numbers: (1) For unknown numbers  
(2) To form formulas or rules

#### Type 1) Use of letters for unknown numbers:

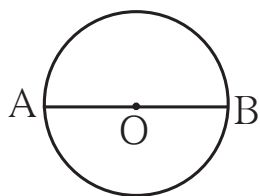
What is the number that is greater than six by seven? The answer is 13. We can work out this. In mathematical terms, it is written as '6+7=13'. But how do we write 'a number is greater than six by seven' using letters? More than a number, means which number? We don't know.

So here we will use the letter 'a' for the unknown number.

- The number greater than a number by seven =  $a + 7$ .
- The number that is four less than a number =  $a - 4$ .
- The sum of a number and 3 =  $a + 3$ .
- If a number is subtracted from 15, the difference =  $15 - a$ .
- 5 times a number =  $a \times 5 = 5a$

- If there is a multiplication of a letter and a number, the number is written before the letter and the multiplication sign is not written.
- For an unknown number, any of the letters a, b, c,...z can be used. This letter is called a 'variable'.

#### Type 2) Use of letters to form a formula or rule:



The accompanying figure shows a circle with centre O. The radius and diameter of the circle are denoted by the letters r and d respectively. The line segments OA and OB are the radii (r) of the circle.

The line segment AB is the diameter (d) of the circle.  
 $\text{seg AB} = \text{seg OA} + \text{seg OB}$



Diameter = Radius + Radius

Diameter = 2 Radius

$d = 2 \times r$

- (2) Complete the following table to learn the rules for addition and multiplication of two numbers.

| Example                                                            | Rule             |                                                                                         |
|--------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------|
| $6 + 3 = 9$ ; $3 + 6 = 9$                                          | $a + b = b + a$  | The sum of the numbers is the same whether they are added in order or in reverse order. |
| $3 + 0 = $ <input type="text"/><br>$5 + 0 = $ <input type="text"/> | $a + 0 = a$      | .....<br>.....                                                                          |
| .....                                                              | $a \times 0 = 0$ | The multiplication of any number and 0 is zero.                                         |
| $8 \times 1 = 8$                                                   | $a \times 1 = a$ | .....<br>.....                                                                          |



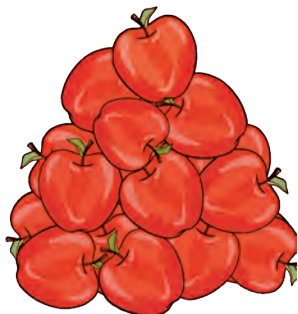
- (1) Write mathematical expressions using letters.
- (i) The sum of a number and 9
  - (ii) A number divided by 8.
  - (iii) Multiplication of a number and 0
  - (iv) 12 is subtracted from a number.
  - (v) 4 times a number
  - (vi) Multiply a number by 1.
- (2) Rewrite the rules and formulas using the letters given in brackets.
- (i) Profit = Selling price - Cost price  
(Profit = P, Selling price = S, Cost price = C)
  - (ii) Loss = Cost Price (C) - Selling Price (S) (Loss = L)



**Understand**

## Equations in One Variable

Pradeep has some apples, Sachin gives him 5 more apples, so he has a total of 12 apples, then how many apples did Pradeep have initially?



Let's assume that Pradeep has 'x' apples. 'x' is a variable.

Number of apples given by Sachin =  apples.

Total number of apples Pradeep has =  apples.

Let us present the above information mathematically as follows.

$\therefore x + 5 = 12$ , Such mathematical arrangement is an 'Equation'.



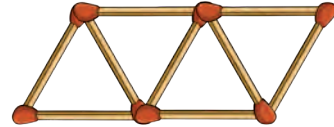
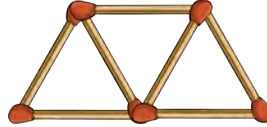
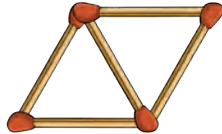
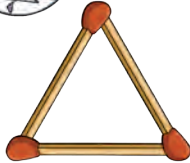
A mathematical arrangement in which only one variable is used in the expressions on the left or right or both sides of the equation is called a 'single variable equation'.

Example.  $a-6=8$ ;  $9=3p$ ;  $\frac{m}{5}=7$ ;  $2x+5=x+7$



**Do it.**

Construct triangles using sticks as shown in the picture and complete the table.



| Number of triangles | Number of sticks<br>( $2 \times \text{number of triangles}$ ) + 1 |
|---------------------|-------------------------------------------------------------------|
| 1                   | $(2 \times 1) + 1 = 3$                                            |
| 2                   | $(2 \times 2) + 1 = 5$                                            |
| 3                   | .....                                                             |

| Number of triangles | Number of sticks<br>( $2 \times \text{number of triangles}$ ) + 1 |
|---------------------|-------------------------------------------------------------------|
| 4                   | .....                                                             |
| 5                   | .....                                                             |
| $n$                 | .....                                                             |



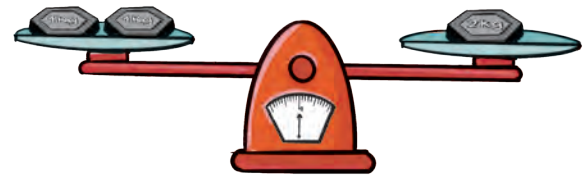
**Understand**

## Solution of Equation

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The scale remains balanced only if the weight on the left and right sides of the scale is equal.

Similarly, an equation is balanced only if the values on both sides of the equation are the same.



Let us see for what value of variable 'y' in the equation  $y+6=7$ , the values of the left and right side of '=' sign become equal.

Let's take the values of y in the equation  $y+6=7$  as 0,1,2,3,...

The right hand side = 7.

i) Let  $y = 0$

Left side =  $y + 6 = 0 + 6 = 6$

$\therefore$  left side  $\neq$  right side

This equation does not balance for the value  $y = 0$ .

$\therefore y = 0$  is not a solution of the given

ii) Let  $y = 1$

Left side =  $y + 6 = 1 + 6 = 7$

$\therefore$  left side = right side

This equation is balance for the value  $y = 1$ .

$\therefore y = 1$  is the solution of the given

A value of the variable that makes given equation balanced or satisfies the equation is called a 'solution of the equation'.





### Illustrative Examples:

1)  $a + 7 = 12$

Solution:  $a + 7 = 12$

$\therefore a + 7 - 7 = 12 - 7$

... Subtract 7 from both sides

$\therefore a + 0 = 5$

$\therefore a = 5$

- 2) Ishwari thought of a number in her mind and when she triples that number, she gets the number 75. What is the number Ishwari thought of in her mind?

Solution: Let the number that Ishwari has in mind be  $x$ .

$\therefore$  Three times that number  $= 3x$

$\therefore 3x = 75$

$\frac{3x}{3} = \frac{75}{3}$  ... Divide both sides by 3.

$\therefore x = 25$

$\therefore$  The number that Ishwari has in mind is 25.



### Practice Set 9.4

- (1) Write equations from the information given below.
  - i) 4 times a number is 24.
  - ii) If one number is subtracted from 37, 12 remains.
  - iii) Shrimayi has some fruits, out of which she gave 8 fruits to Sai. Now she has 15 fruits left.
  - iv) Samarth divides a number by 7, he gets 4 as the quotient.
- (2) Determine which values of the variables in the following equations are the solutions of that equation.
  - i)  $7x = 14$  ( $x = 1, 2, 3, \dots$ )
  - ii)  $p + 6 = 9$  ( $p = 1, 2, 3, 4, \dots$ )
  - iii)  $4 = \frac{y}{8}$  ( $y = 31, 32, 33, \dots$ )
  - iv)  $8 - m = 5$  ( $m = 1, 2, 3, 4, \dots$ )
- (3) Solve the following equations using the properties of equality.
  - i)  $3x = 12$
  - ii)  $x + 4 = 21$
  - iii)  $17 - p = 8$
  - iv)  $\frac{k}{8} = 9$

- (4) Complete the following activity.

- i) If the sum of two consecutive natural numbers is 131, find those numbers.

**Solution:** Let us consider two consecutive numbers  $m$  and  $m+1$ .

But the sum of those two numbers is 131.

$\therefore \square + (m + 1) = 131$

$\therefore \square + 1 = 131$

$\therefore 2m + 1 - 1 = 131 - 1$

(Subtract 1 from both sides)

$\therefore \square = 130$

$\therefore \frac{2m}{2} = \frac{130}{2}$  (Divide both sides by 2)

$\therefore m = \square$

$\therefore m + 1 = 65 + 1 = \square$

$\therefore$  Those two numbers are

$\square$  and  $\square$



- ii) The price of a pant is Rs. 250 more than the price of a shirt and the total price of a pant and a shirt is Rs. 1250. Then find the prices of both.

**Solution :** Let the price of the shirt be  $x$ .

$\therefore$  The price of a pant is  $x + \square$

Write the total price in the form of equation.

$$\therefore x + \square = 1250$$

$$\therefore \square + 250 = 1250$$

$$\therefore 2x = \square \quad (\text{Subtract 250 from both sides})$$

$$\therefore x = \square \quad (\text{Dividing both sides by two})$$

$$\therefore \text{Price of a shirt} = \square \text{ Rs.}$$

$$\therefore \text{Price of a pant} = \square + 250 \text{ Rs.}$$

$$= \square \text{ Rs.}$$

- (5) Write and solve equations from the following verbal examples.

- There are 36 Amla trees in a garden. The remaining are Eucalyptus trees. If there are a total of 150 trees in that garden, then how many eucalyptus trees are there?
- The government made some quintals of food grains available for distribution to the earthquake-affected families in Patan taluka. Out of this, 34 quintals of food grains were distributed to the families in Adul village, leaving 48 quintals of food grains remaining. So how many quintals of food grains were available in total from the government?



### At a glance

- While solving an expression, first solve the parentheses and then solve the expression in the order of division, multiplication, addition and subtraction.
- An 'equation' is an arrangement in which the values of the expressions on the left and right of the '=' sign are equal.
- Adding, subtracting, multiplying any number to both sides, or dividing by a non-zero number to both sides, of an equation does not change the equality.
- A value of the variable that satisfies the equation is called the 'solution of the equation'.



### Exercise 9

- (1) Choose the correct option.

- i) If  $x + 12 = 27$  then  $x = ?$

(a) 39

(b) 15

(c) 25

(d) 29

- ii) If  $23 - n = 12$  then  $n = ?$

(a) 11

(b) 25

(c) 35

(d) 21



iii) If  $12a = 96$  then  $a = ?$

(a) 108

(b) 8

(c) 84

(d) 1152

iv)  $24 \div (3+5) - 2 = \square$

(a) 20

(b) 11

(c) 4

(d) 1

v) Which of the following pairs is there an equality?

(a)  $14 \times 1$ ;  $5-3$

(b)  $6 \times 7$ ;  $14 \times 4$

(c)  $5 \times 3$ ;  $14+1$

(d)  $5 \times 3$ ;  $14-1$

vi) If the sum of a number and its triple is 24, then which of the following is that number?

(a) 12

(b) 6

(c) 3

(d) 108

(2) Complete the equation by placing the correct value in the box below.

(i)  $12 \div 3 = 9 - \square$

(ii)  $15 + 7 = \square \times 2$

(iii)  $48 \div \square = 15 - 12$

(3) Solve the following expressions.

(i)  $8 \times 4 - 17$

(ii)  $9 \div (13 - 10) + 5$

(iii)  $100 \div \{6 + [12 - 4 + (5 \times 3 - 4)]\}$

(4) Solve the following equations using the properties of equality.

(i)  $x + 12 = 25$

(ii)  $6p = 48$

(iii)  $\frac{m}{7} = 7$

(iv)  $27 - t = 14$

(5) Sonal thought of a number in mind and multiplied it by five, the answer was 30. What is that number?

(6) Deepali's age is 5 years less than Asma's age. If the sum of their ages is 27 years, what are their ages?

(7) Students of a school collected some funds to help the flood victims. The amount of the collected funds is four times the number of students. When teachers added Rs. 1500 to it, the total amount became ₹ 5140. Then what is the total number of students of that school?

(8) The number of Neem trees planted near a milk collection centre is 52 more than the number of Peepal trees. If the total number of trees of both types is 170, how many neem trees were planted?



## Funny Numbers

### 1) Smart Cards

- Play the following game with your friend.

|    |    |    |    |
|----|----|----|----|
| 1  | 3  | 5  | 7  |
| 9  | 11 | 13 | 15 |
| 17 | 19 | 21 | 23 |
| 25 | 27 | 29 | 31 |

|    |    |    |    |
|----|----|----|----|
| 2  | 3  | 6  | 7  |
| 10 | 11 | 14 | 15 |
| 18 | 19 | 22 | 23 |
| 26 | 27 | 30 | 31 |

|    |    |    |    |
|----|----|----|----|
| 4  | 5  | 6  | 7  |
| 12 | 13 | 14 | 15 |
| 20 | 21 | 22 | 23 |
| 28 | 29 | 30 | 31 |

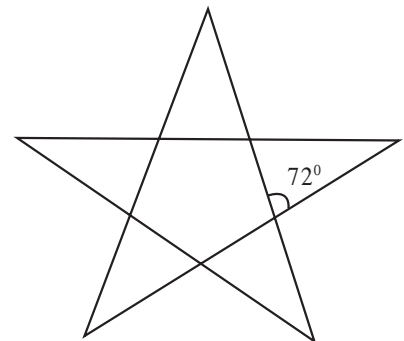
|    |    |    |    |
|----|----|----|----|
| 8  | 9  | 10 | 11 |
| 12 | 13 | 14 | 15 |
| 24 | 25 | 26 | 27 |
| 28 | 29 | 30 | 31 |

|    |    |    |    |
|----|----|----|----|
| 16 | 17 | 18 | 19 |
| 20 | 21 | 22 | 23 |
| 24 | 25 | 26 | 27 |
| 28 | 29 | 30 | 31 |

- Each card has 16 numbers written on it, from 1 to 31.
- Ask your friend to think of his birth date (or any number from 1 to 31).
- Ask your friend to observe carefully which cards contain the number he thought of.
- The trick to identifying the number in your friend's mind is given elsewhere in the book, find it and identify the number in his mind.

### 2) Star

A five-pointed star is given. A regular pentagon is formed inside it. The measure of one angle in the star is given. From this, find and state the measures of all the angles in the star.



**Instruction 1:** The sum of the two angles form a straight line is  $180^\circ$ .

**Instruction 2:** All sides and angles of a regular pentagon are equal in measure.

**Instruction 3:** The sum of the measures of all the angles of a triangle is  $180^\circ$ .

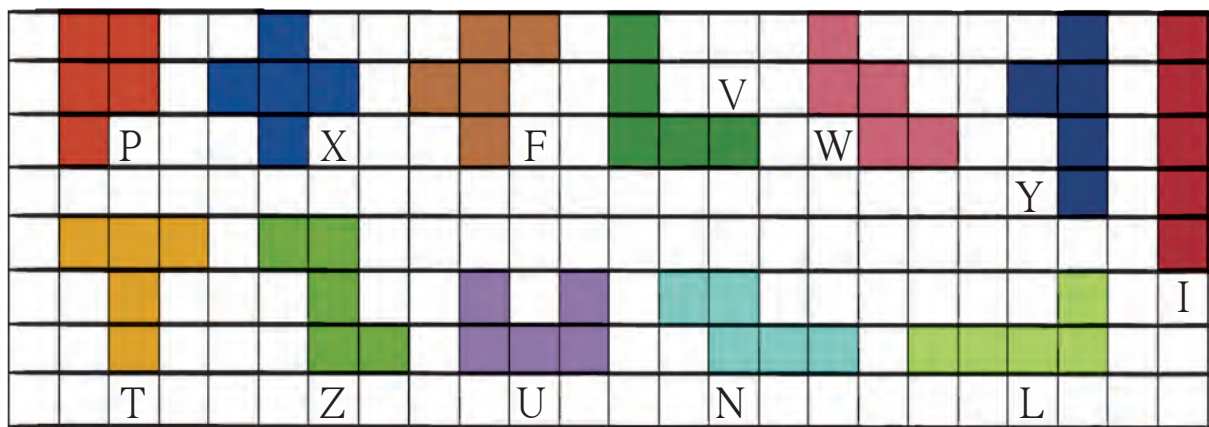
- What is the sum of the measures of all the angles of a pentagon?
- If the star has six points, what difference will there be in the measures of angles?
- If the star has more than six points, what will be the measure of each angle of the figure (polygon) formed inside it? Can you find the formula for the sum of the measures of all its angles? Try it out.

**3. Pentominoes:** A shape formed by connecting 5 squares, of the same size, side-by-side.

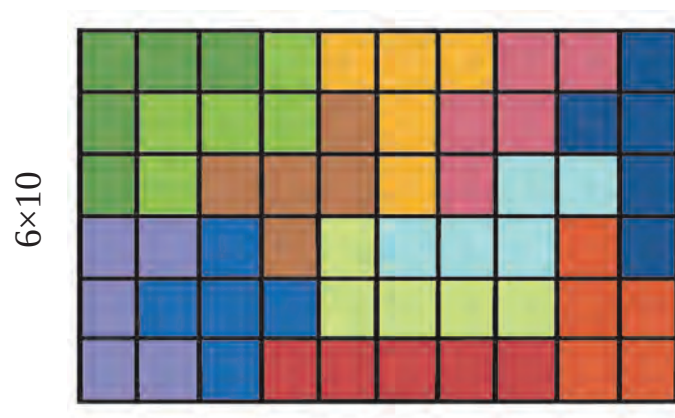
Take a square or rectangular grid paper.

Cut out pentominoes of different shapes from it as follows. The number of possible ways to form a pentomino is given below.

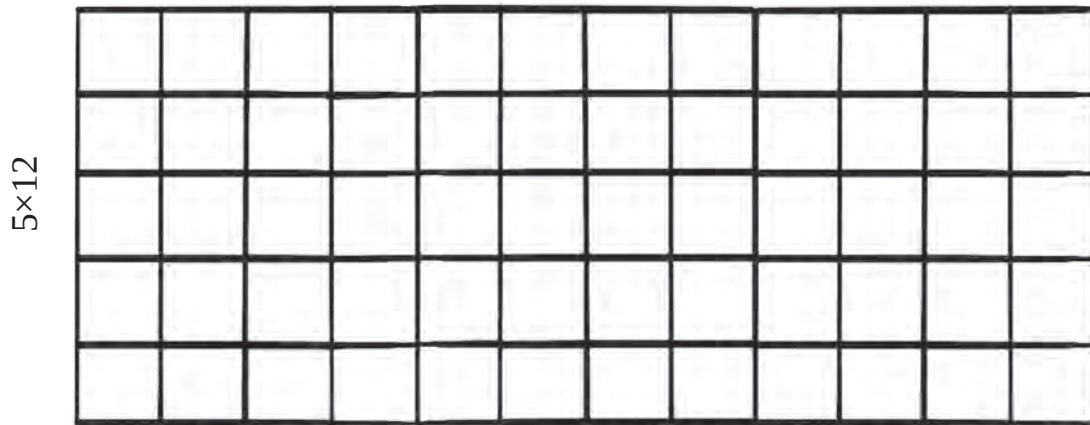
To help you remember a pentomino, English letter is given according to its shape.



**Following 6×10 rectangle is completed by using above pentamines.**

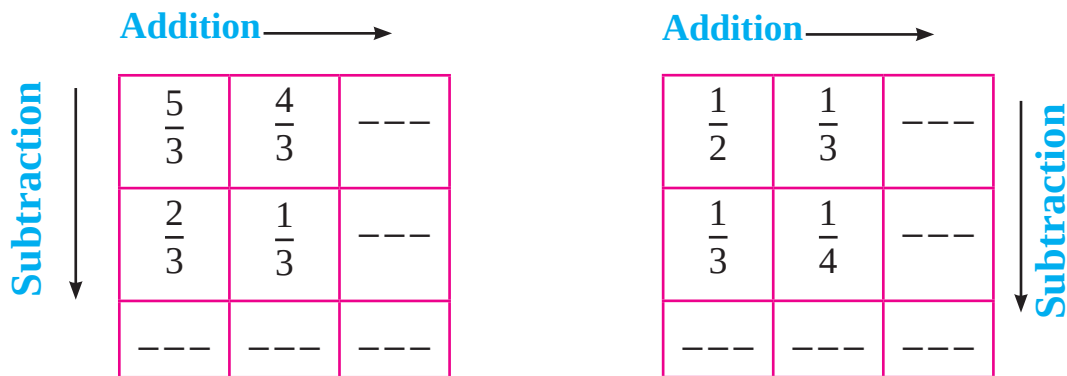


Complete the following  $5 \times 12$  rectangle as above :



In this way, take a  $4 \times 15$  and  $3 \times 20$  grid and make a pentominoes structure and enjoy the game. Note the area of each rectangle and the pentominoes placed in it (Tessellation- No Overlap, No Gaps). Understand the correlation between these two. Look at the dimensions of each rectangle. Do you see any similarity in them? Think about it. Discuss it.

#### 4. Addition-Subtraction Grid.



- For which other fractions can such grids be made? Create them and verify your answers..
- Make such a grid for any numbers other than fractions.







## 11 Decimal Fractions



### Revision

Previously we learned that a fraction is written as a quotient:  $\frac{\text{Numerator}}{\text{Denominator}}$

Observe the following fractions. Identify those fractions whose denominators are 10, 100, 1000, 10,000.

$$\frac{7}{5}, \frac{4}{10}, \frac{15}{100}, \frac{287}{119}, \frac{438}{1000}, \frac{27}{10}, \frac{9}{10000}, \frac{5}{8}, \frac{10}{17}, \frac{5}{100}, \frac{89}{1000}, \frac{45}{10}$$

The denominators of the following fractions are 10, 100, 1000, ...

$$\frac{4}{10}, \frac{15}{100}, \frac{27}{10}, \frac{438}{1000}, \frac{9}{10000}, \frac{5}{100}, \frac{89}{1000}, \frac{45}{10}$$

### Decimal Fraction



### Understand

- Fractions whose denominators are 10, 100, 1000,... are called decimal fractions. While writing these fractions, just as the decimal system has unit's ten's, hundred's, thousand's places in the order, on the right side of unit's place one tenth  $\frac{1}{10}$ , one hundredth  $\frac{1}{100}$ , one thousandth  $\frac{1}{1000}$  places are there.
- One tenth means taking exactly 1 part out of 10 equal parts of a whole.
- While writing numbers, a symbol (.) is placed after the units place in the integer part to indicate the end of the integer part. This symbol is called the 'decimal symbol'.
- The mix fraction  $5\frac{3}{10}$  is written as 5.3. It is read as 'five integer three by ten' or 'five decimal point three'.
- While writing  $\frac{5}{100}$  we write the digit 0 in the decimal place, keeping in mind that there is nothing in that place. It is written as 0.05 and read as 'zero integer point zero five'.



- Just as the numbers on a scale are written in decimal notation, we see price tags of goods and meter readings at petrol pumps written in decimal notation.
- Find other places in daily life where the decimal system is used.

### Expanded forms of decimal fractions

- The place values of the digits in the number 54.632 are given in the table below.

| Position                                               | Ten's place        | Unit's place     | Decimal point | One-Ten's place               | One-hundredth's place           | One-thousandth's place            |
|--------------------------------------------------------|--------------------|------------------|---------------|-------------------------------|---------------------------------|-----------------------------------|
|                                                        | 10                 | 1                | .             | $\frac{1}{10}$                | $\frac{1}{100}$                 | $\frac{1}{1000}$                  |
| Digit                                                  | 5                  | 4                | .             | 6                             | 3                               | 2                                 |
| Place value                                            | $5 \times 10 = 50$ | $4 \times 1 = 4$ | .             | $6 \times \frac{1}{10} = 0.6$ | $3 \times \frac{1}{100} = 0.03$ | $2 \times \frac{1}{1000} = 0.002$ |
| <b>Reading:</b> Fifty-four decimal point six three two |                    |                  |               | <b>Writing :</b> 54.632       |                                 |                                   |



- Expanded form:  $54.632 = 5 \times 10 + 4 \times 1 + 6 \times \frac{1}{10} + 3 \times \frac{1}{100} + 2 \times \frac{1}{1000}$   
 $= 50 + 4 + \frac{6}{10} + \frac{3}{100} + \frac{2}{1000}$

### Illustrative Examples:

- $3.342 = 3 \times 1 + 3 \times \frac{1}{10} + 4 \times \frac{1}{100} + 2 \times \frac{1}{1000} = 3 + \frac{3}{10} + \frac{4}{100} + \frac{2}{1000}$
- $45.305 = 4 \times 10 + 5 \times 1 + 3 \times \frac{1}{10} + 0 \times \frac{1}{100} + 5 \times \frac{1}{1000}$   
 $= 40 + 5 + \frac{3}{10} + 0 + \frac{5}{1000}$



### Understand

### Comparison of Decimal Fractions

While comparing decimal fractions.

- First, compare the integers. For example, 24.27, 32.48 are compared as  $24.27 < 32.48$ .
- If the integers are equal, then the digits in the first decimal place after the decimal point should be compared. For example, 9.42 and 9.72 are compared as  $9.42 < 9.72$
- If the digits in the first decimal place are also the same, then the digits in the second decimal place should be compared. For example, 8.78, 8.71 are compared as  $8.78 > 8.71$ .



## Practice Set 11.1

- (1) Write the following fractions as decimal fractions.
 

|                      |                        |                       |                       |
|----------------------|------------------------|-----------------------|-----------------------|
| i) $\frac{16}{100}$  | ii) $\frac{275}{1000}$ | iii) $\frac{6}{100}$  | iv) $\frac{269}{100}$ |
| v) $\frac{15}{1000}$ | vi) $\frac{235}{10}$   | vii) $\frac{7}{1000}$ | viii) $6\frac{3}{10}$ |
- (2) Write the place value of the underlined digits.
 

|                   |                   |                    |                   |                 |
|-------------------|-------------------|--------------------|-------------------|-----------------|
| i) 34. <u>2</u> 6 | ii) 8. <u>8</u> 8 | iii) 0. <u>8</u> 1 | iv) 0. <u>4</u> 0 | v) <u>6</u> .64 |
|-------------------|-------------------|--------------------|-------------------|-----------------|
- (3) Determine the smaller and larger numbers.
 

|              |                  |                 |                |               |
|--------------|------------------|-----------------|----------------|---------------|
| i) 0.2 , 0.4 | ii) 70.08 , 70.4 | iii) 6.6 , 6.58 | iv) 7.4 , 7.35 | v) 0.76 , 0.8 |
|--------------|------------------|-----------------|----------------|---------------|
- (4) Write the following numbers in ascending and descending order.
 

|                         |                           |
|-------------------------|---------------------------|
| i) 8.6, 8.59, 8.09, 8.8 | ii) 7.8, 7.66, 8.06, 8.68 |
|-------------------------|---------------------------|

### Conversion of common fractions to decimal fractions



#### Understand

- If the denominator of a common fraction is 10 or a multiple of 10, then it can be written as a decimal fraction.  
We can write. For example i)  $\frac{7}{10} = 0.7$  ii)  $\frac{79}{100} = 0.79$  iii)  $\frac{45}{1000} = 0.045$
- While converting common fractions to decimal fractions, the denominator has to be 10, 100, 1000,... If it is not in this form, we multiply numerator and denominator by suitable number and make denominator as 10, 100, 1000,.....
- If denominator of a common fraction cannot be converted as 10, 100, 1000,..... the fraction can be converted into decimal fractions by dividing the numerator and denominator by the same number.

### Converting decimal fractions to common fractions



#### Understand

- The fraction 32.46 is converted into a common fraction as  $32.46 = \frac{3246}{100}$
- Here, there are two digits after the decimal point, so two zeros are written after 1 in the denominator of the common fraction and the number is written in the numerator without considering the decimal point.

Example :

|                                |                                   |
|--------------------------------|-----------------------------------|
| i) $0.7 = \frac{7}{10}$        | ii) $0.17 = \frac{17}{100}$       |
| iii) $0.045 = \frac{45}{1000}$ | iv) $25.382 = \frac{25382}{1000}$ |



## Illustrative Examples:

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$$1) \frac{3}{4} = \frac{3 \times 25}{4 \times 25} = \frac{75}{100} = 0.75$$

$$2) \frac{23}{125} = \frac{23 \times 8}{125 \times 8} = \frac{184}{1000} = 0.184$$

$$3) \frac{16}{400} = \frac{16 \div 4}{400 \div 4} = \frac{4}{100} = 0.04$$

$$4) \frac{9}{20} \text{ Kilogram}$$

$$= \frac{9 \times 50}{20 \times 50} \text{ Kilogram}$$

$$= \frac{450}{1000} \text{ Kilogram} = 0.450 \text{ Kilogram}$$

**Writing half, quarter, three quarters, one and a quarter etc. as decimal fractions**

|                   |                 |                                                 |                   |       |
|-------------------|-----------------|-------------------------------------------------|-------------------|-------|
| Half              | $\frac{1}{2}$   | $\frac{1 \times 5}{2 \times 5}$                 | $\frac{5}{10}$    | 0.5   |
| Quarter           | $\frac{1}{4}$   | $\frac{1 \times 25}{4 \times 25}$               | $\frac{25}{100}$  | 0.25  |
| Three Quarter     | $\frac{3}{4}$   | $\frac{3 \times 25}{4 \times 25}$               | $\frac{75}{100}$  | 0.75  |
| One and a Quarter | $1 \frac{1}{4}$ | $\frac{5}{4} = \frac{5 \times 25}{4 \times 25}$ | $\frac{125}{100}$ | 1.25  |
| One and a half    | $1 \frac{1}{2}$ | .....                                           | .....             | ..... |

## Addition and subtraction of decimal fractions



### Understand

- While adding and subtracting decimal fractions, the decimal fractions should be placed vertically so that the decimal point is below the decimal point.

(1) Add 0.3 and 0.4.

$$\rightarrow 0.3 = \frac{3}{10}, 0.4 = \frac{4}{10}$$

$$0.3 + 0.4 = \frac{3}{10} + \frac{4}{10} = \frac{7}{10} = 0.7$$

$$0.3 + 0.4 = 0.7$$

|   | Unit |   | Decimal |
|---|------|---|---------|
|   | 0    | . | 3       |
| + | 0    | . | 4       |
|   | 0    | . | 7       |

(2) Subtract 0.7 from 1.8.

$$\rightarrow 1.8 = \frac{18}{10}, 0.7 = \frac{7}{10}$$

$$1.8 - 0.7 = \frac{18}{10} - \frac{7}{10} = \frac{11}{10} = 1.1$$

$$1.8 - 0.7 = 1.1$$

|   |      |   |         |
|---|------|---|---------|
|   | Unit |   | Decimal |
|   | 1    | . | 8       |
| - | 0    | . | 7       |
|   | 1    | . | 1       |



3)  $3.64 + 5.4 = ?$

|   | Units |   | Tenths | Hundredths |
|---|-------|---|--------|------------|
| → | 1     |   |        |            |
|   | 3     | . | 6      | 4          |
| + | 5     | . | 4      | 0          |
|   | 9     | . | 0      | 4          |

$3.64 + 5.4 = 9.04$

4)  $20 - 11.35 = ?$

|   | Units | Units         |   | Tenths        | Hundredths |
|---|-------|---------------|---|---------------|------------|
| → |       | 9             |   | 9             |            |
|   | 1     | <del>10</del> |   | <del>10</del> | 10         |
| - | 2     | 0             | . | 0             | 0          |
|   | 1     | 1             | . | 3             | 5          |
|   | 0     | 8             | . | 6             | 5          |

$20 - 11.35 = 8.65$



(1) Add the following numbers.

(i)  $24.11 + 3.80$

(ii)  $14.01 + 1.1 + 1.98$

(iii)  $8.85 + 0.51$

(iv)  $2.3 + 18.84$

(2) Subtract the following numbers.

(i)  $0.86 - 0.34$

(ii)  $1.75 - 0.41$

(iii)  $6.42 - 3.3$

(iv)  $1.7 - 0.9$

(v)  $4 - 1.7$

(iv)  $7.63 - 4$

(3) Write as a common fraction, improper fraction or mixed fraction.

(i) 0.9

(ii) 0.07

(iii) 0.191

(iv) 0.049

(v) 4.36

(vi) 0.015

(vii) 4.56

(viii) 5.84





### At a glance

- If the denominator of a common fraction is 10 or a multiple of 10, then that fraction is called a decimal fraction.
- While adding or subtracting decimal fractions, arrange them so that the decimal point is under the decimal point and add or subtract as usual.
- Uses of Decimal Fractions: Converting a common fraction into a decimal fraction makes comparisons easier. The decimal system works easily when doing calculations in a store or using a calculator. Decimal fractions are also used in scientific experiments or in engineering for very precise measurements.



### Exercise 11

- (1) Fill in the blanks.
  - i) The fraction 0.8 is written as a common fraction as .....
  - ii) The integer part of the number 15.9 is .....
  - iii) The digit in the ten's place in 171.9 is.....
  - iv) The place value of 8 in 9.8 is .....
  - v) The place value of 5 in the number 33.52 is.....
- (2) Write the following decimal fractions in common fraction and reduce them to their simplest form.
  - i) 0.2
  - ii) 0.62
  - iii) 0.605
  - iv) 0.017
  - v) 5.46
- (3) Solve.
  - i)  $23.84 + 12.37$
  - ii)  $5.617 + 4.594$
  - iii)  $12.78 + 10.25$
  - iv)  $61.7 - 45.2$
  - v)  $9.47 - 6.82$
  - vi)  $17.93 - 15.82$
- (4) Write the place value of each digit in the number 246.135.
- (5) Anita practices running for police recruitment. She runs 3.27 km every morning and 2.8 km in the evening. How many kilometres does she run in a day?







## 12 Profit and Loss



### Understand

Like every year, this year too, a 'Green Fair' was organized on the school ground in Mehekari village. The villagers, students, teachers and parents together decided environmental protection as the main topic of the fair.



### Cost Price and Selling Price

**Teacher:** Yesterday's 'Green Fair' was very good. Congratulations to everyone! Shraddha, how much did you spend on purchase of seedlings?

**Shraddha:** Sir, I had purchased seedlings worth ₹1250.

**Teacher:** This is your cost price, right? Raghav, have all the plants you brought been sold?

**Raghav:** Yes sir. We had made an attractive arrangement of all the plants.

**Muskan:** And sir, we had written down the feature of each plant.

**Teacher:** Very good! How many rupees do you collect by selling these plants?

**Shraddha:** Sir, ₹1560/-

**Teacher:** Oh wow! This is your selling price.



Riya: Sir, that means we earned more than we spent.

Teacher: Well done! That means your selling price is greater than the cost price. That means you made a benefit. In business, we call benefit as profit.

Shraddha: How much profit did we make, sir?

Teacher: Profit is obtained by subtracting the cost price from the selling price. Calculate it.

Raghav:  $1560 - 1250 = 310$

$\therefore$  Profit = ₹ 310

**Profit = Selling Price - Cost Price**

Riya: But what if you have received less money than the cost price, sir?

Muskan: Oh, wouldn't that have been a loss to you?

Teacher: This means that if the selling price is less than the cost price, there is a loss.

**Loss = Cost price - Selling price**

| Profit                                                                                  | Loss                                                                                   |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| When selling price > cost price, there is profit<br>Profit = selling price - cost price | When selling price < cost price, there is a loss.<br>Loss = cost price - selling price |



### Mathematical Discussion

1. Which factors need to be considered while determining the selling price to make a profit?
2. Which tricks does the shopkeeper use to increase the sale of goods?
3. If a customer does not have full payment, will you, as a seller, still give him the item?

### Illustrative Examples:

- 1) A fruit seller bought 16 dozen bananas for ₹480 and sold all the bananas at ₹ 45 per dozen. Did he make a profit or a loss in this transaction? How much?

→ Purchase price of 16 dozen bananas = ₹ 480

Selling price of 16 dozen bananas =  $16 \times 45 = ₹ 720$

Here selling price > cost price

$\therefore$  Profit is made.

**Profit = Selling price - Cost price**

$\therefore$  Profit =  $720 - 480 = 240$

The fruit seller made a profit of ₹240.



- 2) A vegetable seller bought 342 kg of tomatoes for ₹8,208. Out of this, 318 kg of tomatoes were sold at ₹25 per kg and 24 kg of tomatoes were spoiled. Did he make a profit or a loss in this transaction? How much?

→ Purchase price = ₹8,208

Selling price of 318 kg tomatoes =  $318 \times 25 = ₹7950$

Here the selling price is < cost price

∴ Loss occurred.

**Loss = Cost price - Selling price**

∴ Loss =  $8,208 - 7,950 = 258$

The vegetable seller suffered a loss of ₹258.

### Activity

- Give the experience to students of buying and selling food, toys, handicrafts, etc. at the school's fun fair.



- 1) Complete the table below.

| Ex.  | Cost price (₹) | Selling price (₹) | Profit / Loss | How much ? (₹) |
|------|----------------|-------------------|---------------|----------------|
| i.   | 650            | 720               |               |                |
| ii.  | 1,390          | 1,215             |               |                |
| iii. | 2,750          | 2,560             |               |                |
| iv.  | 8,195          | 8,266             |               |                |

- 2) A shopkeeper bought 150 kg of organic jaggery for ₹7,500. If he sold it for ₹8,250, how much profit did he make?
- 3) Rakhma bought 72 bunches of coriander from the wholesale market for ₹864. Out of them, 8 bunches were spoiled. She sold the remaining bunches for ₹15 each. Did she make a profit or a loss? How much?
- 4) 10 women from a women's self-help group started a venture to make papads. They spent ₹20,000 to make papads. They sold the prepared papads in the market and earned ₹26,500. From the profit, ₹1,500 was given for school material for children in the village. The remaining amount was divided equally among the members.
- What was the total cost to prepare papad?
  - What was the total amount they received from the sale?
  - What was the profit?
  - How much money did they give for social work?
  - How much money was left for distribution among the members?
  - How much did each member receive?
- 5) Taking the purchase price as ₹1,860
- Frame and solve example for profit.
  - Frame and solve example for loss.



## Total purchase price, profit and loss



### Understand



Akash is a progressive farmer. This year he decided to grow fenugreek (Methi) organically. He spent ₹850 on seeds. Instead of chemical fertilizers, he spent ₹1,250 on manure and organic pesticides. He paid ₹1,650 as wages to the workers. He spent ₹800 on transportation to transport 450 bundles. When he went to the market, he put up a signboard and advertised "Organic Methi". He sold each bunch at ₹25. Did Akash make a profit or a loss in this transaction? How much?

To figure out this, you will need to calculate the total cost price and selling price.

1. Calculate the total expenses on the crop (cost price).

Total cost price = expenses on Seeds + manure and organic pesticides + wages + transportation cost.

$$\therefore \text{Total cost price} = 850 + 1,250 + 1,650 + 800$$

$$\therefore \text{Total cost price} = ₹4,550$$

2. Calculate the selling price.

Selling price = Total bundles  $\times$  Price per bundle

$$\therefore \text{Selling price} = 450 \times 25$$

$$\therefore \text{Selling price} = ₹11,250$$

3. Let's determine whether there is a profit or a loss.

Here, the profit in this transaction was made because the selling price  $>$  the cost price.

Profit = Selling price - Cost price

$$\therefore \text{Profit} = 11,250 - 4,550$$

$$\therefore \text{Profit} = ₹6,700$$

$$\therefore \text{Akash made a profit of ₹6,700 from selling organic fenugreek.}$$

In a purchase and sale transaction, the cost of manufacturing the item and all other expenses incurred for it have to be included in the cost price before selling it. This is called the 'total cost price'. Total cost price = Original cost price + Other expenses.



### Discuss.

1. If you treat customers angrily, will they come back? What can you do to improve your behaviour so that customers come back to you?
2. What will you feel if you make a profit? What will you learn from a loss?
3. What would be the purpose of charitable organization selling goods on a 'no profit, no loss' basis?





## Illustrative Examples:

- Sangeeta spent ₹10,800 including GST to buy 12 sarees of the same price. She spent ₹750 on transportation and made a profit of ₹5,250 by selling all the sarees. What is the selling price of one saree?

→ Total cost price = Original cost price + Other expenses

$$\therefore \text{Total cost price} = 10,800 + 750 = ₹11,550$$

$$\text{Profit} = \text{Selling price} - \text{Total cost price}$$

$$\therefore \text{Selling price} = \text{Profit} + \text{Total cost price}$$

$$\therefore \text{Selling price} = 5,250 + 11,550 = 16,800$$

$$\text{Selling price of one saree} = \text{Total selling price} \div \text{Number of sarees}$$

$$\therefore \text{Selling price of a saree} = 16,800 \div 12 = ₹1,400$$

$$\therefore \text{The selling price of a saree is ₹1,400.}$$

## Activity

### Market survey

- Go to the nearest market and talk to the shopkeepers. Observe and obtain information based on the following points. From the information obtained, express your opinion on the impact of each point on the profit and loss of the shopkeepers.
  - Arrangement for the presentation of goods.
  - Timely sale of the goods.
  - Availability of the item in demand.
  - Quality of the item.
  - Advertising and providing information about the product.
  - Selection of a suitable place for sale.



- Iqbal bought an old car for ₹2,75,000. He spent ₹65,000 on repairs. He sold the car for ₹3,89,000. Did he make a profit or a loss? How much?
- Kusum bought sacks from a wholesaler by spending ₹4,750. She lost ₹450 by selling them. For how much did she sell the sacks?
- Sara earned ₹1,285 by selling painting materials. She made a profit of ₹375. What was the purchase price of the materials?
- Priya spent ₹2,505 on materials and ₹455 on packing while making handicrafts. She sold all the items for ₹3,480. Did she make a profit or a loss? How much?
- Taking the selling price ₹1,355
  - Frame and solve a profit example.
  - Frame and solve an example of a loss.





### At a glance

- A profit occurs when the selling price is more than the cost price.
- Profit = Selling price - Cost price
- A loss occurs when the selling price is less than the cost price.
- Loss = Cost price - Sale price
- Total cost price = Original cost price + Other expenses



### Exercise 12

(1) Choose the correct option.

- If the cost price is ₹418 and the profit is ₹72, the selling price is ₹ ....  
(a) 346                      (b) 418                      (c) 490                      (d) 360
- If the selling price is ₹1275 and the loss is ₹84, the cost price is ₹ ....  
(a) 191                      (b) 1359                      (c) 1275                      (d) 1191

(2) Fill in the blanks.

- If Eco-friendly bags were purchased for ₹357 and sold for ₹418, then profit is ...
- If the transportation cost is ₹289 and the cost price is ₹3628, the total cost price is ₹ ....
- If the cost price is ₹143 and the selling price is ₹128, then the loss is ₹ ....

(3) Solve.

- Prajwal bought a bicycle for ₹7,450 and later sold it for ₹6,820. Did he make a profit or a loss? How much?
  - Naresh sold toys in a fair for ₹1,265 at a loss of ₹345. Calculate the cost price of toys.
  - A bookseller bought 118 books for ₹5074. He sold all those books at ₹50 each. Did he make a profit or a loss? How much?
- (4) Julie bought 72 compass boxes at ₹145 each. Her transportation cost to get them from the shop was ₹628. If she sells each compass box at ₹162, does she make a profit or a loss? How much?
- (5) The village youth group purchased 357 tree pots at ₹247 each under the 'Village Green Campaign'. ₹629 was spent on transportation. They sold each pot at ₹289. They decided to donate the profit from the sale to the village library.
- What was the total cost price?
  - What was the selling price?
  - How much amount was donated to the library?
  - What social values are seen in this initiative along with financial transactions?







## Revision

Write the correct number in the box.

i)  $\frac{4}{5} = \frac{4 \times \boxed{\phantom{00}}}{5 \times \boxed{\phantom{00}}} = \frac{\boxed{\phantom{00}}}{100}$

ii)  $0.07 = \frac{\boxed{\phantom{00}}}{100}$

## Percentage



## Understand



I got  $\frac{45}{50}$  marks in mathematics.

I got  $\frac{45}{60}$  marks in mathematics.



Even if their scores are same, you must compare the total marks earned against the maximum marks to determine who actually performed better.

Sakhi's Marks :  $\frac{45}{50}$

$$\frac{45}{50} = \frac{45 \times 2}{50 \times 2} = \frac{90}{100}$$

Sunny's Marks :  $\frac{45}{60}$

$$\frac{45 \div 15}{60 \div 15} = \frac{3}{4} = \frac{3 \times 25}{4 \times 25} = \frac{75}{100}$$

Sakhi got 90 marks out of 100 and Sunny got 75 marks out of 100. Therefore, Sakhi got more marks than Sunny. Here, the denominator of both the ratios is 100. This makes it easy to compare. Ratio in which the denominator is 100 is called percent. This is written as 'percent' or using symbol %.

The symbol % is read as 'percent'. Percent is also called percentage or per cent. Sakhi's marks are 90% and Sunny's marks are 75%, so Sakhi got 15% more marks than Sunny.

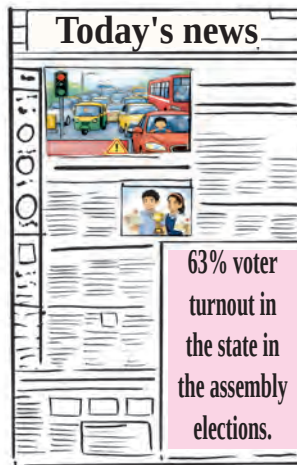
The word 'percent' is derived from the Latin word 'per centum'. 'Per centum' means 'per hundred' (per 100).



## Discuss.

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Observe the pictures and answer the following questions.



- What is the % discount on ready-made garments?
- What percent of votes were cast in the assembly elections?

The advertisement has announced a 40% discount on ready-made clothes. 40% discount means that if a dress costs ₹ 100, then there will be a discount of ₹ 40 and the customer will get the dress for ₹ 60.

## Converting a fraction to a percent



## Understand

### Illustrative Examples:

- 1)  $\frac{3}{5}$  means what percent ?

To make the denominator of the above fraction 100, multiply the denominator, 5, by 20. To make an equivalent fraction, the numerator also needs to be multiplied by 20.

$$\frac{3}{5} = \frac{3 \times 20}{5 \times 20} = \frac{60}{100} = 60\% \text{ (Sixty percent)}$$

- 2) Convert 1.3 to percent.

$$1.3 = \frac{13}{10} = \frac{13 \times 10}{10 \times 10} = \frac{130}{100} = 130\%$$

- 3) What percent is 0.25?

$$0.25 = \frac{25}{100} = 25\%$$

- 4)  $\frac{5}{8} = \frac{5 \times 100}{8 \times 100} = \frac{500}{800} = \frac{500 \div 8}{800 \div 8} = \frac{62.5}{100} = 62.5\%$



## Convert percentages to fractions

### Illustrative Examples:

1) Convert 50% to a fraction.

$$50\% = \frac{50}{100} = \frac{50 \div 50}{100 \div 50} = \frac{1}{2}$$

2) Convert the 49 percent to a fraction.

$$49 \text{ percent } \frac{49}{100}$$

3) Convert  $\frac{5}{4}\%$  to fraction.

$$\begin{aligned} \frac{5}{4}\% &= \frac{5}{4} \div 100 = \frac{5}{4} \times \frac{1}{100} \\ &= \frac{5}{400} = \frac{1}{80} \end{aligned}$$

4) Convert  $8\frac{1}{2}\%$  to a fraction.

$$\begin{aligned} 8\frac{1}{2}\% &= \frac{17}{2}\% = 8.5\% \\ &= \frac{8.5}{100} = \frac{85}{1000} = \frac{17}{200} \end{aligned}$$



(1) Write the following ratios using the word 'percent' and the symbol '% '.

i)  $\frac{71}{100}$

ii)  $\frac{45}{100}$

iii)  $\frac{186}{100}$

iv)  $\frac{19}{100}$

(2) Complete the following activities.

i)  $\frac{1}{2} = \frac{1 \times \boxed{\phantom{00}}}{2 \times \boxed{\phantom{00}}} = \frac{\boxed{\phantom{00}}}{\boxed{\phantom{00}}} = \boxed{\phantom{00}}\%$

ii)  $\frac{5}{4} = \frac{5 \times \boxed{\phantom{00}}}{4 \times \boxed{\phantom{00}}} = \frac{\boxed{\phantom{00}}}{\boxed{\phantom{00}}} = \boxed{\phantom{00}}\%$

(3) Convert the following percentage into fractions.

i) 64 percent    ii) 91%    iii) 175%    iv)  $16\frac{2}{3}\%$     v) 100%

(4) Convert the following fractions into percentage.

i)  $\frac{2}{5}$     ii)  $\frac{3}{8}$     iii)  $\frac{5}{4}$     iv)  $\frac{1}{16}$     v)  $\frac{736}{800}$





## Understand

### To find expected percentage of given number

- 1) What is '30 percent of 90'?
- 30 percent of 90 =  $90 \times 30\%$

$$= 90 \times \frac{30}{100} = 27$$

- 2) If Sameera got 70% marks in 800 marks exam, how many marks did she get in total?

$$\begin{aligned} 70\% \text{ of } 800 &= 800 \times 70\% \\ &= 800 \times \frac{70}{100} = 560 \end{aligned}$$

### To find what percent a number is of another number

- 1) Out of 300 pages in the book, 180 pages were read. What percent pages were read?

180 out of 300 pages, means  $\frac{180}{300}$

$$\frac{180}{300} = \frac{180 \div 3}{300 \div 3} = \frac{60}{100} = 60\%$$

60 % pages were read.

- 2) What percent of 80 is 32?
- Suppose,  $x\%$  of 80 = 32

$$80 \times \frac{x}{100} = 32$$

$$\therefore \frac{x}{100} = \frac{32}{80}$$

$$\therefore x = \frac{32}{80} \times 100 = 40$$

$$\therefore 40\% \text{ of } 80 = 32$$

### Word Problems :

- 1) The price of a sowing machine is ₹1,05,000. If 35% subsidy is given on this, then how much subsidy will be given on the sowing machine?

Price of the sowing machine = ₹1,05,000

Percentage subsidy = 35%

Subsidy = Price of the sowing machine  $\times$  Percentage

$$= 1,05,000 \times 35\%$$

$$= 1,05,000 \times \frac{35}{100}$$

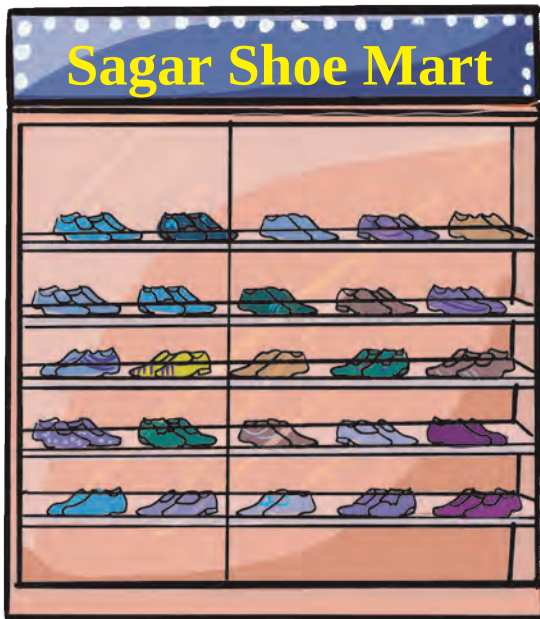
$$= 1,050 \times 35$$

$$= \mathbf{36,750}$$

$\therefore$  Subsidy on sowing machine is ₹36,750

- 2) Rohan and Rajat went to a market to buy sports shoes worth ₹500. Rohan bought the shoes from Sagar Shoe Mart. The shopkeeper gave him a discount of ₹125. Rajat bought the shoes from Ramesh Shoe Mart. The shopkeeper gave him a discount of 30%. Who got the shoes at a lower price?





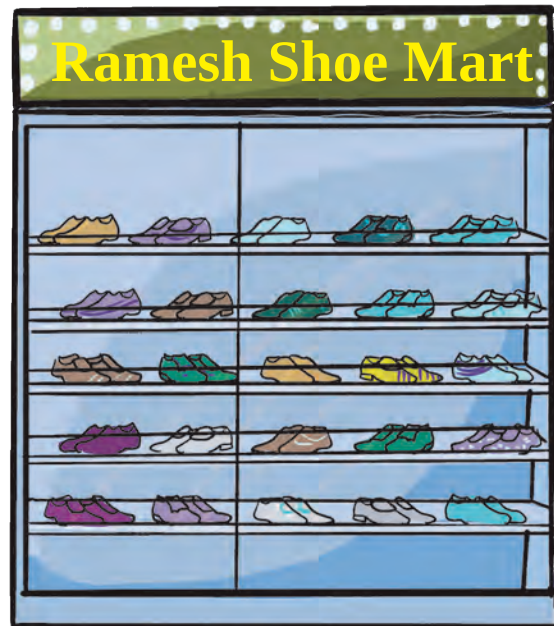
Rohan bought shoes from Sagar Shoe Mart.

The shopkeeper gave him a discount of ₹ 125.

$$\text{C.P.} = ₹ 500$$

$$\text{Discount} = ₹ 125$$

$$\begin{aligned} \text{S.P.} &= 500 - 125 \\ &= ₹ 375 \end{aligned}$$



Rajat bought shoes from Ramesh Shoe Mart.

The shopkeeper gave him a 30% discount.

$$\text{C.P.} = ₹ 500$$

$$\text{Discount} = 30\%$$

$$\therefore \text{Discount} = 500 \times \frac{30}{100} = 5 \times 30 = ₹ 150$$

$$\text{S.P.} = 500 - 150 = ₹ 350$$

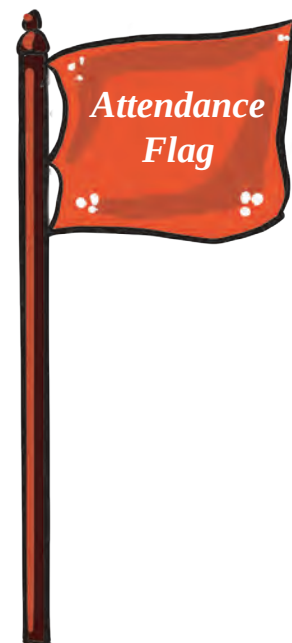
On comparison, it is noticed that Rajat got the shoes at a lower price than Rohan.



**Do it.**

Each day, the class with the highest proportion of attendance hoists the attendance flag in front of their classroom. Determine, from the table below, which class will hoist the attendance flag today.

| Class   | Total strength | Today's Attendance | Percentage Attendance |
|---------|----------------|--------------------|-----------------------|
| Fifth   | 50             | 45                 |                       |
| Sixth   | 60             | 48                 |                       |
| Seventh | 60             | 42                 |                       |
| Eighth  | 70             | 56                 |                       |



- Calculate the percentage attendance of each class.
- Compare the percentage attendance of all classes.
- Which class will hoist the attendance flag today?

### Activity

- Look at the board that shows today's attendance at your school and decide which class should be awarded the 'attendance flag'.



- Find the given percentage of the given number?
  - 80% of 125
  - 37% of 67
  - 125% of 300
  - 87% of 1200
- In each of the following pairs of numbers, calculate what percent the first number is of the second number?
  - 4, 10
  - 44, 88
  - 23, 92
  - 60, 300
  - 215, 250
- 384 out of 400 students voted in the school cabinet election.
  - What was the voting percentage in the election?
  - What is the percentage of students who did not vote?
- The price of a tractor is Rs. 5,50,000 and 45% subsidy is given on it. How much subsidy is given on the tractor?
- Jowar was sown in 75% of a field with an area of 19500 square meters. In how many square meters of the field was jowar sown?
- Kasturi bought one bag of organic fertilizer for Rs.1000. She got 15% discount on it. How much discount will she get on two such bags?



### Mathematical Discussion

- Learn about balanced diet and discuss its components with your teacher.



### At a glance

- A percentage is a fraction whose denominator or a ratio whose second term is 100.
- 'Percentage' is also called 'percent' or 'per cent'. It is represented by the symbol '%'. '%' is read as 'percent'.
- Percentages are used in transactions: profit and loss, grants received, analysis of student scores, results, interest rates, and inflation rates.







### Exercise 13

- 1) Choose the correct option.
  - (i) 10% of 500 = .....  
(a) 10            (b) 50            (c) 5            (d) 100
  - (ii) The number 12 is ..... percent of 40?  
(a) 15            (b) 20            (c) 25            (d) 30
- 2) Solve the following examples and write the correct symbol among  $<$ ,  $>$ ,  $=$  in the box.
  - (i) 15 percent of 60        10 percent of 10
  - (ii) 20% of 10            10% of 20
  - (iii) 10 percent of 250        20 percent of 125
  - (iv) 30% of 160            40% of 150
- (3) The number of malnourished people in a country decreased from 1,20,00,000 to 90,00,000 in 5 years. Then find the percentage by which the number of malnourished people decreased in these 5 years.
- (4) Out of 14000 sugarcane seedlings brought by a farmer from a nursery, 700 seedlings were bad.
  - i) How many seedlings were good?
  - ii) What percent of the seedlings were good?
  - iii) What percentage of the seedlings were bad?
- (5) Doctor has advised meena to take 25gm protein every day. From one glass milk she gets 8gm protein. What percent of daily protein requirement does she get from one glass milk ?
- (6) In an election, Deepti got 11484 votes, which was 44% of the total votes. Trupti got 26% of the total votes. Out of the total votes, how many people did not vote for either of them?
- (7) A cricket team played 20 matches last year. Out of them, it lost 40% matches and won all the remaining matches. How many matches did it win?
  - i) What changes need to be made in the example above to make the answer '15 matches won'? Do that, rewrite the example, solve it and check.
  - ii) If you want to increase the winning percentage of matches by 10, what change would you make in the example? Write it.



## Remember

$\frac{\%}{\text{oo}}$  is the percentage symbol.  
But it should be %. It has  
an extra zero at the bottom.  
Percent means how many out  
of every 100, right ?

Does symbol  
 $\frac{\%}{\text{oo}}$  have a different  
meaning ?

Yes,  
exactly

The salinity of  
seawater on the west  
coast of India  
is 35  $\frac{\%}{\text{oo}}$



Oh, do you know  
about  $\frac{\%}{\text{oo}}$  symbol ?

Yesterday, when my brother  
was studying, I saw symbol  
 $\frac{\%}{\text{oo}}$  in his Geography book. It  
is used for "how many out of  
a thousand".

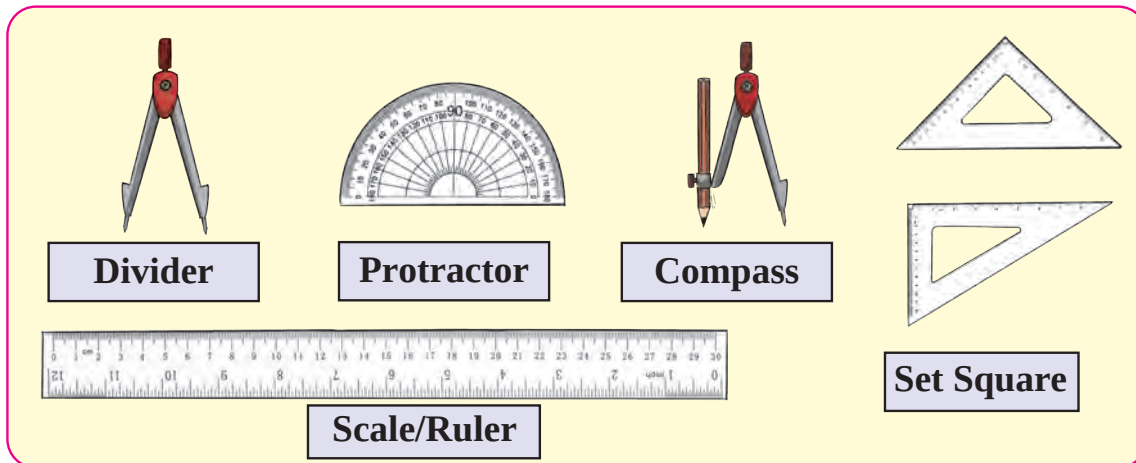
Per hundred – %

Per thousand –  $\frac{\%}{\text{oo}}$



Revision

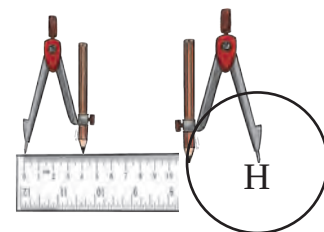
## Identification of compass box



Use the tools in your compass box to complete the following activities.

- 1) Take a point on a paper. Name it H. Now, with the help of a ruler, can you show all the points that are 4 cm away from H? If not, keep a distance 4 cm in the compass and place the tip of the compass at point H and rotate the pencil.

What shape do you get? A circle! We can draw a circle using a compass.



Point H is called the centre of the circle. And the distance taken is called the radius. The distance of each point on the circle from point H is 4 cm. Show the centre, radius and diameter of this circle.

- 2) What would you use if you want the exact length of the given line segment? You can measure the length of the given line segment using a ruler.

Here the length of line segment AB is 8 cm. It is written as  $l(AB) = 8 \text{ cm}$



A. \_\_\_\_\_ .B

Measure and check the length of line segment AB using a divider and a measuring scale.

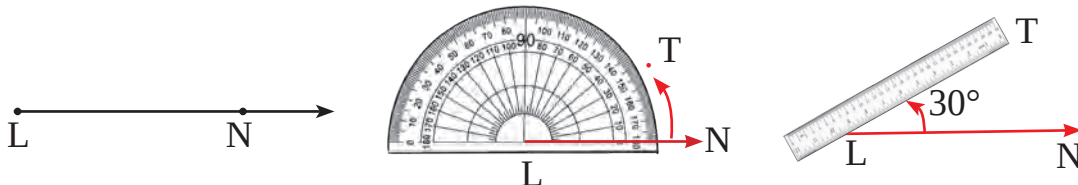


**Divider :** The tool shown on the side is called a divider. It is used to measure the distance between two points. For this, a measuring scale is also used.

### 3) Drawing angles using a protractor.

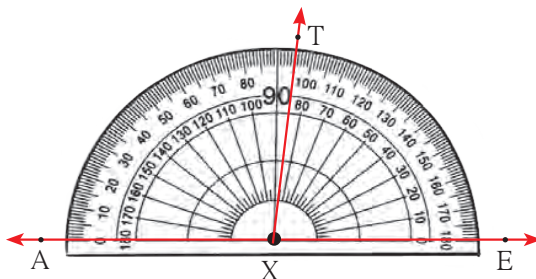
**Example.** Draw  $\angle TLN$  of measure  $30^\circ$ .

- (1) Draw the ray LN.
- (2) Place the centre of the protractor at L and take point T at  $30^\circ$  mark so that the bottom line lies on LN.
- (3) Draw ray LT.
- (4)  $\angle TLN$  is an angle of measure  $30^\circ$ .  $m\angle TLN = 30^\circ$

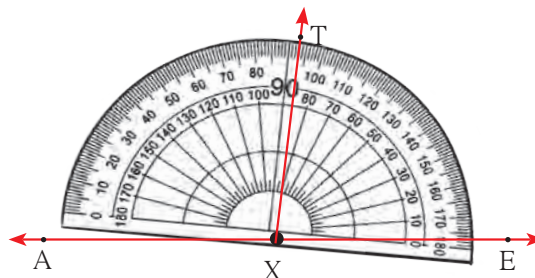


### Mathematical Discussion

- Can you draw  $45^\circ$  angle without using a protractor? Can you use set square for that? Try drawing it.
- Which other angles can be drawn without using a protractor? Draw them in your notebook.
- Below are two pictures of angle measurements. Find what is wrong with them.



$m\angle TXE = 95^\circ$



$m\angle TXE = 87^\circ$



### Practice Set 14.1

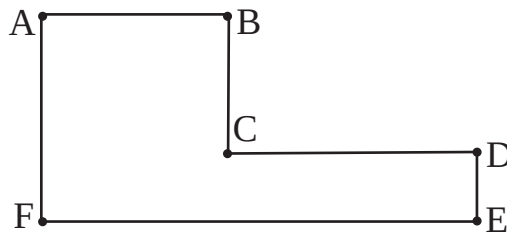
- (1) Measure and write the lengths of all the line segments in the figure below.

The length of line segment BC =  $l(BC)$

= ..... cm,

The length of line segment AB =  $l(AB)$

= ..... cm



- (2) Draw a line segments of the following measurements.

i) Seg AR = 6 cm.    ii) Seg BT = 4 cm.    iii) Seg DS = 9 cm

- (3) Draw angles of the following measurements.

i)  $m\angle BRT = 70^\circ$

ii)  $m\angle XYZ = 105^\circ$

iii)  $m\angle ABC = 140^\circ$



## Mathematical Game

- Ask your friend to measure a line segment on a paper. Estimate its length. Then measure the actual length. If your answer is very close to the actual length, you get 1 point, otherwise 0. Then you draw a line segment and ask your friend to guess. The one with the more points wins.
- This game can also be played by dividing the class into two groups, estimating the angle measure and then actually measuring it.

| Segment                                                                           | Estimated length | Actual length | Marks |
|-----------------------------------------------------------------------------------|------------------|---------------|-------|
|  | 2                | 2.1           | 1     |
|                                                                                   |                  |               |       |
|                                                                                   |                  |               |       |

## References in history



The given picture is of the musical pillars in the Vitthal Temple in Hampi (Karnataka). These pillars in this temple, built in the 15th century, are of uniform length. What would have been used to measure length in the past when the measuring scale was not invented? Collect information about it.







## Understand

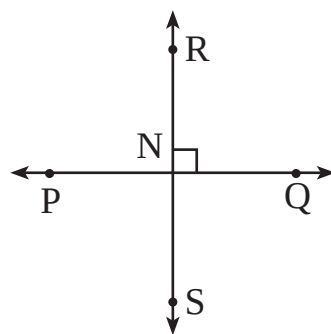
# Geometrical Constructions

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## Perpendicular Line :

In the figure to the side, line PQ and line RS intersect each other at point N. Measure each angle that is near point N. Are these angles right angles?

Since these lines intersect each other at right angles, they are said to be perpendicular to each other. This is represented by the symbol 'Line PQ  $\perp$  Line RS'. It is read as 'Line PQ Perpendicular Line RS'.



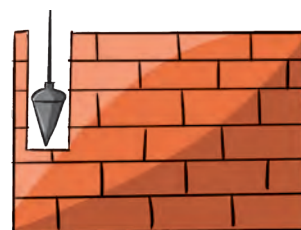
Just as two lines are perpendicular to each other, two rays, two line segments can be perpendicular to each other. The edges of your notebook are perpendicular to each other.



## Think about

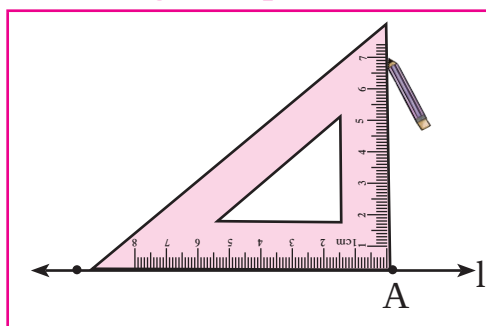
### How do you measure your height?

- When we stand straight, what is the angle we make with the ground?
- Are we standing perpendicular to the ground?
- Take a piece of paper from a notebook. Fold the paper in the middle. Fold the paper in the middle again as shown in the figure. Draw lines on both the folds. These lines are perpendicular to each other.
- Find out which tool is used to build a wall perpendicular to the ground ?



## I) Drawing a perpendicular at a point on a line to that line

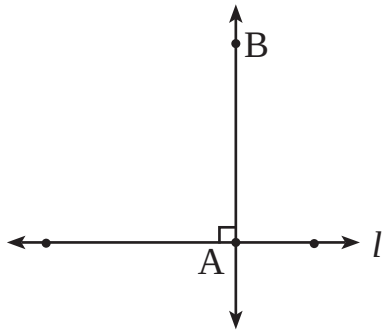
### (1) Using Set square



- 1) To draw a perpendicular at point A on line  $l$ , place the set square on line  $l$  such that one of the edges of forming the right angle is on line  $l$ .
- 2) Move the set square in such a way that the vertex forming the right angle of the set square is at A.





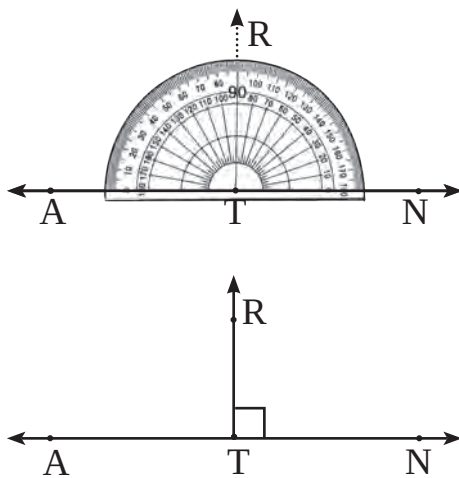


3) Draw line AB along the edge of the set square.

4) Measure the angle between line  $l$  and line AB

5) Line  $AB \perp$  Line  $l$ .

## (2) Using a protractor



T is a point on the line AN. To draw a perpendicular at point T

1) Place the protractor so that the centre point is at T.

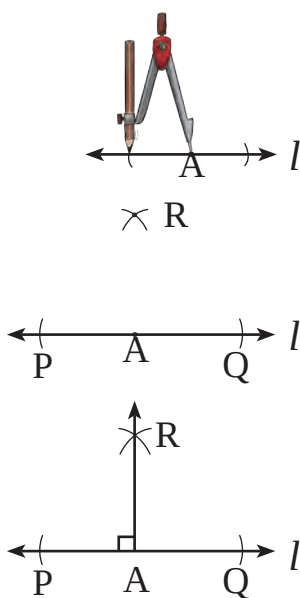
2) Take point R at the  $90^\circ$  mark of the protractor.

3) Join point T and point R

4) Measure  $\angle RTA$  and  $\angle RTN$

5)  $TR \perp AN$

## (3) Using a compass



A is a point on the line  $l$ . To draw a perpendicular to the line  $l$  at point A.

1) Put the tip of the compass at A, draw two arcs intersecting the line. Name the points of intersection as P and Q.

2) In compass take a distance greater than half of PQ.

3) Place the tip of the compass at point P and point Q and draw two intersecting arcs on the same side of the line.

4) Name the intersection point as R.

5) Join point A and point R.

6)  $AR \perp$  line PQ



**Do it.**

- Now you can draw a perpendicular at a point on a line to that line. Using this, draw a rectangle and a square. Which properties of rectangle and square do you need to use for this?

## II) Drawing a perpendicular to a line from a point outside the line.

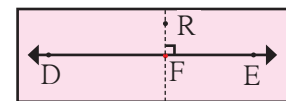
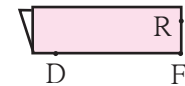
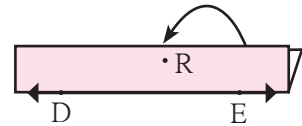
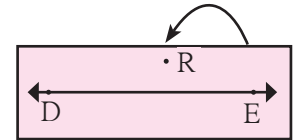
### 1) By folding a paper

The point R is outside the line DE.

To draw the perpendicular to line DE from point R :

- 1) Fold the paper on line DE.
- 2) Fold the paper again at point R so that two parts of line DE coincide with each other.
- 3) Name the bottom point of the fold as F.
- 4) Measure  $\angle RFE$ .

$RF \perp$  line DE



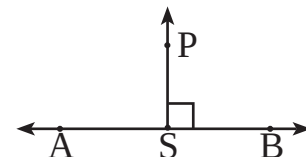
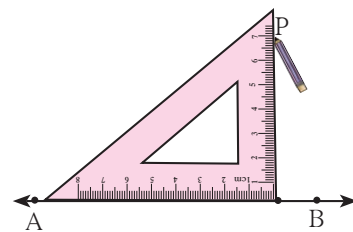
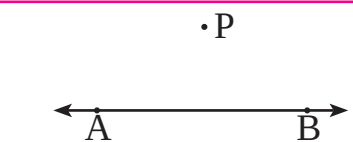
### 2) Using Set square

The point P is on the upper side of the line AB.

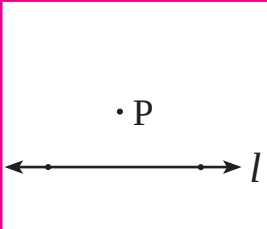
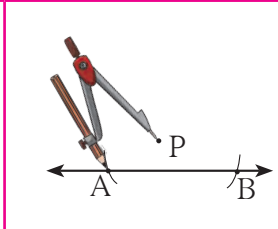
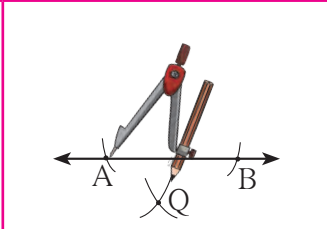
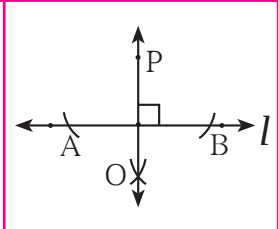
To draw a perpendicular from P to the line AB

- 1) Place the edge of the set square, forming right angle, on line AB.
- 2) Move the set square so that point P lies on the other edge of the set square.
- 3) Name the point of the set square making right angle and on line AB as S.
- 4) Draw the line PS.
- 5)  $\angle PSB$  is a right angle.

$PS \perp$  line AB



### 3) Using a compass and ruler

|                                                                                                         |                                                                                                                                                            |                                                                                                                                                                    |                                                                                                                             |
|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
|                        |                                                                           |                                                                                  |                                          |
| <p>The point P is above the line <math>l</math>. To draw a perpendicular to the line from point P :</p> | <p>1) Put the tip of the compass at point P and draw two arcs intersecting the line at a convenient distance. Name the intersection points as A and B.</p> | <p>2) In compass take distance more than half of AB. Place the compass at point A and point B and draw two arcs intersecting each other on opposite side of P.</p> | <p>3) Name the intersection point as Q.<br/>4) Join the points P, Q.<br/>5) <math>PQ \perp</math> line AB. Verify this.</p> |

### Practice Set 14.2

- 1) Draw a line  $y$ . Take a point D on it. Using compass, draw a line  $n$  perpendicular to line  $y$  from point D.
- 2) Draw the line PQ. Take a point T on the line. Using a set square, draw a perpendicular to the line PQ from point T.
- 3) Draw the line DF. Take two points B and M outside the line. Draw perpendiculars to the line DF through point B and point M using a compass and ruler.
- 4) Draw a line  $m$ . Take points P, Q on the opposite sides of the line. Draw perpendiculars to the line  $m$  through points P and Q. Measure angles formed.
- 5) Draw a square using the tools in the compass box. Measure its sides.



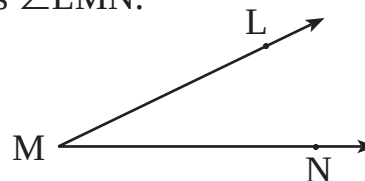
Do it.

### III) To draw an angle of a given measure.

For example, draw  $\angle HAT$  of the same measure as  $\angle LMN$ .

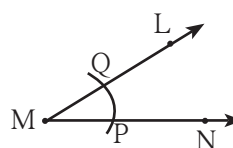
#### (1) With the help of tracing paper.

Place a tracing paper on a figure of any angle and trace the angle. Draw it in your notebook.



#### (2) With the help of compass and straightedge.

1) Take a sufficient distance in the compass and place the compass at point M and draw arcs on side MN and side ML of  $\angle LMN$ . Name the points of intersection P and Q



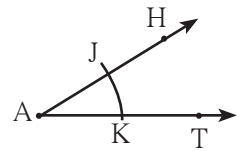
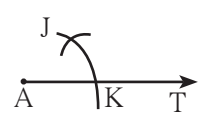
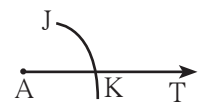
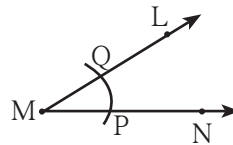
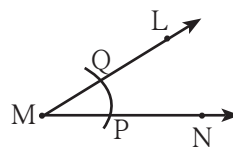
2) Draw ray AT.

3) Place tip of the compass at A and draw an arc of same radius as above. Let it intersect AT in K.

4) Place the tips of the compass at P and take a distance upto point Q. Without changing the distance between the tips, place a tip of the compass at point K and draw another arc intersecting the previous arc at J.

5) Draw ray AH passing through point A and point J.

5) Measure the  $\angle HAT$  and check if it is equal to  $\angle LMN$ .

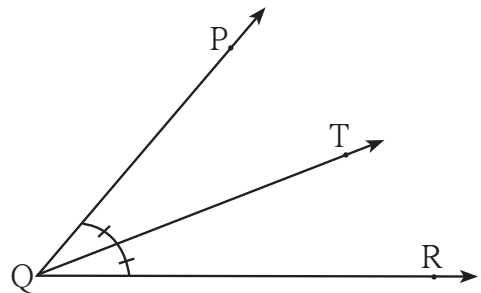


#### IV) Drawing the angle bisector of a given angle.

- A ray that divides a given angle into two equal parts is called the angle bisector of that angle.

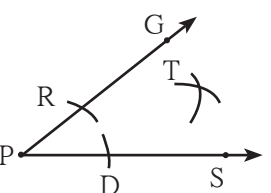
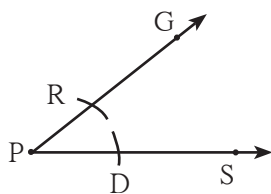
##### i) Drawing angle bisectors using trace paper.

Take a trace paper. Draw an angle of any measure on the trace paper. Fold the paper through the vertex of the angle so that two arms of the angle coincide with each other. What happens to the angle because of the fold? observe. The fold formed divides the angle into two equal parts. The fold is the bisector of that angle. In the figure, ray QT is the bisector of  $\angle PQR$ . Draw angles of various sizes on paper and draw angle bisectors by folding the paper.



##### ii) Drawing the angle bisector with the help of a compass.

Example. Draw angle  $\angle GPS$ . Draw the bisector of this angle.



1) Draw  $\angle GPS$

2) Place the tip of the compass at point P.

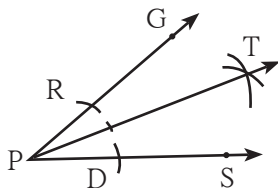
3) Draw arcs on ray PG and ray PS by taking the appropriate distance in the compass.

4) Name the intersection points as R and D

5) Place the tip of the compass at point R and draw an arc inside the angle. Maintaining the same distance between the tips of the compass. Place the tip of the compass at point D and draw another arc intersecting the previous arc.

6) Name the intersection point of the arcs as T.





7) Draw the ray PT.

8) Ray PT is the bisector of  $\angle GPS$

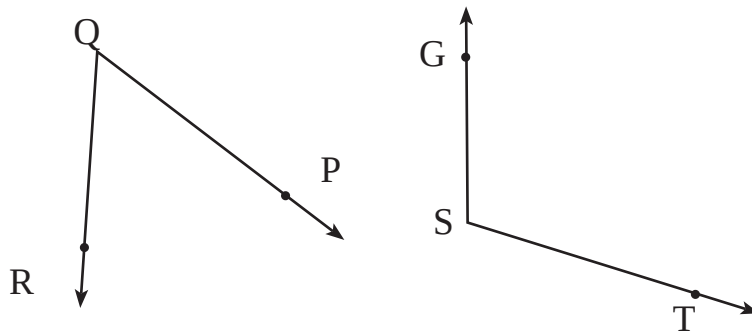
9) Are the measures of  $\angle GPT$  and  $\angle TPS$  the same ? Check this.



- (1) Using the tools in the compass box, draw the angles of the following measure. Use a compass and a ruler to bisect them.

(i)  $110^\circ$       (ii)  $30^\circ$       (iii)  $45^\circ$       (iv)  $90^\circ$

- (2) Using compass and ruler, draw angles equal to the measures of the angles given below.



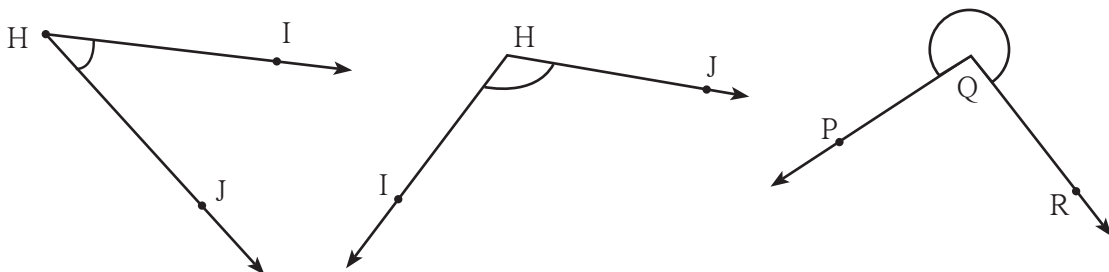
### At a glance

- A line segment of a given length can be drawn using a ruler and a divider.
- An angle can be measured using a protractor and an angle of a given measure can be drawn.
- Lines that intersect at right angles are perpendicular to each other.
- The ray that divides a given angle into two equal parts is called the angle bisector of that angle.

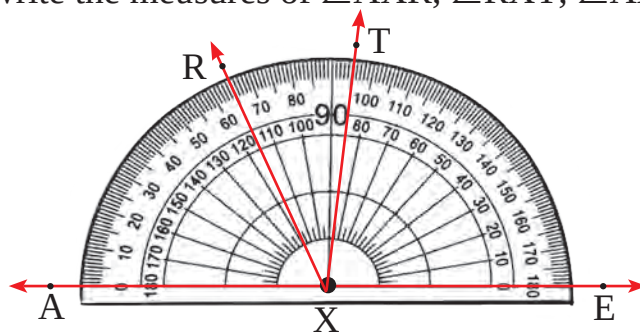


### Exercise 14

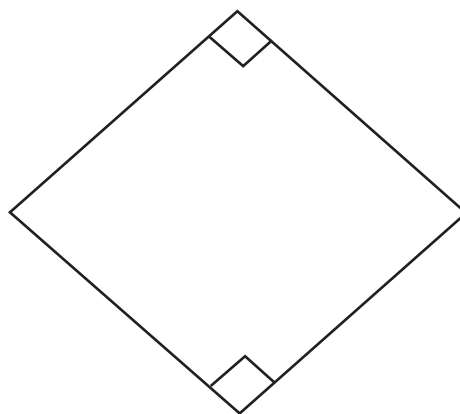
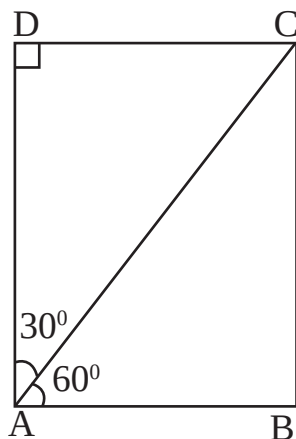
- (1) Take an acute angle and bisect it.
- (2) Draw  $\angle RTG$  of measure  $105^\circ$  and bisect it.
- (3) Measure and write the measures of the angles given below.



- (4) Draw a circle using a compass and draw a 'line-drawing' (Rakha chitr) using other tools in the compass box.
- (5) Measure and write the measures of  $\angle AXR$ ,  $\angle RXT$ ,  $\angle AXT$  and  $\angle RXE$ .



- (6) Draw the following figures using only set squares.



- (7) Complete the table below.

|                                     |  |  |  |
|-------------------------------------|--|--|--|
| Figure of angle                     |  |  |  |
| Approximate measure of the angle    |  |  |  |
| Exact measure by actual measurement |  |  |  |

ICT tool Geogebra Link : <https://www.geogebra.org/classic> Practice drawing angles of different measures using various tools in Geogebra.

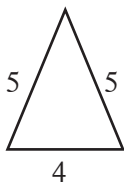




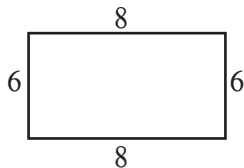
### Revision

### Perimeter

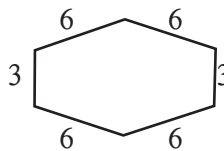
- Add up the length of all the sides in each figure and write them in the blank space below. The length of the sides are in centimetres.



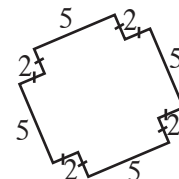
.....



.....



.....



.....

The sum of the length of all the sides of a closed figure is the perimeter of that figure.

### Formula for the perimeter of a rectangle



#### Understand

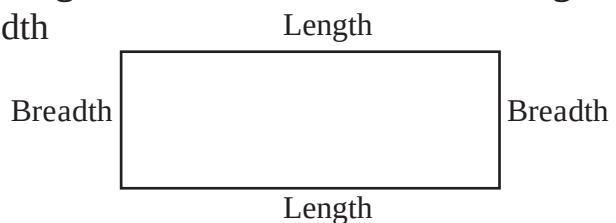
A rectangle is given in the figure. Let us see how the formula for its perimeter is derived.

**Perimeter of a rectangle = Sum of the length of all the sides of the rectangle**

$$= \text{length} + \text{breadth} + \text{length} + \text{breadth}$$

$$= (2 \times \text{length}) + (2 \times \text{breadth})$$

$$= 2 \times (\text{length} + \text{breadth})$$



**Perimeter of a rectangle =  $2 \times (\text{length} + \text{breadth})$**

### Illustrative Examples:

- The length of rectangle LMNO is 8.5 cm and the breadth is 5.5 cm. Find the perimeter of this rectangle.

$$\begin{aligned} \rightarrow \text{Perimeter of rectangle LMNO} &= 2 \times (\text{length} + \text{breadth}) \\ &= 2 \times (8.5 + 5.5) \\ &= 2 \times 14 \\ &= 28 \text{ cm} \end{aligned}$$

**The perimeter of rectangle LMNO is 28 cm.**

- 2) Write the dimensions of the rectangular objects given in the table below and verify the formula for the perimeter of a rectangle.

| Rectangular objects | Length | Breadth | Sum of all sides | $2 \times (\text{length} + \text{breadth})$ |
|---------------------|--------|---------|------------------|---------------------------------------------|
| Black board         |        |         |                  |                                             |
| Notebook            |        |         |                  |                                             |
| Mathematics book    |        |         |                  |                                             |

## The formula for the perimeter of a square

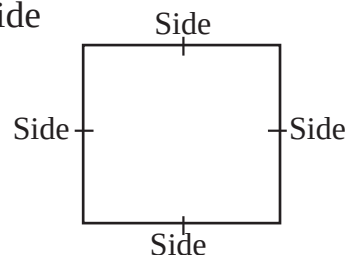


### Understand

All sides of a square have the same length.

Perimeter of a square = Sum of the length of all the sides of the square  
 $= \text{Side} + \text{Side} + \text{Side} + \text{Side} = 4 \times \text{length of side}$

**Perimeter of a square =  $4 \times \text{length of side}$**



### Illustrative Examples:

- 1) How many meters of wire will be needed to fence a square plot with a side length of 20 meters with three layers?

→ To build a single-layer fence, you need to find the perimeter of the square.

$$\begin{aligned} \text{Perimeter of the square} &= 4 \times \text{length of the side} \\ &= 4 \times 20 \\ &= 80 \text{ meters} \end{aligned}$$

The perimeter of the square is 80 meters. The fence is to be built in three layers.

Therefore, we multiply it by three.

$$3 \times 80 = 240 \text{ meters}$$

240 meters wire will be required to build a three-layer fence around the square.

- 2) The perimeter of a rectangle with length 28 cm and breadth 20 cm is equal to the perimeter of a square. What is the side of that square?

$$\begin{aligned} \text{Perimeter of rectangle} &= 2 (\text{length} + \text{width}) \\ &= 2 (28 + 20) = 96 \text{ cm} \end{aligned}$$

In the example, the perimeter of the rectangle is equal to the perimeter of the square



Perimeter of the square = Perimeter of the given rectangle

$$4 \times \text{Side} = 96$$

$$\therefore \text{Side} = \frac{96}{4}$$

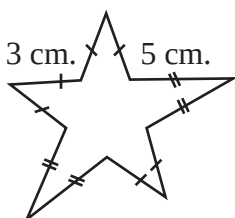
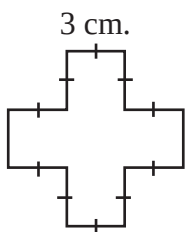
$$\therefore \text{Side} = 24$$

The side of the square is 24 cm.



### Discuss.

- Discuss how to find the perimeters of the following figures. Length of sides are in cm.



### Understand

### Perimeter of a triangle

The perimeter of a triangle is the sum of the length of all three sides of the triangle.

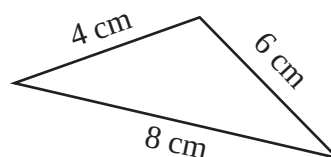
### Illustrative Example:

Find the perimeter of a triangle with sides of length 4 cm, 6 cm, 8 cm.

Perimeter of a triangle = Sum of the length of all the sides of the triangle

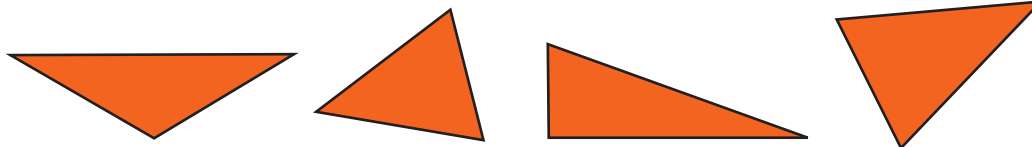
$$= 4 + 6 + 8$$

$$= 18 \text{ cm}$$



### Do it.

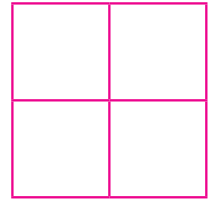
- Cut four triangular flags of different sizes from a coloured paper. Measure the sides of each flag and calculate the perimeter.



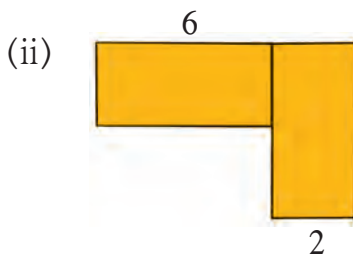
### Think about

- What is the unit of perimeter of a figure when some sides are given in centimetres and some sides in meters?

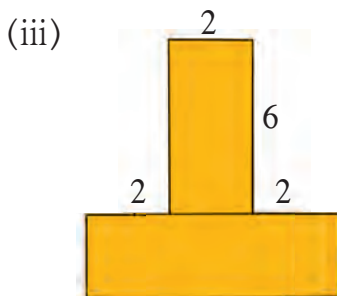
- (1) Find the perimeter of a rectangular plot with length 20 meters and breadth 10 meters.
- (2) The perimeter of a rectangle is 64 cm. If its length is 17 cm, then what is its breadth?
- (3) The perimeter of a triangle is 55 cm and two of its sides are 20 cm and 14 cm. Find the length of the third side.
- (4) A rectangle with sides of 5 cm and 3 cm is formed by bending a wire. If a square is formed using the same wire, what will be the length of the side of that square?
- (5) Take a square piece of cloth one meter long and make four handkerchiefs of the same size as shown in the figure. How much total length of lace will be required to lace all the handkerchiefs?
- (6) If the following figures are made using a rope of length 36 cm, what will be the length of each side?
  - a) Square
  - b) Rectangle
- (7) Take a rectangular piece of paper 6 cm long and 4 cm wide. Cut it into two equal parts. Arrange the two pieces without overlapping as shown in the figure and find the perimeter.



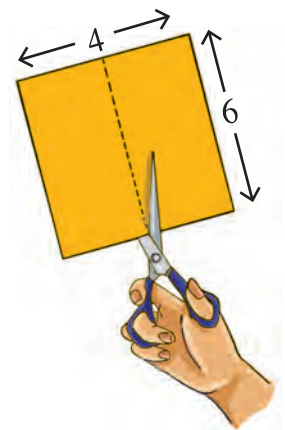
Perimeter = ..... cm



Perimeter = ..... cm



Perimeter = ..... cm



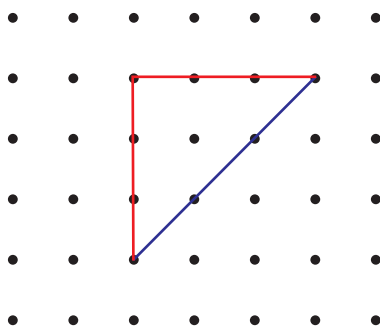


## Mathematical Discussion

- In the figure below, the distance between two points on a horizontal or vertical line is 1 unit.



The perimeter of the triangle is 9 units



The perimeter of the triangle is more than 9 units

What do you think ?

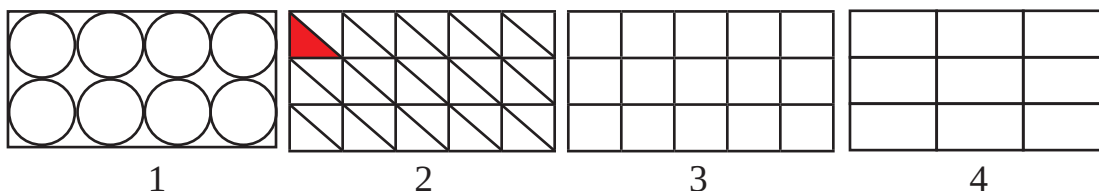


Do it.

## Area

Seema and Suraj cut several pieces of the same size and shape (circle, right triangle, square, rectangle) from a card sheet. Using these pieces, they wanted to measure the surface of the table.

Seema first placed circular pieces on the table, closely fitting them together. On another separate table, triangular pieces were placed so that the entire surface of the table was covered. Then Suraj covered the surface of the table first with square and then rectangular pieces.



Oh Suraj, the circle shape didn't completely cover the surface of the table. Some space remained uncovered!

Yes, The surface of the table is completely covered with shapes like triangles, squares and rectangles.



- The measurement of the space occupied by a flat surface of a figure is the 'area' of that figure.
- The area of Table 2 is 30 triangle units.
- The area of Table 3 is ..... square units.
- The area of Table 4 is..... rectangle unit.



## Understand

## Unit of area

To measure a plane figure, a plane figure is used as the unit.

To measure a plane figure, a square with side of 1 unit is used as the standard unit. This is the most convenient unit. Square unit is a standard unit of area.

- If we take the side in cm, then its area is written in square centimeter, for which a square with a side of 1 cm is taken as the standard unit. This is abbreviated as 1 sq.cm.

| Region                                                                                                                                                | Unit of Side | Unit of Area      | Brief writing |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------------------|---------------|
| <br>Ex. floor, book, notebook                                       | centimeter   | square centimeter | sq.cm.        |
| <br>Ex. open space around the house, construction area, playground | meter        | square meter      | sq.m.         |
| <br>Ex. Taluka, District, State                                    | kilometer    | square kilometer  | sq.km.        |

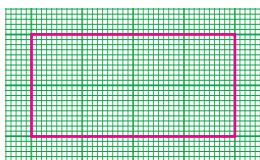




## Area of a rectangle

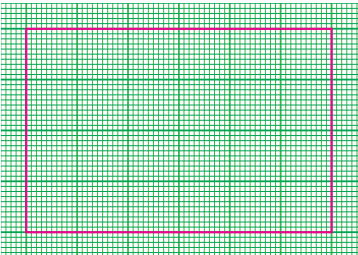
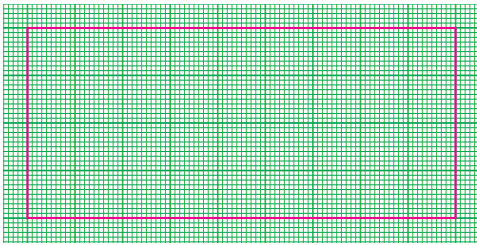


### Understand



The length of this rectangle is 4 cm and the breadth is 2 cm. This rectangle contains 8 squares of size 1 sq.cm. Therefore, the area of this rectangle is 8 sq.cm.

Complete the table below.

| Rectangular Region                                                                  | Number of squares | Area      | Length | Breadth | Area = length $\times$ breadth |
|-------------------------------------------------------------------------------------|-------------------|-----------|--------|---------|--------------------------------|
|    | 24                | 24 sq. cm | 6 cm   | 4 cm    | $6 \times 4$                   |
|  | .....             | .....     | 9 cm   | .....   | .....                          |

Area of rectangle = length  $\times$  breadth

| Square Region       | Number of squares | Area    | Side  | Side  | Side $\times$ Side |
|---------------------|-------------------|---------|-------|-------|--------------------|
| Square with sides 3 | 9                 | 9 sq.cm | 3 cm  | 3 cm  | $3 \times 3$       |
| Square with sides 4 | .....             | .....   | ..... | ..... | .....              |

- Since all four sides of a square are equal, it is clear that area of a square = side  $\times$  side.

Area of a square = side  $\times$  side



### Illustrative Examples:

- \* Find the area of a rectangle whose length is 8 cm and breadth is 7 cm?  
Area of rectangle = length  $\times$  breadth  
 $= 8 \times 7$   
 $= 56 \text{ sq. cm}$

- \* Find the area of a square with side 6 cm?  
Area of a square = side  $\times$  side  
 $= 6 \times 6$   
 $= 36 \text{ sq. cm}$



**Do it.**

Take the measurements of the objects in the following table and complete the table.

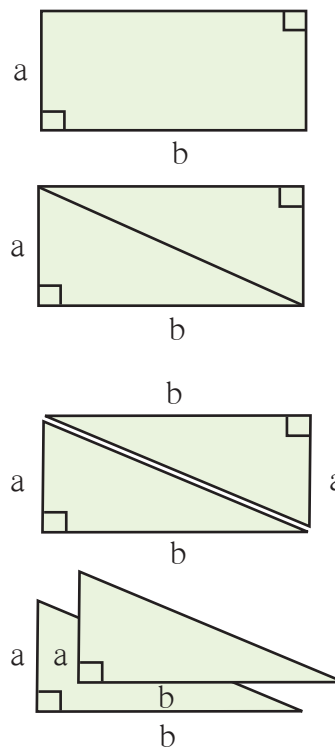
| Figure / Object           | Length | Breadth | Area |
|---------------------------|--------|---------|------|
| Classroom floor           |        |         |      |
| Classroom Window          |        |         |      |
| Any surface of a cupboard |        |         |      |
| A tile                    |        |         |      |



**Understand**

### Area of triangle

- Take a rectangular sheet of paper.
- It's length is  $b$  and breadth is  $a$ .
- Cut it on one of its diagonals and divide it into two right triangles.
- See if the two right triangles formed are match exactly. Since they are matching, the areas of both triangles are equal.
- From the figure it is observed that,  
The area of rectangle is equal to the sum of areas of these two right angled triangles.  
 $2 \times \text{Area of a right angled triangle} = \text{Area of a rectangle}$



$$\therefore 2 \times \text{Area of a right angled triangle} = a \times b$$

$$\therefore \text{Area of a right angled triangle} = \frac{(a \times b)}{2}$$

a and b are sides forming right angle.

Area of a right angled triangle =

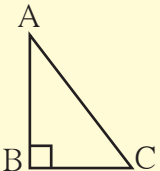
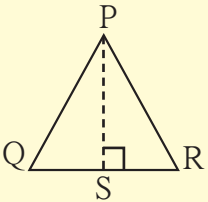
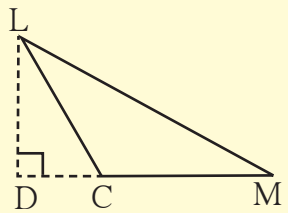
$$\frac{\text{product of the length of the sides forming the right angle}}{2}$$

In a right angled triangle, if one of the two sides forming the right angle is considered as the base, then the other side forming the right angle is the height.

$$\therefore \text{Area of a right angled triangle} = \frac{\text{base} \times \text{height}}{2}$$

**If  $\triangle ABC$  is any triangle, take any one of the sides as the base. The length of the perpendicular drawn from the vertex opposite to that side to the base is the height of that triangle.**

### Triangle: Base and Altitude

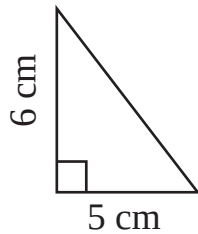
| Type of triangle                                                                    | Base | Height |
|-------------------------------------------------------------------------------------|------|--------|
|   | BC   | AB     |
|  | QR   | PS     |
|  | CM   | LD     |

$$\text{Area of a triangle} = \frac{\text{base} \times \text{height}}{2}$$



### Illustrative Examples:

- 1) If the sides of a right angled triangle that form the right angle are 5 cm and 6 cm, find the area of the triangle.



Area of right angled triangle =  
product of the length of the sides  
forming the right angle

$$\begin{aligned} &= \frac{5 \times 6}{2} \\ &= \frac{30}{2} \\ &= 15 \text{ sq.cm} \end{aligned}$$



**Do it**

- Cut out triangular pieces of paper of various sizes, measure their bases and heights, and find their areas.

### Illustrative Example:

- 1) A hall is 5 meters long and 4 meters wide. A square carpet of side 3 meters is laid on the floor in that hall. Find the area of the region not covered by the carpet.

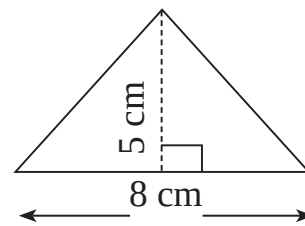
→ Length of the hall = 5 meters,  
breadth = 4 meters

$$\begin{aligned} \text{Area of the hall} &= \text{Length} \times \text{Breadth} \\ &= 5 \times 4 = 20 \text{ sq.m.} \end{aligned}$$

$$\text{Area of the square carpet} = \text{Side} \times \text{Side} = 3 \times 3 = 9 \text{ sq.m.}$$

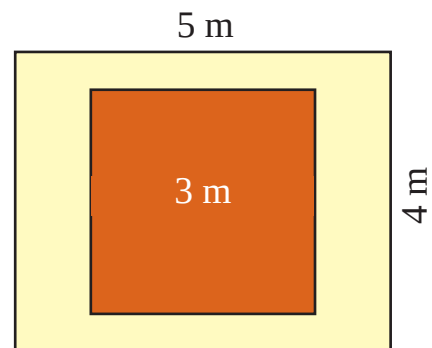
$$\begin{aligned} \therefore \text{Area of the region not covered by the carpet} \\ &= \text{Area of the hall} - \text{Area of the carpet} \\ &= 20 - 9 = 11 \text{ sq.m.} \end{aligned}$$

The area of the region not covered by the carpet is 11 sq.m.



Area of triangle

$$\begin{aligned} &= \frac{\text{base} \times \text{height}}{2} \\ &= \frac{8 \times 5}{2} \\ &= \frac{40}{2} \\ &= 20 \text{ sq.cm} \end{aligned}$$



- 2) Sunitabai planted flower plants in square flower beds of side 4 meters each in four corners of a land measuring 12 meters long and 10 meters wide. Find the area of the land without flowers beds.

Length of land = 12 meters, breadth of land = 10 meters

**Area of land** = length  $\times$  breadth

$$= 12 \times 10 = 120 \text{ sq. m.}$$

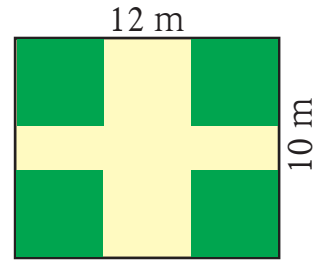
Side of each square bed = 4 m

Area of one flower bed = side  $\times$  side =  $4 \times 4 = 16 \text{ sq. m.}$

$\therefore$  Area of one flower bed =  $4 \times 16 \text{ sq.m} = 64 \text{ sq. m}$

$\therefore$  Area of land without flower beds = **Area of land** – **Area of four flower beds**  
 $= 120 - 64 = 56 \text{ sq. m.}$

$\therefore$  The area of land without flower beds is 56 sq. m.



**Understand**

**Complete the table below :**

| Figure    | No. | Length (cm) | Breadth (cm) | Perimeter (cm) | Area (sq. cm.) |
|-----------|-----|-------------|--------------|----------------|----------------|
| Rectangle | 1   | 5           | 4            | 18             | 20             |
|           | 2   | 8           | 2            | 20             | 16             |
|           | 3   | 6           | 4            | ....           | ....           |
|           | 4   | 9           | 1            | ....           | ....           |

- From the above table, it can be seen that, rectangles with the same perimeter can have different areas.

**Complete the table below :**

| Figure | No. | Side (cm) | Perimeter (cm) | Area (sq. cm.) |
|--------|-----|-----------|----------------|----------------|
| Square | 1   | 3         | 12             | 9              |
|        | 2   | 4         | 16             | 16             |
|        | 3   | 5         | ....           | ....           |
|        | 4   | 6         | ....           | ....           |

- From the table above, it can be seen that if the perimeter of a square is increased, the area increases.





(1) Find the area of the triangle.

i) Base 8 cm, height 5 cm

ii) Base 7 m, height 4 m

(2) Given the length of the sides of a rectangle below, find the perimeter and area of the rectangle.

i) 8 cm, 3 cm

ii) 3.5 m, 2 m

iii) 6 m, 3.5 m

iv) 9.5 cm, 5.3 cm

(3) Given the length of the side of a square below, find the perimeter and area of the square.

i) 10 cm

ii) 14 meters

iii) 5.5 meters

(4) A rectangular mango orchard is 100 meters long and 50 meters wide. If each mango tree takes 25 sq. meters of space, then how many mango trees can be planted in the orchard at most?



### At a glance

- The sum of the length of all the sides of a closed figure is the perimeter of that figure.
- The unit of the perimeter of any figure is same as the unit of its side.
- The area of a figure is the measurement of the space occupied by that figure on a flat surface.

$$\text{Area of a triangle} = \frac{\text{base} \times \text{height}}{2}$$

$$\text{Area of a right angled triangle} = \frac{\text{Product of the length of the sides forming the right angle}}{2}$$



### Exercise 15

(1) Choose the correct option.

i) What is the perimeter of a square with side length 7 cm?

a) 49 cm

b) 24 cm

c) 28 sq. cm.

d) 28 cm

ii) If the length of a rectangle is 5 cm and the breadth is 2 cm, what are its perimeter and area respectively?

a) 14 sq. cm., 10 cm.

b) 14 cm, 10 sq. cm.

c) 10 cm, 14 sq. cm

d) 24 cm, 20 sq. cm

iii) A square piece of paper of any size is folded in half. When cut along the fold, two rectangles are formed. Accordingly, which of the following statements is always true? Identify it.

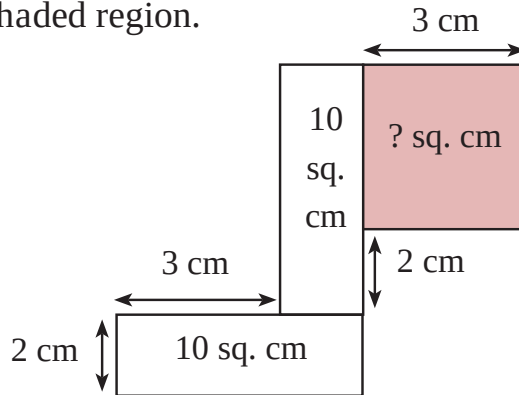
a) The area of every rectangle is greater than the area of the square.

b) The perimeter of the square is greater than the sum of the perimeters of two rectangles.

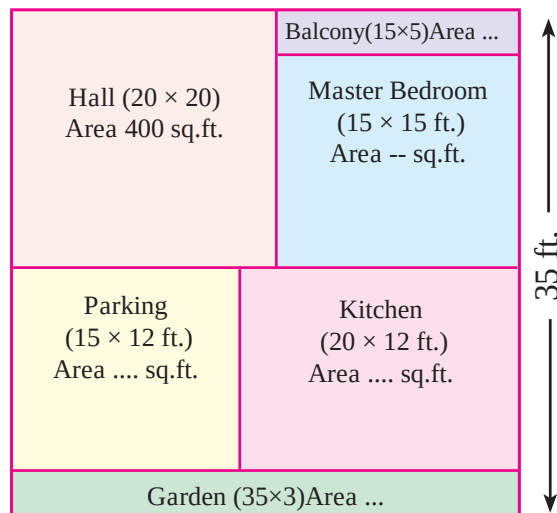




- c) The perimeter of square is smaller than the sum of perimeters of two rectangles.
- d) The area of a square is always three times the sum of the areas of the two rectangles.
- (2) Given the length of the sides of a triangle, find the perimeter.  
i) 12 m, 7 m, 5 m ii) 6.5 cm, 7 cm, 5 cm
- (3) Given the length and breadth of three rectangles below, which rectangle has the largest and smallest area?  
i) 9 m, 6 m ii) 17 m, 3 m iii) 4 m, 14 m
- (4) In the figure below, the length of some sides and areas are given. Find the area of the shaded region.



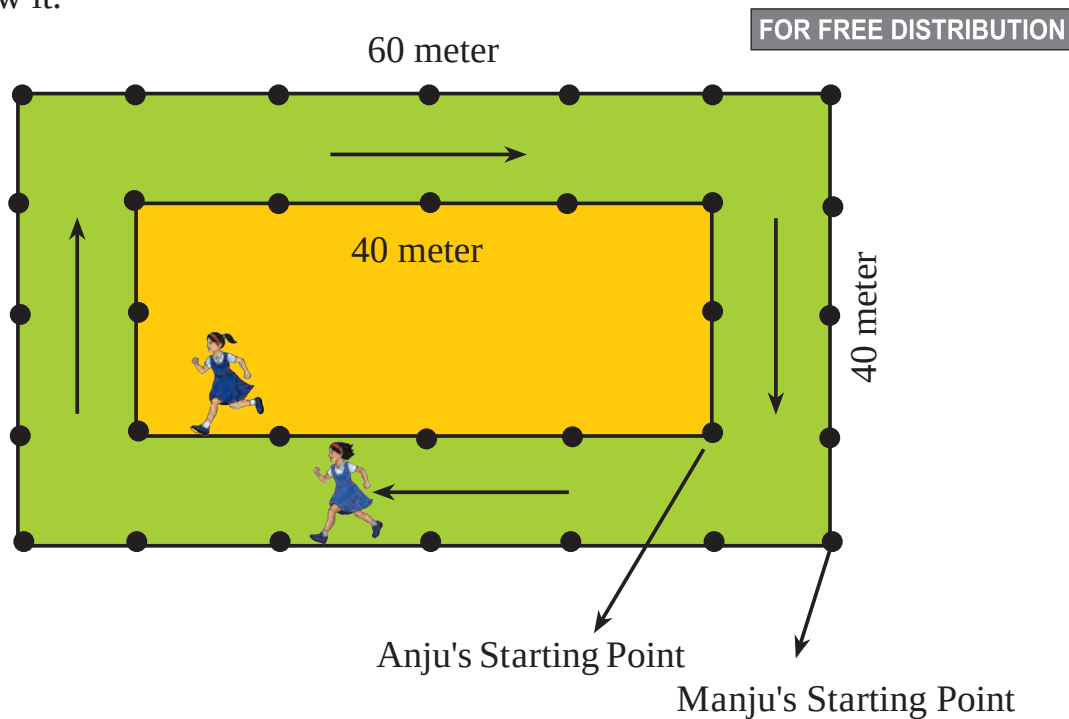
- (5) Frame two examples for each of the following statements.
- When the perimeter of a rectangle is increased, the area increases.
  - When the perimeter of a square is increased, the area increases.
  - Increasing the perimeter of a rectangle does not necessarily increase its area.
- (6) The plan of Amir's house is given below. This plan is in a square plot. Observe the plan.



- Find the missing areas in it.
- Find the area of Amir's entire house.



- (7) Look at the following picture and find the answers to the questions given below it.



Two tracks are drawn in a rectangular field, one on the inside and one on the outside. Anju started running on the inside track and Manju on the outside track in the same direction. Anju completed 7 laps and Manju completed 4 laps before they both stopped. Discuss the total distance covered by both and find the answers to the following questions.

- How much distance did Anju cover?
- How much distance did Manju cover?
- Who among the two covered the greater distance?

### Activity

- Find the area of any square figure in the classroom or at home.





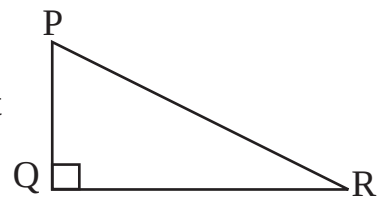
**Write the types of triangles:**

- i) Types of triangles based on angles \_\_\_\_\_
- ii) Types of triangles based on sides \_\_\_\_\_

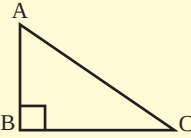
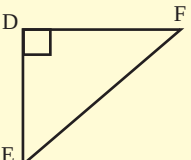
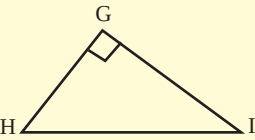
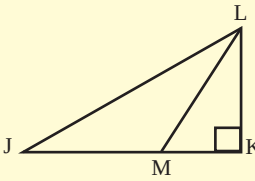
### Baudhayana - Pythagoras Theorem



- A triangle in which one angle is a right angle is called a **right angled triangle**.
- In a right angled triangle, the side opposite the right angle is the hypotenuse.
- In the adjacent  $\Delta PQR$ , side PR is the hypotenuse.
- Side PQ and side QR are the sides that form the right angle.



**Complete the table below :**

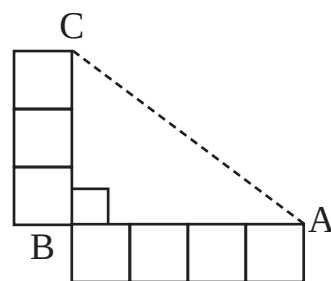
| Sr. No. | Figure                                                                              | Right angle  | Hypotenuse | Sides that form right angle |
|---------|-------------------------------------------------------------------------------------|--------------|------------|-----------------------------|
| 1       |  | $\angle ABC$ | Side AC    | Side AB , Side BC           |
| 2       |  |              |            |                             |
| 3       |  |              |            |                             |
| 4       |  |              |            |                             |



### Do it.

- (1) Cut several square pieces of paper or cardboard with side one unit each.

Place 4 pieces of these squares in straight line. Name their end points A and B. Place 3 pieces perpendicular to seg BA at point B as shown next to each other. Name the top point C. Finally, Join point A and point C.



- (2) Place the pieces on side AB to form a square with the length of side AB.

- (3) Place the pieces on side BC to form a square of length BC.

Total square pieces on both sides forming the right angle =  $9 + 16 = 25$  .....(1)

- (4) Make a large square by placing all the pieces on both sides on side AC. What do you notice?

Total 25 pieces were used to form the square on hypotenuse AC.

Total square pieces on hypotenuse = 25..... (2)

Area of the square on hypotenuse AC = 25 square units

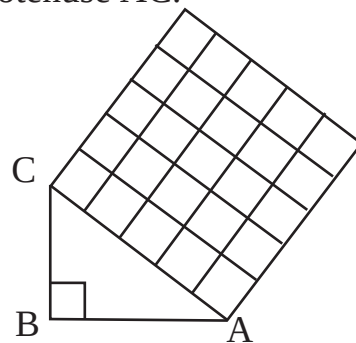
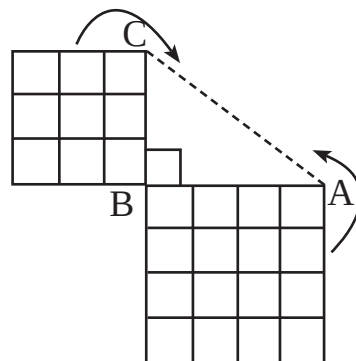
From statements 1 and 2, it can be seen that,

$$9 + 16 = 25$$

$$\text{Square of 3} + \text{Square of 4} = \text{Square of 5}$$

$$[l(BC)]^2 + [l(AB)]^2 = [l(AC)]^2$$

**That is, the area of the square drawn on the hypotenuse of a right angled triangle is equal to the sum of the areas of the squares drawn on the other two sides.**



**Baudhayana – Pythagoras Theorem :** In a right angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.  

$$(\text{hypotenuse})^2 = (\text{side 1})^2 + (\text{side 2})^2$$

The above theorem was discovered by the famous Indian mathematician Baudhayana in the 8th century BC. The Greek mathematician Pythagoras formulated this theory and proved it; but the original idea belonged to the Indian sage Baudhayana

\* Verify the Baudhayana-Pythagoras theorem using the graph paper.



## References in history

**Baudhayana:** Baudhayana was an ancient Indian mathematician. He had proposed the concept of Pythagoras theorem long before, especially in his book 'Shulvasutra', the theorem related to right-angled triangle is explained. Its period is considered to be 700 BC. Baudhayana is said to have been born in approximately the 8th century BC.

The most important contribution of the 'Shulvasutras' to Geometry is the statement known as the Pythagoras theorem.

**Pythagoras (570-495 BC):** He was a great ancient Greek mathematician and philosopher. His most important recognition in the field of mathematics is the 'Pythagoras theorem'. His mathematical studies included Pure Numerology, namely Arithmetic, Applied Statistics, Geometry and Astrology.

### Illustrative Examples:

- 1) In right-angled  $\triangle PQR$ ,  $m\angle Q = 90^\circ$ ;  $l(QR) = 5\text{cm}$ ,  $l(PQ) = 12\text{cm}$ . What is the length of side PR?

→ In right-angled  $\triangle PQR$ ,  $m\angle Q = 90^\circ \therefore$  side PR is a hypotenuse.

According to Baudhayana-Pythagoras theorem,

$$(\text{hypotenuse})^2 = (\text{side 1})^2 + (\text{side 2})^2$$

$$[l(PR)]^2 = [l(PQ)]^2 + [l(QR)]^2$$

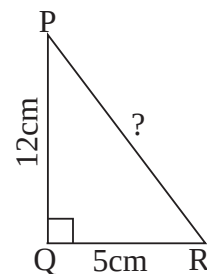
$$\therefore [l(PR)]^2 = (12)^2 + (5)^2$$

$$\therefore [l(PR)]^2 = 144 + 25$$

$$\therefore [l(PR)]^2 = 169$$

$$\therefore l(PR) = 13 \dots\dots (13)^2 = 169$$

$\therefore$  The length of seg PR is 13 cm.



- 2) In right-angled  $\triangle XYZ$ ,  $m\angle Y = 90^\circ$ ;  $l(XY) = 6\text{cm}$ ,  $l(XZ) = 10\text{cm}$ . What is the length of side YZ?

In right-angled  $\triangle XYZ$ ,  $m\angle Y = 90^\circ$  so side XZ is the hypotenuse.

By Boudhayana-Pythagoras theorem,

$$(\text{hypotenuse})^2 = (\text{side 1})^2 + (\text{side 2})^2$$

$$[l(XZ)]^2 = [l(XY)]^2 + [l(YZ)]^2$$

$$\therefore (10)^2 = (6)^2 + [l(YZ)]^2$$

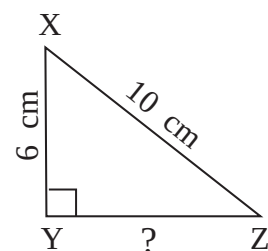
$$\therefore 100 = 36 + [l(YZ)]^2$$

$$\therefore 100 - 36 = [l(YZ)]^2$$

$$\therefore 64 = [l(YZ)]^2$$

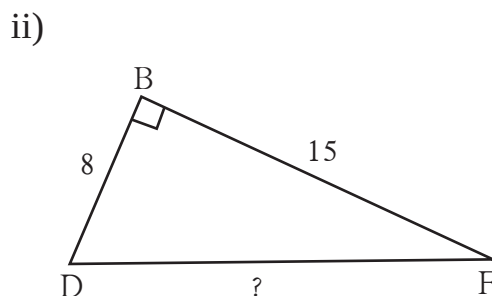
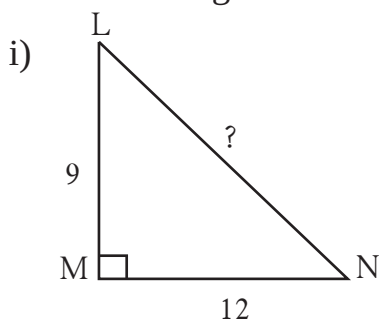
$$\therefore 8 = l(YZ) = \dots\dots (8)^2 = 64$$

$\therefore$  Length of seg YZ = 8 cm.



## Practice Set 16.1

- (1) Look at the figures below and find the length of the unknown side.



- (2) A person walks 12 m. east from his house. There he turns north at a right angle and stops after walking 9 m. How far is the person from his original place?
- (3) A ladder 35 ft. long reaches a window 21 ft. above the ground. Find the distance between the base of the wall and the lower end of the ladder.



### Understand

### Baudhayana - Pythagorean triplet

If the square of the largest of three natural numbers is equal to the sum of the squares of the remaining two numbers, then those numbers are called 'Baudhayana-Pythagorean triplets'. For example, the set of numbers (3, 4, 5).

A triangle is a right-angled triangle if its sides follow the Baudhayana-Pythagoras triplet. In any right-angled triangle, the hypotenuse is the longest side of that triangle.

### Illustrative Examples:

- (1) Is the number set (7, 24, 25) a Baudhayana-Pythagorean triple?

→ Let us square each of the numbers in (7, 24, 25).

Square of 7 =  $7 \times 7 = 49$ ; Square of 24 =  $24 \times 24 = 576$ ;

Square of 25 =  $25 \times 25 = 625$

(i) The square of the largest number =  $25^2 = 625$

(ii) Sum of squares of smaller numbers =  $49 + 576 = 625$

From (i) and (ii)  $7^2 + 24^2 = 25^2$

$\therefore$  (7, 24, 25) is a Baudhayana-Pythagorean triplet.

- (2) The sides of a triangle are 11, 12, 15. From this, determine whether it is a right-angled triangle or not.

→ In (11, 12, 15)  $11^2 = 11 \times 11 = 121$

$12^2 = 12 \times 12 = 144$

$15^2 = 15 \times 15 = 225$

(i) The sum of squares of smaller numbers =  $121 + 122 = 121 + 144 = 265$

(ii) The square of larger number =  $15^2 = 225$





- ∴ The sum of squares of smaller numbers  $\neq$  The square of larger number.  
 ∴ This number group (11, 12, 15) is not a Baudhayana-Pythagorean triplet.  
 The triangle with sides 11, 12, 15 is not a right angled triangle.



- (1) Determine which are the Baudhayana-Pythagoras triplets in the number set given below.  
 i) 4, 5, 6                  ii) 2, 6, 7                  iii) 15, 20, 25  
 (2) The sides of some triangles are given below. Find out which ones are right-angled triangles?  
 i) 11, 60, 61              ii) 40, 20, 30.              iii) 9, 40, 41



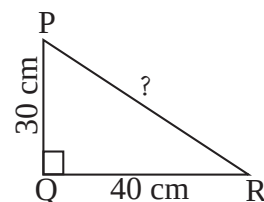
### At a glance

- Baudhayana - Pythagorean Theorem
- In a right-angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.
- $(\text{hypotenuse})^2 = (\text{side 1})^2 + (\text{side 2})^2$
- If the sides of a given triangle form a Baudhayana-Pythagorean triplet, then the triangle is a right-angled triangle.



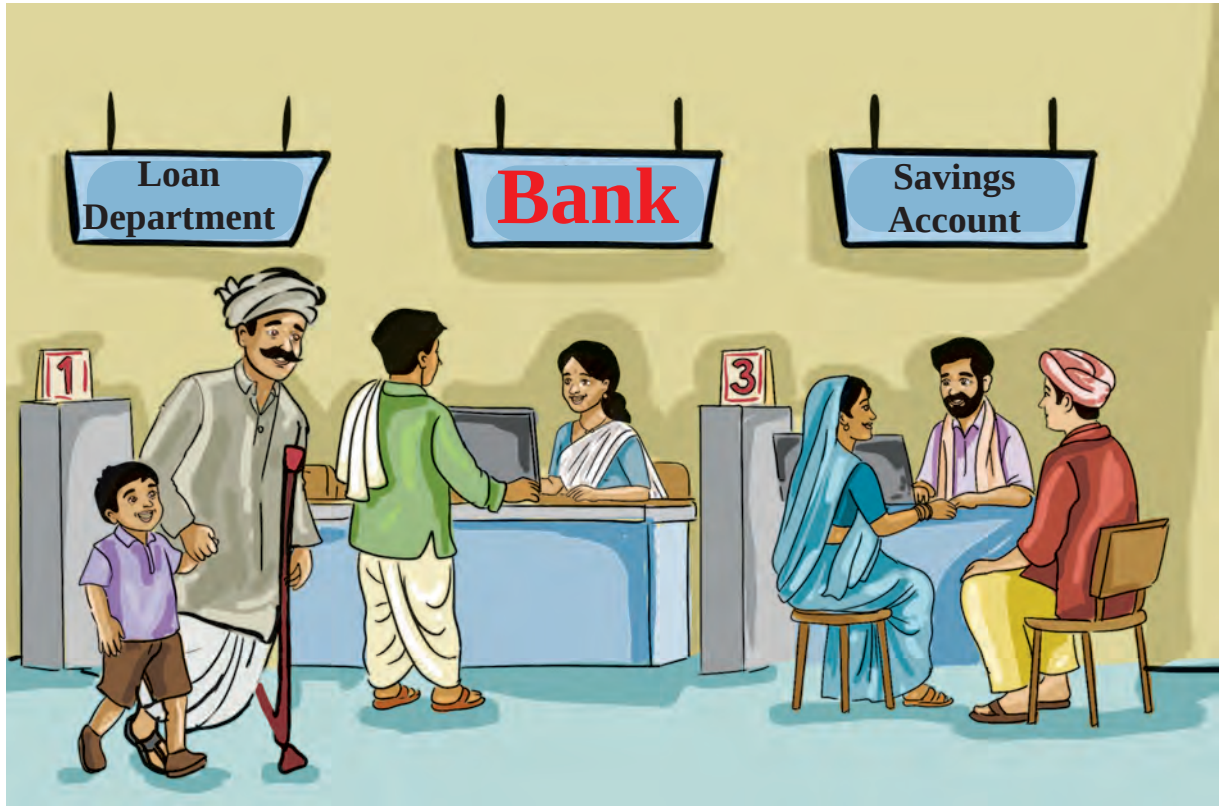
### Exercise 16

- (1) Choose the correct option.  
 i) The hypotenuse of a right-angled triangle is 41 cm. and one side forming the right angle is 40 cm, then what is the length of the other side?  
 a) 81 cm                  b) 21 cm                  c) 9 cm                  d) 1 cm  
 ii) A ladder 17 m. long is placed diagonally against a wall at the distance of 8 m. from it, at what distance above the ground will the top of the ladder be?  
 a) 15 m                  b) 23 m                  c) 32 m                  d) 25 m  
 iii) Two pillars of height 20 m and 27 m respectively are standing on the ground at the distance of 24 m from each other. What is the distance between their upper ends?  
 a) 7 m                  b) 24 m                  c) 25 m                  d) 26 m  
 (2) Observe the adjacent figure and find the length of the unknown side.  
 (3) Determine whether the set of numbers (6, 8, 10) is a Baudhayana-Pythagorean triplet or not.  
 (4) Find Baudhayana-Pythagorean triplets.





### Mathematical Discussion



**Observe the picture and discuss it in class based on the points given below.**

- 1) Name the banks you know.
- 2) What types of work are done in bank?
- 3) What is the importance of bank in daily life?
- 4) Have you ever been to a bank?

### Bank



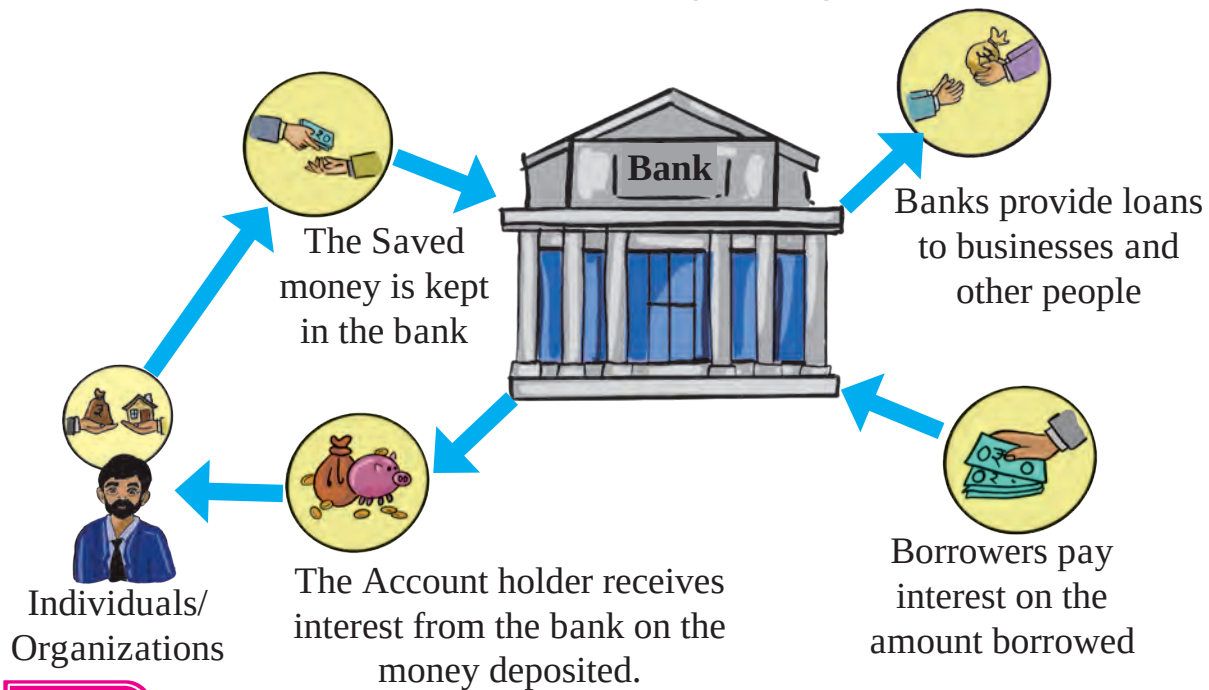
### Understand

- Money transactions are carried out in banks. Banks are given official approval and license by the government to carry out financial transactions.
- Bank is an authorized and licensed financial institution. Bank provides financial transaction services to its customers. The money deposited in the bank remains safe.
- In India, the banking system is regulated by the Reserve Bank of India (RBI).



## Mathematical Discussion

Discuss in class how bank works using the diagram below.



### Activity

- 1) Get information about bank by visiting a bank in your area with your teachers.
- 2) Create a model bank at school with the help of teachers and demonstrate some of the bank's functions.
- 3) Prepare a table based on different countries and their currencies.
- 4) Record how banking-related tasks are carried out in your household.

**Types of Banks:** Central banks, Commercial banks, Co-operative banks, Industrial development banks and Nationalised Banks etc.



## Understand

### General Functions of Bank

- 1) **Accepting deposits:** The bank accepts deposits from individuals or institutions. The bank pays interest at a fixed rate on deposits like Fixed Deposit, Recurring Deposit and so on.
- 2) **Lending:** The bank provides the following types of loans to the account holders. Such as short-term loans (duration up to 1 year), medium term loans (duration up to 5 years), long term loans (duration more than 5 years). The bank charges interest on the loans given.
- 3) **Locker facility:** The bank provides locker facility to keep valuables and important documents safe.
- 4) **E-Banking Facilities:** The bank provides many facilities like ATM facility, debit cards, credit cards, net banking, mobile banking, UPI-QR code.





## Understand

# Opening a bank account



**Bank Employee** : You need a passport size photograph with the application.

**Woman** : Yes. A passport size photograph is attached to the application.

**Bank Employee** : For your ID proof, attach a self-attested Xerox copy of any one of the following documents: Aadhaar card, PAN card, or voter ID card.

**Woman** : Yes. It is attached with the application.

**Bank Employee** : For proof of residential address, attach a xerox copy of any one of the following: electricity bill, ration card, driving license, passport, residence certificate.

**Women** : Yes. It is attached with the application.

**Bank Employee** : Your mobile number.

**Women** : It is written on the application.

**Bank Employee** : You will have to go through the KYC (Know Your Customer) process. You can also do e-kyc from home.

**Women** : Yes. I will do it online through mobile phone.

**Bank Employee** : The documents have been verified and completed. Your new account ready. Your account number is 4150089xxx.

**Women** : Thank you!





## Internet usage

**E-KYC** stands for Electronic Know your Customer. The e-KYC process is paperless. This process is completed digitally and does not require physical documents. This process is important to prevent things like **money laundering or fraud**.



Visit : <https://www.npci.org.in/what-we-do/e-kyc-services/e-kyc-setu-system>



### Understand

| Account type                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1) Savings Account                                                                                                                                                                                                                                                                                                                    | 2) Current Account                                                                                                                                                                                                                                                                                                                                                              |
| <ul style="list-style-type: none"><li>Any person can open a savings account.</li><li>The purpose of this account is to help people develop the habit of saving.</li><li>There are some restrictions on withdrawing money from this account.</li><li>The bank pays interest at a certain rate on the balance in the account.</li></ul> | <ul style="list-style-type: none"><li>Business organizations, traders, or organizations or individuals who have to conduct large amounts of daily financial transactions can open a current account.</li><li>There are no restrictions on depositing or withdrawing money from this account.</li><li>The bank does not pay any interest on the amount in the account.</li></ul> |

- Get information about other types of accounts in the bank.

## Deposit Slip/Challan



### Understand

**Challan** - While depositing money in a bank, a form containing the following information in a specific format is called Challan (1) Account holder's name (2) Account number (3) Date (4) Amount details (5) Amount in words (6) Signature of the payer. The bank staff records the amount deposited in the bank in the passbook. Keep your passbook updated regularly. Also, keep it safe.



### Do it.

Jayesh Kumar has come to the bank to deposit ₹8,325 into his savings account. He has twelve ₹500 notes, ten ₹200 notes, two ₹100 notes, two ₹50 notes, one ₹20 note and one ₹5 coin. Help him fill in the sample challan.



|                                           |            |                    |                 |               |       |
|-------------------------------------------|------------|--------------------|-----------------|---------------|-------|
| <b>GRAMIN BANK</b><br>At/po. Arodi, Jalna |            | <b>GRAMIN BANK</b> |                 | Branch xxx    |       |
|                                           |            |                    |                 | Date DD/MM/YY |       |
| Name of the account holder _____          |            |                    |                 | 1             |       |
| Account Number _____                      |            |                    |                 | 2             |       |
| Mobile Number _____                       |            |                    |                 | 3             |       |
| <b>Cheque / Cash Details</b>              |            |                    |                 |               |       |
| Bank and Branch                           | Cheque No. | Cheque Issuer      | Cash Details    | Rs.           | Paise |
|                                           |            |                    | 500 x _____     | _____         |       |
|                                           |            |                    | 200 x _____     | _____         |       |
|                                           |            |                    | 100 x _____     | _____         |       |
| Not Necessary                             |            |                    | 50 x _____      | _____         |       |
|                                           |            |                    | 20 x _____      | _____         |       |
|                                           |            |                    | 10 x _____      | _____         |       |
|                                           |            |                    | 5 x _____       | _____         |       |
|                                           |            |                    | Total           | _____         |       |
| Amount in Words _____                     |            |                    |                 | 5             |       |
| Signature of payer _____                  |            |                    | XXXX<br>Cashier |               |       |

## Cheque



### Understand

FOR FREE DISTRIBUTION

A cheque is an order to pay money to a specific person or transfer it. Cheques are only available to account holders. The bank charges additional fees for the cheque book.

**Activity: A sample check is given below. Fill in the information in it.**

- ① Date    ② Name of the person to whom the money is to be paid  
 ③ Amount in words    ④ Amount in figures    ⑤ Signature

|                                           |  |                    |  |           |
|-------------------------------------------|--|--------------------|--|-----------|
| <b>GRAMIN BANK</b><br>At/po. Arodi, Jalna |  | <b>GRAMIN BANK</b> |  | _____     |
|                                           |  |                    |  | 1         |
| Pay _____                                 |  |                    |  | 2         |
| Rupees _____                              |  |                    |  | 3         |
| Saving A/C XXXXXXXX                       |  |                    |  | 4         |
| _____                                     |  |                    |  | 5         |
| XXXXXXXX    XXXX XXXX                     |  |                    |  | Signature |





## Charging Interest



### Understand

The bank pays depositors a certain amount of money as compensation for keeping their money in the bank. The bank also charges borrowers a certain amount of money as compensation for lending money. This amount is called interest. The amount deposited in the bank or given by the bank to the borrower is called **principal**.

**Amount:** Amount is the money returned by the bank to the depositor along with interest at the end of the term or the money that the borrower has to return to the bank along with interest while repaying the loan.

Amount = Principal + Interest.

While charging interest on a deposit or loan, its rate is given for every 100 rupees. The interest rate is stated per period. Interest rate is stated as p.c.p.a., that is, the interest paid for every hundred rupees for each year.

The amount of time for which the principal is used is called the **period**.

### Simple Interest

In this class, we will consider interest charged for only one year. This is simple interest.



### Do it.



Radhika took a loan of ₹25,000 from a bank to start a vegetable business. The bank charged an interest at the rate of 8 p.c.p.a. After a year, Radhika returned ₹27,000 to the bank.

How much extra amount did Radhika pay while returning the money to the bank ?

₹25,000

₹27,000

–

₹2,000

Principal

Rate of interest

Period

Amount

Interest Charged

**Amount = Principal + Interest**

Principal =  -  Rate of Interest : p.c.p.a 8 (Percent per annum 8)  
Means for every 100 rupees, for one year interest  
Interest =  -  is ₹8.



(1) Complete the following activity.

| Sr. No. | Principal | Interest | Amount   |
|---------|-----------|----------|----------|
| 1       | ₹ 5,800   | ₹ 380    |          |
| 2       |           | ₹ 1,050  | ₹ 13,550 |
| 3       | ₹ 35,500  |          | ₹ 39,760 |

(2) If the principal is ₹8,000 and the interest is ₹640, then what will be the amount?

### Illustrative Examples:

If Kiran deposits ₹18,000 in a bank at the rate of 5 p.c.p.a., how much interest will Kiran get in one year?

→ Principal = 18000, Period 1 year, Interest rate 5 p.c.p.a.  
Simple interest (x) = ?

Let us assume that the interest will be x rupees on the principal of ₹18,000. An interest of ₹5 is given on the principal of Rs 100. Let us take the ratio of interest to principal. By expressing this ratio in two forms, we get the equation.



$$\frac{x}{18,000} = \frac{5}{100}$$

$$\therefore \frac{x}{18,000} \times 18,000 = \frac{5}{100} \times 18,000 \quad (\text{Multiply both sides by } 18,000)$$

$$\therefore x = 5 \times 180$$

$$\therefore x = 900$$

$\therefore$  Kiran will get interest of ₹ 900 for one year on ₹18,000.



### At a glance

- **Principal :** The amount deposited in the bank or borrowed from the bank is called principal.
- **Period :** The period is the amount of time for which a deposit is made in a bank or a loan is taken.
- **Interest rate :** The interest rate is the interest on Rs 100 for one year.
- **Interest :** Interest is an excess amount received on deposits in the bank or the excess amount paid by borrowers to the bank while repaying the amount borrowed from the bank.



### Exercise 17

- Choose the correct option.
  - If for one year period the principal is ₹24,500 and the interest is ₹1,830, then what will be the amount?  
 a) ₹ 22,670      b) ₹ 26,530      c) ₹ 26,330      d) ₹ 27,230
  - If Ruben deposits Rs. 42,000 in a bank for one year at the rate of 6 p.c.p.a., how much interest will he get?  
 a) ₹ 2,500      b) ₹ 2,550      c) ₹ 2,820      d) ₹ 2,520
- If Shrushti keeps ₹ 35,000 in a bank at the rate of 8% per annum, how much interest will she get in one year?
- If Pradeep deposits ₹ 48,380 at the rate of 6 p.c.p.a. in a bank for one year, how much interest will Pradeep get from the bank?
- Nancy took a loan of ₹ 18,650 at the rate of 10 p.c.p.a. from a bank to buy a sewing machine. Find the amount that Nancy will have to pay to the bank after one year?
- Mohanrao took a loan of ₹ 55,750 from a bank at the rate of 12 p.c.p.a. to build a water tank in his field. How much total amount will Mohanrao deposit to the bank to become debt-free after one year?
- If a loan of ₹1,30,000 is taken from a bank at the rate of 11 p.c.p.a. for one year to build a gymkhana for the families in the Sahvihar housing society, how much interest will the society will have to pay to the bank in one year? Also, how much total amount will they have to pay to the bank?





## 18 Data Representation



### Revision

The marks obtained by 15 students of a class out of 10 in a test in History are as follows:

8, 2, 4, 5, 2, 3, 6, 2, 6, 1, 9, 10, 7, 6, 6

Out of 15, 4 students got 6 marks and 3 students got 2 marks. There is one student each who got 1, 3, 4, 5, 7, 8, 9, 10 marks. Here, the marks obtained by the students in the test are the 'data'. Each of these numbers (the marks obtained) is the 'score'.

A table can be used to present this information.

| Marks in Examination<br>(Score) | Tally<br>Marks | Total Number of students<br>(Frequency) |
|---------------------------------|----------------|-----------------------------------------|
| 1                               |                | 1                                       |
| 2                               |                | 3                                       |
| 3                               |                | 1                                       |
| 4                               |                | 1                                       |
| 5                               |                | 1                                       |
| 6                               |                | 4                                       |
| 7                               |                | 1                                       |
| 8                               |                | 1                                       |
| 9                               |                | 1                                       |
| 10                              |                | 1                                       |
| Total Number of students:       |                | N = 15                                  |

### Frequency Table



### Understand

Frequency tables are used to present the given information in a simple and easy manner to understand. In this,

- There are three columns: Score, Tally marks and Frequency.
- The scores are written in ascending order in the first column.
- The second column shows the Tally marks.
- In the third column, we write the number of tally marks (frequency) for each score.





**Do it.**

- 1) When a die is thrown 30 times, the numbers obtained on the surface each time are given below.



1, 5, 3, 6, 2, 1, 4, 3, 5, 6,  
2, 1, 4, 3, 5, 6, 2, 1, 4, 3,  
5, 6, 2, 1, 4, 3, 5, 6, 2, 1.

**Complete the frequency table from the given data.**

| Score | Tally Marks | Frequency |
|-------|-------------|-----------|
| 1     |             | 6         |
| 2     |             |           |
| 3     |             |           |
| 4     |             |           |
| 5     |             |           |
| 6     |             |           |
|       |             | N = 30    |

The sum of frequencies in a table is represented by the letter 'N'. This sum is equal to the total number of **observations (score)** in the given data.

- 2) Complete the table below by collecting the blood group information of the students in your class.

| Blood Group | Tally Marks | Frequency |
|-------------|-------------|-----------|
| A+          |             |           |
| A-          |             |           |
| B+          |             |           |
| B-          |             |           |
| AB+         |             |           |
| AB-         |             |           |
| O+          |             |           |
| O-          |             |           |
|             |             | N =       |



- (1) The number of saplings planted by 30 students of the Green Army in a school is given in the frequency table below. Complete the following table.



| Number of saplings | Tally Marks | Frequency |
|--------------------|-------------|-----------|
| 1                  |             |           |
| 2                  |             | 6         |
| 3                  |             | 12        |
| 4                  |             |           |
|                    |             | N =       |

- (2) The number of buffaloes owned by 40 families in the village Vani is as follows.

1, 6, 5, 4, 3, 2, 7, 2, 3, 4, 6, 5, 3, 2, 6, 4, 5, 5, 3, 2, 3, 4, 2, 5, 4, 3, 7, 6, 5, 4, 1, 2, 3, 2, 6, 4, 6, 5, 4, 2. Prepare a frequency table from this data.

- (3) Given a number below, prepare a frequency table of the digits after the decimal point.

3.14159265358979323846264338327950288419716939937517

- (4) Shailaja gave the following names to the birds in the forest for her survey: M-Peacock, S-Stork, K-Cuckoo, W-Woodpecker, G-Eagle. The survey she conducted is as follows.

M, M, S, K, S, G, G, G, G, W, M, G, G, S, S, S, K, S, S, S, S, K, M, S, S, S, M, W, S, S, S, W, S, S, S, S, S, M, M, S, S, S, K, M, S, S, W, G, G, G.

Prepare a frequency table of birds in the forest based on the above information.



### Understand

## Pictograph

In today's modern era, information is easily available, but it is easier to understand if it is presented in a systematic manner. While managing information, use of pictographs, bar charts and infographics makes it easier to understand quickly.



### Do it.

Complete the following activities and answer the questions.



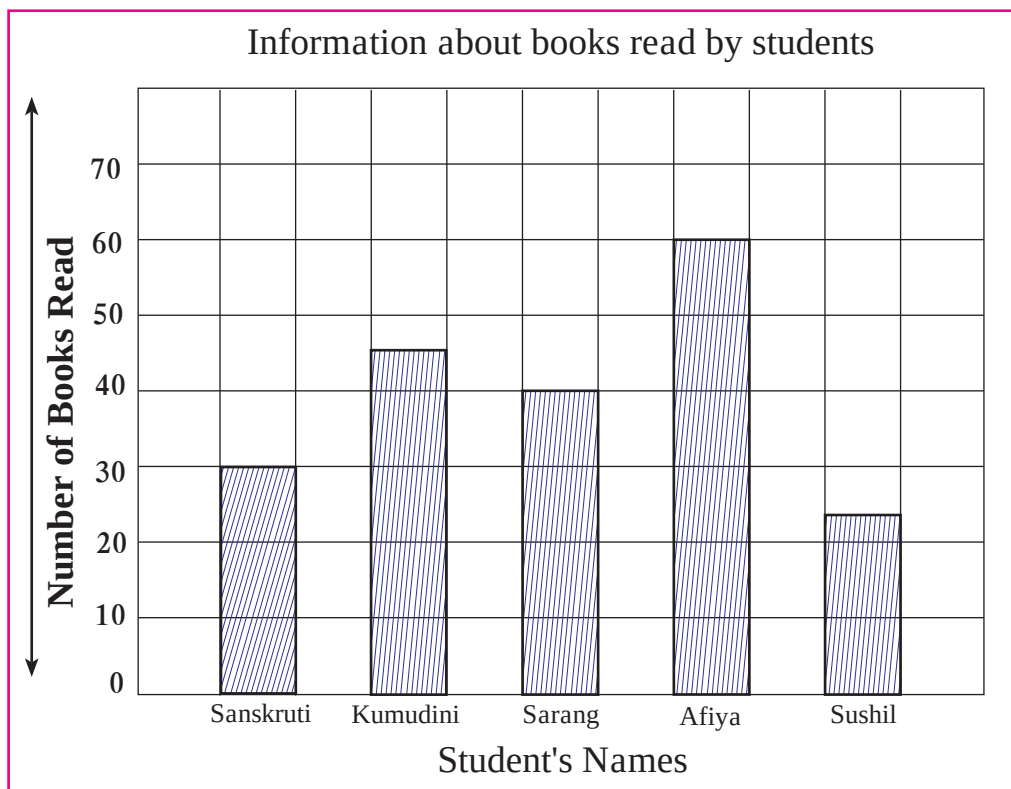


| Names of student | Number of books read | Pictograph                                                                         | Scale  = 5 Books |
|------------------|----------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Sanskruti        |                      |  |                                                                                                     |
| Kumudini         | 45                   |                                                                                    |                                                                                                     |
| Sarang           |                      |  |                                                                                                     |
| Aafiya           | 60                   |                                                                                    |                                                                                                     |
| Sushil           | 24                   |                                                                                    |                                                                                                     |

1. Who has read the maximum number of books?
2. How many more books did Aafia read than Kumudini?
3. Can we draw pictures to show the books Sushil has read?

The above information is easy for us to understand. But we could not accurately show the number of books read by Sushil. For this, if we draw a bar graph of the above pictograph, we can show the number of books read by Sushil.

**Scale : 1 unit = 10 books**



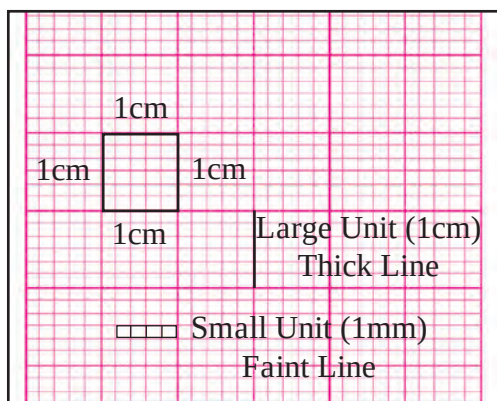
We will use graph paper to show this accurately.





## Understand

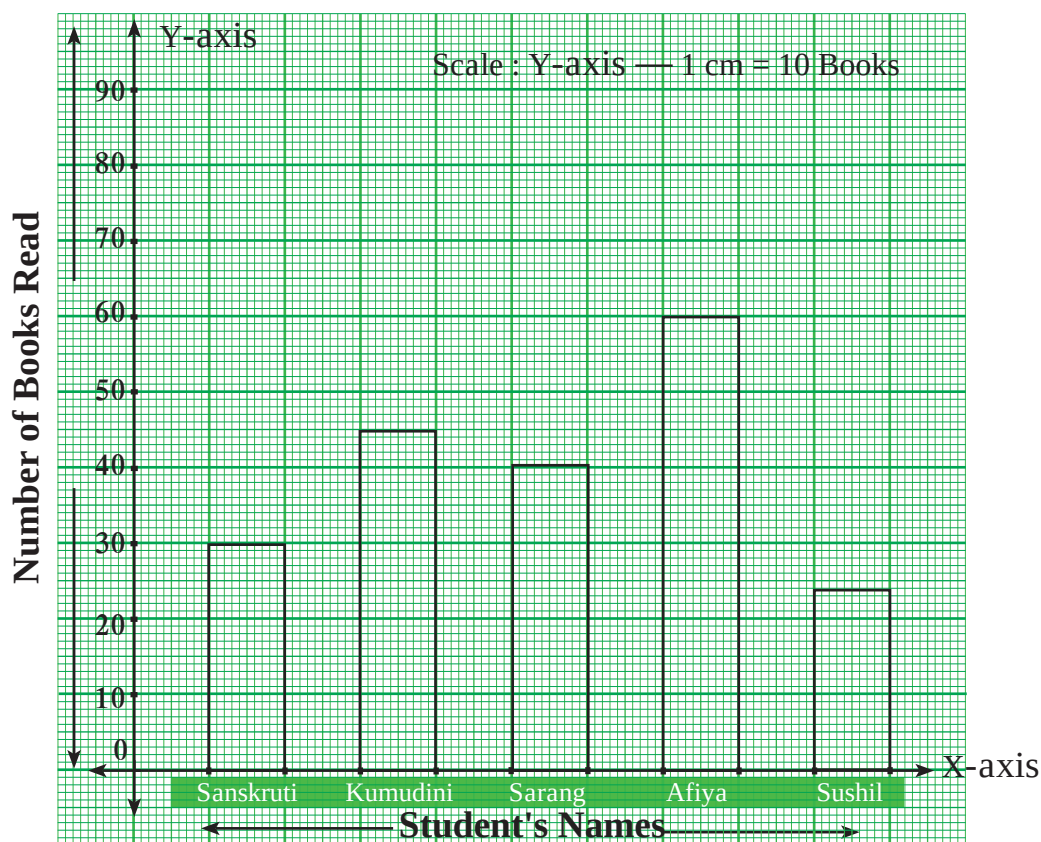
## Introduction to graph paper



Look at the graph paper shown here. It has some thick lines and some faint lines. The thick lines represent larger units. The faint lines represent smaller units that are divided into equal parts. This design makes it easier to show the height of the column with the correct proportions.

## Bargraph

Now let us use a graph to show the number of books Sushil has read.



The graph above, as shown on the paper, usually takes a horizontal line at the bottom as the 'base'. This is called the X-axis. A line perpendicular to this line is drawn on the left side of the paper and is called the Y-axis.

The items to be plotted are displayed at equal distances on the X-axis. The numbers associated with each item are displayed in a vertical column next to its name. This column is of the appropriate height according to the scale and parallel to the Y-axis.



When numerical information is displayed using vertical or horizontal columns, it is called a bar chart.

**Characteristics of a bar graph:**

- The information is displayed in vertical or horizontal rectangular columns. All columns are of the same breadth.
- The height of columns depends on the frequency of the item.
- A bar chart makes it easier to compare given items.

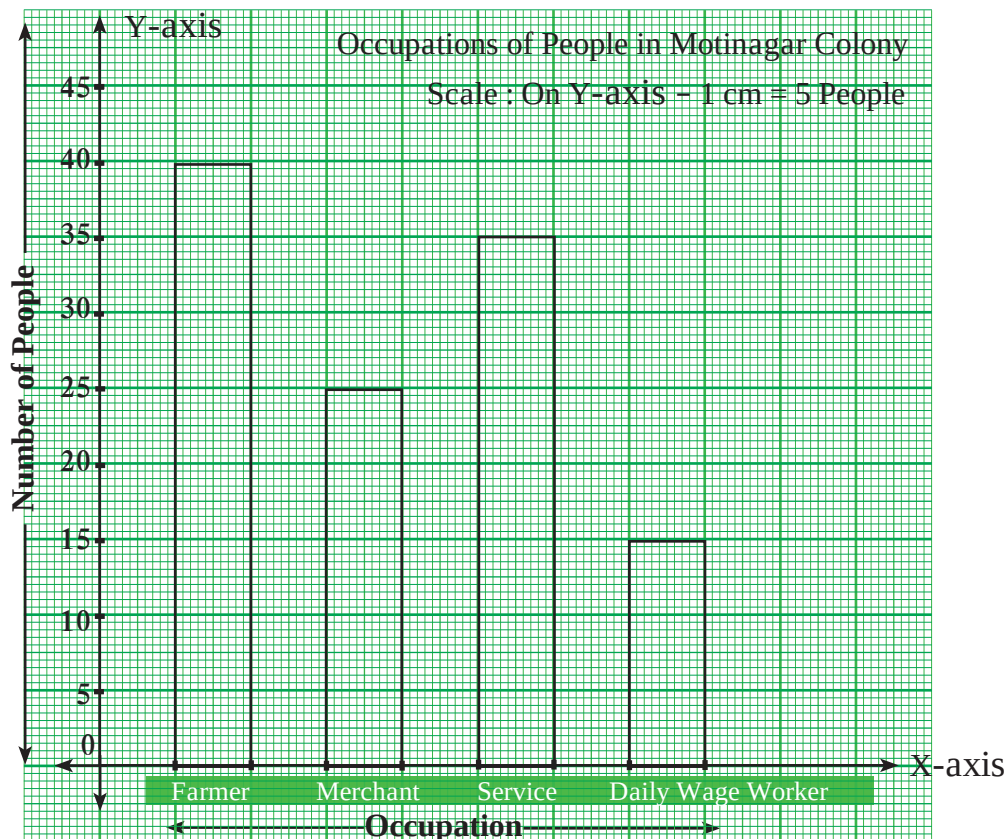
**Illustrative Examples:**

The occupations of people living in Motinagar colony are given in the table below.

| Occupation        | Farmer | Merchant | Service | Daily Wage Worker |
|-------------------|--------|----------|---------|-------------------|
| Number of Persons | 40     | 25       | 35      | 15                |

Perform the following steps to draw a bar graph based on the above information.

1. First draw X axis and Y axis.
2. Consider the number of individuals to determine the scale.
3. Since the number of persons in the given information is shown in multiples of 5, take the scale 1cm = 5 persons on the Y axis.
4. Write the scale in the right corner.
5. Draw columns for each occupation with height according to the number of people, on the X axis.

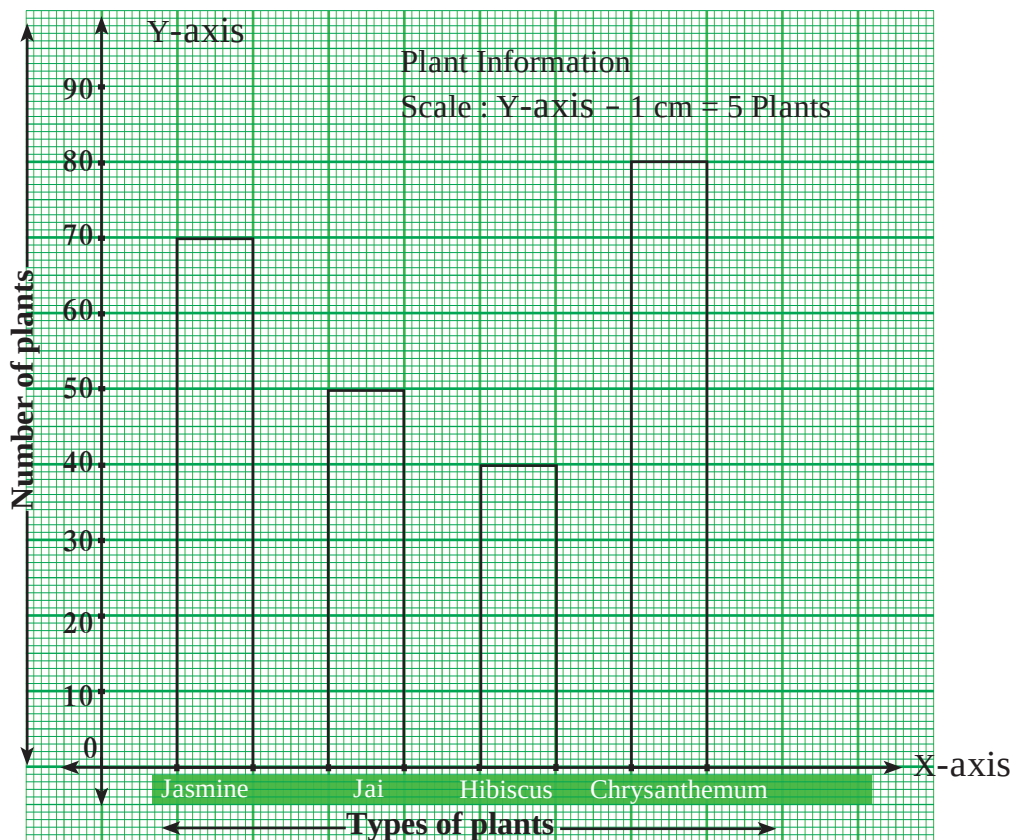




1. If the numbers are large, how can we take the scale?
2. Is the graph drawn to scale according to the size of the graph paper?
3. What happens if people's occupations are taken on the Y axis and their number is taken on the X axis?
4. Prepare questions on the above graph. (e.g. How many people are engaged in agricultural business in Motinagar?)



- (1) The following bar graph shows the information about the plants in the nursery. Observe the graph and answer the questions.



- i) What information is shown on the X and Y axes?
- ii) What kind of flower plants are there in the nursery?
- iii) Which plant has the highest number?
- iv) How many more are the number of chrysanthemum plants than Jasmine plants?
- v) What is the total number of plants in the bar graph?



- (2) The average estimated lifespan of some animals is given below. Draw a bar graph showing this information with appropriate scale.  
Ox- 28 years, Horse - 30 years, Dog - 22 years, Cat - 25 years,  
Goat - 15 years, Cow- 22 years.
- (3) The table below shows the monthly expenditure of John's family on various items. (Scale: On the Y axis take 1 cm = ₹500)

| Items        | House Rent | Grocery | Education | Electricity Bill | Traveling | Savings |
|--------------|------------|---------|-----------|------------------|-----------|---------|
| Expenses (₹) | 4000       | 4500    | 2500      | 800              | 1000      | 2200    |

Based on this information, draw a bar graph according to the given scale.

- (4) The number of students participating in variety entertainment programmes in a school's gathering is shown in the table below. Draw a bar graph based on this information.

| Program            | Drama | Dance | Singing | Playing | One-act play |
|--------------------|-------|-------|---------|---------|--------------|
| Number of Students | 24    | 40    | 16      | 8       | 4            |

- (5) A survey was conducted on students in a village. What they would like to do in their free time is given. Draw a bar graph based on this information.

| Activity           | Sports | Reading | Listening to Music | Drawing |
|--------------------|--------|---------|--------------------|---------|
| Number of Students | 10     | 25      | 10                 | 20      |



## Understand

## Infographics

Infographics is the presentation of information in an artistic, aesthetic and attractive way. They use text, images, diagrams, graphs, symbols, and colours to present complex information in a simple and easy-to-understand manner.

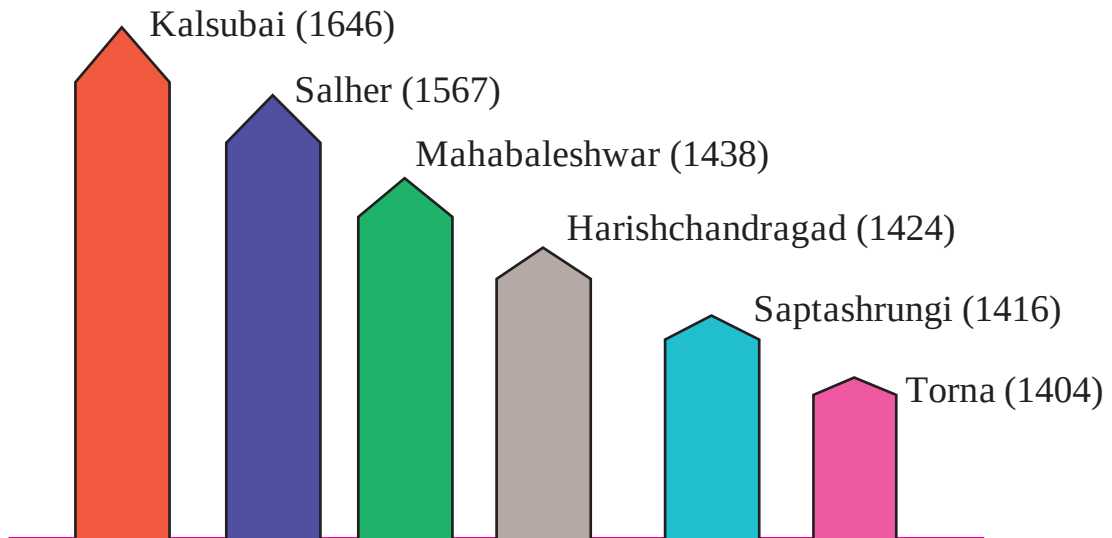
For example, the height (in meters) above sea level of some peaks in the Sahyadri Mountains of Maharashtra is given in the table below.

| Peaks of Mountain | Height in meters |
|-------------------|------------------|
| Kalsubai          | 1646             |
| Salher            | 1567             |
| Mahabaleshwar     | 1438             |
| Harishchandragad  | 1424             |
| Saptashrungi      | 1416             |
| Torna             | 1404             |

In which district and taluka of Maharashtra are the peaks in the table located? What is the temperature of that place?

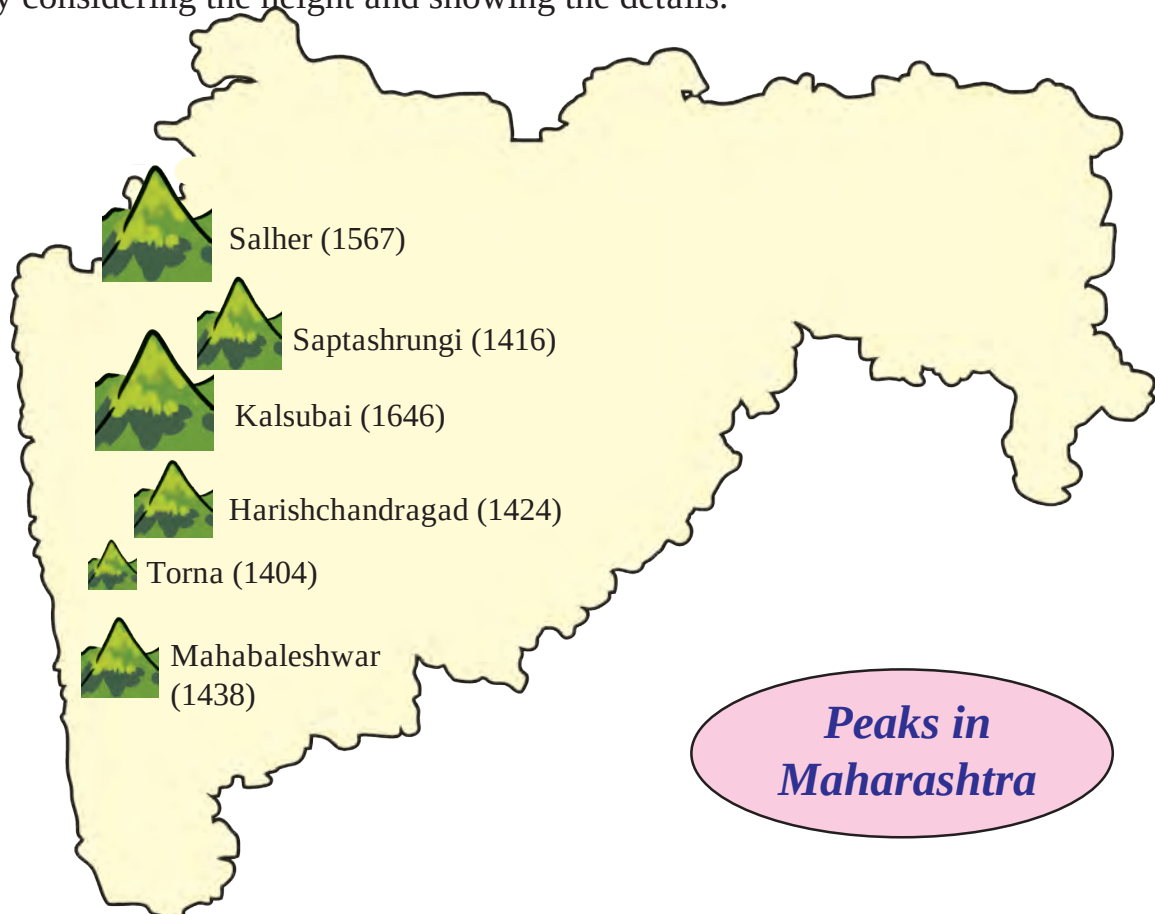
To display the information more effectively using infographics, a bar graph of the information given in the table above is drawn as follows.





In this, vertical columns are used to show mountains. The purpose of the above column diagram is only to show the height. Therefore, all the columns are kept of uniform width. So that only the height can be compared; but the width cannot be estimated.

Infographics can be used to reduce such limitations. For example, showing the mountains of Maharashtra. Going further, we can show the shape of the mountains. To make it more realistic, we can present the infographic more effectively by considering the height and showing the details.





It is very important to take proper precautions while creating or reading infographics.

### Features of Infographics:

- Easy to understand
- Attractive
- Increases memory capacity
- Understands in less time
- Layout in the form of pictures, symbols, graphs



#### At a glance

- Frequency is a measure of how often a score is obtained.
- When numerical information is displayed using columns, it is called a bar graph.
- Infographics are the presentation of information in an artistic, aesthetic and attractive way.

#### References in history

When scientists provide information about planets and stars in space, they often use infographics. In ancient times, people used to draw pictures on walls. Later this became the basis of pictograms. The world's first infographic was created in Scotland in 1786 by William Placker. For the first time, financial information was shown in a bar graph.

#### Activity

- Collect bar graphs and pictographs from newspapers or magazines showing various information, read it and prepare questions based on the graphs.



#### Exercise 18

- (1) The travel time from Pune to Mumbai by different modes of transport is as follows:

| Means of transportation  | car | bus | Railway | Airplane |
|--------------------------|-----|-----|---------|----------|
| Time required (in hours) | 4   | 5   | 3       | 1        |

Draw a bar graph based on the above information.



- (2) The time (in minutes) allotted for daily exercise by 20 people in an ashram is given.

25, 30, 40, 20, 15, 35, 45, 50, 25, 30, 60, 55, 20, 15, 40, 35, 50, 45, 30, 25

Prepare a frequency table based on the above information.

- (3) A question was asked to 30 students in a class, "In what way do you like to learn science?" Draw a bar graph based on the following information.

| Learning method    | By experimenting | By looking at the picture/ charts | By way of Story / example | Question-and-answer method | Group discussion |
|--------------------|------------------|-----------------------------------|---------------------------|----------------------------|------------------|
| Number of students | 10               | 7                                 | 5                         | 4                          | 4                |

- (4) The frequency table of prime numbers between 1 and 50 is given below. Complete the table.

| Score    | Tally marks | Frequency |
|----------|-------------|-----------|
| 1 to 10  |             | 4         |
| 11 to 20 |             |           |
| 21 to 30 |             |           |
| 31 to 40 |             |           |
| 41 to 50 |             |           |
|          |             | N = 15    |

- (5) Information about the hobbies of each of the 25 senior citizens in a society was collected. The information is as follows.

Singing, Sports, Drawing, Reading, Sports, Sports, Singing, Reading, Drawing, Sports, Singing, Reading, Sports, Sports, Drawing, Sports, Singing, Reading, Sports, Sports, Drawing, Singing, Reading, Sports, Sports

Based on the above information, prepare a frequency table and draw a bar graph of that information.

6. Pooja, Umesh, Pradeep, Madhuri have 3, 1, 4, 5 postage stamps respectively. Draw an infographic of this information.

### Activity

- Collect information about the plants in your school's garden. Try to draw a bar graph from that information.





## 1. Sir Srinivas Ramanujan



(22/12/1887 – 26/04/1920)

World-renowned mathematician Srinivasa Ramanujan was born in Erode, then in the Madras province. He worked on number theory, mathematical analysis, and infinite series. He became world-famous for proving approximately 4000 theorems.

Ramanujan's birthday is 22-12-1887. December 22 is celebrated as 'National Mathematics Day'.

### Ramanujan Square

- Add up the group of four numbers in the square formed by his date of birth.
- Does the sum of horizontal, vertical and diagonal lines add up the same?
- Can you make a square like this with your date of birth?
- Search on internet and make it.

|    |    |    |    |
|----|----|----|----|
| 22 | 12 | 18 | 87 |
| 88 | 17 | 9  | 25 |
| 10 | 24 | 89 | 16 |
| 19 | 86 | 23 | 11 |

## 2. Brahma Tower (Tower of Hanoi)

- You can make this game at home as shown in the picture. On a wooden or cardboard board, arrange three wooden sticks (bamboo sticks or empty sketch pens) and coloured wooden or cardboard discs of different sizes in ascending order.



Initial State



Final State

### Rules of the game:

- 1) Only one disc can be picked up from any stick at a time and placed on one of the other sticks.
  - 2) A larger disk cannot be placed on a smaller disk.
  - 3) The game ends when all the discs on stick-1 are moved to stick-3 in the original order.
- First, play this game using only three discs. Count the total number of times you move discs to move all three discs to stick-3. To do this, observe the moves in the picture below.



Increase the number of discs one by one and write the number of moves taken in a table.

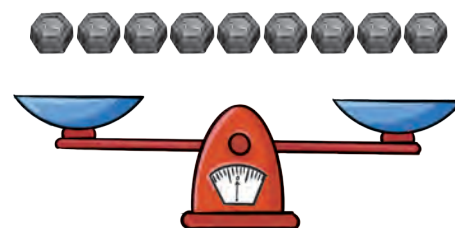
| I                      |                        | My friend               |                        |
|------------------------|------------------------|-------------------------|------------------------|
| Number of discs<br>(n) | Number of moves<br>(t) | Number of discs<br>(n1) | Number of moves<br>(t) |
| 3 discs                |                        | 3 discs                 |                        |
| 4 discs                |                        | 4 discs                 |                        |
| 5 discs                |                        | 5 discs                 |                        |
| 6 discs                |                        | 6 discs                 |                        |
| 7 discs                |                        | 7 discs                 |                        |

### Observations and Records:

1. Find the minimum number of moves required each time the number of discs is increased.
2. Can a formula be prepared showing the relationship between the number of disks 'n' and the moves 't'?
3. Find out more about this game and discuss it in class.
4. Did you like this game?

### 3. Identify lightweight object using pan balance

- You have nine identical looking objects.
- One of them is light in weight.
- How many times will you use pan balance to identify this lightweight object?

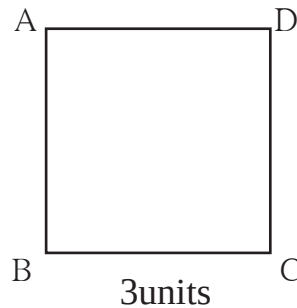
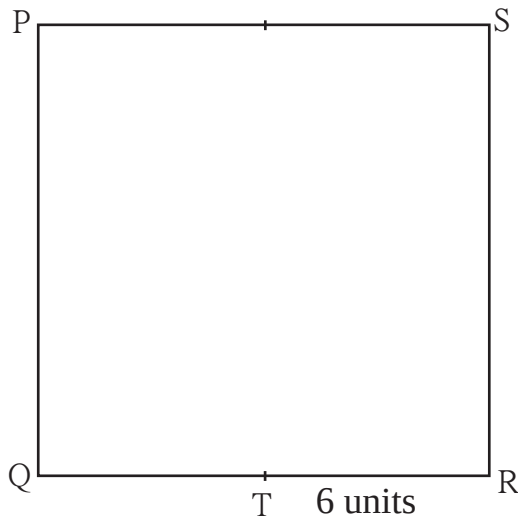


### Solution:

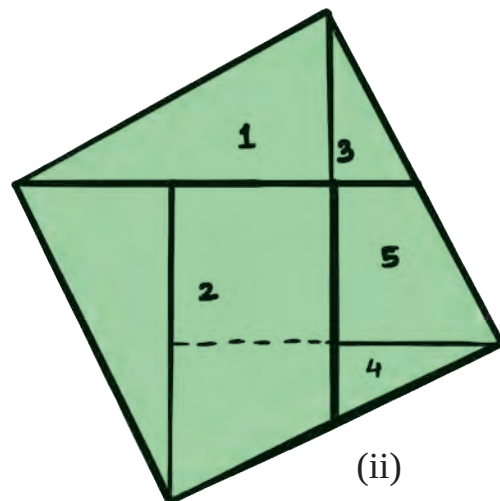
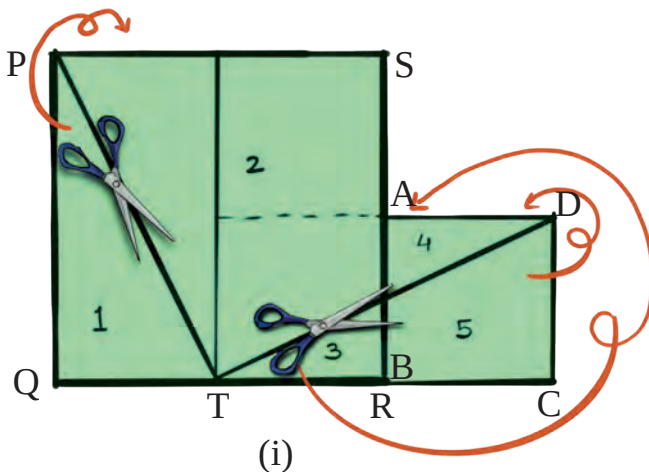
1. Four objects were placed in each of the two pans.
  2. If a pan has gone up, then one of the four objects in that pan will be lighter in weight.
  3. Now, if two objects of these four objects are placed in two pans, and one of the pans goes up, then one of the two objects in that pan will be lighter in weight.
  4. Finally, when these two objects are placed in two pans, the pan with the lower weight will go up. Here, the pan balance is used three times.
- If the pan balance is to be used twice, how will you find the lightweight object? Find the solution and share it with the class.
  - When can a lightweight object be found using the pan balance only once?

(The trick on page 75 of Magical Mathematics 1 : Add up the first number on the cards that contain your friend's number. This sum is the number your friend has in mind.)

#### 4) Sum of areas of squares



1. Cut out a square PQRS with side 6 units and a square ABCD with side 3 units (half of 6). T is the midpoint of QR.
2. Glue both squares together.
3. Draw two line segments TP and TD from T.



4. Cut TP and TD as shown in the figure.
5. Join the cut pieces as shown in figure (ii).
6. A new square has been formed by joining two squares, that is, the sum of the areas of the two squares is equal to the area of the newly formed square. In this way, our ancient mathematicians discovered new theories by playing with figures.

#### Question:

1. Find the relationship between the area and perimeter of two squares and the newly formed square.
2. Side BC is half of side QR. If it is not half, will a new square be formed?

## 1. Three Dimensional Shapes

### Practice set 1.1

1)

| Pyramid/<br>Prism            | Quadrilateral |               | Pentagon |          |
|------------------------------|---------------|---------------|----------|----------|
| Bottom face                  | Quadrilateral | Quadrilateral | Pentagon | Pentagon |
| Number of Vertices           | 8             | 5             | 10       | 6        |
| Number of edges              | 12            | 8             | 15       | 10       |
| Number of vertical faces     | 4             | 4             | 5        | 5        |
| Number of top / bottom faces | 2             | 1             | 2        | 1        |

2) Cylinder, Sphere, Cone, Cuboid

### Exercise - 1

- 1) i. B- Cylinder    ii. D- Cone  
      iii. A- Square
- 2) i. Rectangular Pyramid  
      ii. Sphere  
      ii. Rectangular Prism  
      iv. Cylinder

## 2. Lines and Angles

### Practice set 2.1

- 1) i. Like points D, E, O  
      ii. Like seg. OE  
      iii. ray EB, ray OA  
      iv. Collinear points – D, E, O, B  
      v. Line DE  
      vi. Non-collinear points -A, O, E

### Practice set 2.2

- 1) E, L, H
- 2) i. Acute angle    ii. Zero angle  
      iii. Right angle  
      iv. Obtuse angle  
      v. Straight angle  
      vi. Reflex angle
- 4) i. 2  
      ii. Acute angle, right angle, obtuse angle  
      iii. Zero angle

### Exercise - 2

- 1) i. d -4                      ii. b – acute angle  
      iii. a - O                  iv. a -Line
- 2) i. e                      ii. a  
      iii. b                      iv. c                      v. d
- 3) i. Like ray DA, ray EB  
      ii. Line PQ, Line CF  
      iii. Like ray DE, ray CE  
      iv. ray EB, ray EQ, ray EF, ray EP  
      v. obtuse angle, ray EB and ray EQ

## 3. In the World of Numbers

### Practice set 3.1

- 1) i. Forty nine crore thirty eight lakh sixty seven thousand five hundred twenty one.  
      ii. Twenty eight crore forty one lakh fifty seven thousand six hundred forty nine.





iii. Two crore thirty seven lakh fifty thousand one hundred forty five

iv. Six crore seventy nine lakh eighty one thousand forty nine.

- 2) i.  $<$                       ii.  $>$   
    iii.  $=$                      iv.  $<$
- 3) i. 57,500                ii. 1,38,67,193
- 4) ₹ 2,56,000, ₹ 82000
- 5) ₹ 20,75,003

### Practice set 3.2

- 1) i.  $<$                       ii.  $<$   
    iii.  $<$                      iv.  $<$   
    v.  $>$                       vi.  $>$
- 2) i. 11                        ii. -9  
    iii. -5                    iv. 8  
    v. -95                    vi. 0
- 3) i. 133                      ii. 8  
    iii. 185                   iv. -320  
    v. -9                      vi. 9

### Exercise - 3

- 1) i. c- an integer    ii. b -<
- 2) i. false                ii. true  
    iii. false            iv. false
- 3) 85,93,21,793    87,29,35,104  
    87,39,27,401    89,51,23,307
- 6) ₹ 1,57,64,346, ₹ 19,08,822

### 4 Patterns

#### Practice set 4.1

- 1) i. 1                        ii. 2  
    iii. 2                    iv. 3
- 2) (2,3) (4,5) (6,7) (10,11) (19,20)

- 3) 3, 11, 19, 31, 41, 43
- 4) i. 3, 5 twin prime, difference is 2  
    ii. 7, 9 not twin prime, 9 not a prime number.  
    iii. 11, 13 twin prime, difference is 2  
    iv. 23, 29 not twin prime, difference is 2  
    v. 29, 31 twin prime, difference is 2  
    vi. 41, 47 twin prime, difference is 2.

### Practice set 4.2

- 1) 1, 1, 1                      2) 8, 9, 10
- 3) 11, 13, 15                4) 14, 16, 18
- 5) 28, 36, 45                6) 49, 64, 81
- 7) 216, 343, 512    8) 34, 55, 89
- 9) 128, 256, 512    10) 243, 729, 2187

### Exercise - 4

- 1) 26, 37
- 3) 126, 217, 344
- 4) 1, 3, 6, 10, 15 tenth number 55
- 5) Triangular number series
- 6) 1, 4, 9, 16, Square number series
- 7) 2500

### 5 G.C.D. and L.C.M.

#### Practice set 5.1

- 1) i.  $69 = 2 \times 23$   
    ii.  $27 = 3 \times 3 \times 3$ ,  
    iii.  $78 = 2 \times 3 \times 13$   
    iv.  $126 = 2 \times 3 \times 3 \times 7$



- 2) i. 1, 2, 3, 6, 9, 18  
 ii. 1, 2, 4, 7, 8, 14, 28, 56  
 iii. 1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 20, 24, 30, 40, 60, 120  
 iv. 1, 2, 4, 8, 16, 32, 64, 128, 256,  
 v. 1, 67

### Practice set 5.2

- 1) i. 12      ii. 1      iii. 13  
      iv. 12      v. 6  
 2) 3 m      3) 30 m      4) 12 flowers

### Practice set 5.3

- 1) i. 48                      ii. 120  
      iii. 288                iv. 180  
      v. 180  
 2) i. 120                    ii. 60  
      iii. 50                   iv. 50  
      v. 12 minutes  
      vi. 24  
      vii. 60 minutes (9.00 a.m.)

### Exercise - 5

- 1) i.  $b - 3$                 ii.  $a - 1$   
 2) i. true                  ii. false  
      iii. true                iv. false  
 3) i.  $91 = 7 \times 13$   
      ii.  $160 = 2 \times 2 \times 2 \times 2 \times 2 \times 5$   
      iii. 89  
 4) i. 21      ii. 1      iii. 60  
 5) i. 176      ii. 49      iii. 48  
 6) i. 2      ii. 12      iii. 18 inch  
      iv. 1) 21 fruits      2) 9 boxes

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## 6. Fractions

### Practice set 6.1

- 3) i.  $\frac{10}{10}$       ii.  $\frac{13}{12}$       iii.  $\frac{26}{15}$   
      iv.  $\frac{11}{10}$       v.  $\frac{3}{3}$       vi.  $\frac{6}{35}$   
 4)  $\frac{3}{5}$

### Exercise - 6

- 1) i.  $\frac{7}{7}$       ii.  $\frac{1}{8}$       iii.  $\frac{6}{7}$   
      iv.  $\frac{5}{14}$       v. 0      vi.  $\frac{5}{5}$   
 2) i.  $\frac{1}{3}$       ii.  $\frac{12}{10}$       iii.  $\frac{134}{21}$   
      iv.  $\frac{4}{4}$       v.  $\frac{3}{8}$       vi. 2  
 3)  $\frac{2}{5}$                                       4)  $\frac{1}{20}$   
 5)  $\frac{5}{4}$                                       6) No

## 7. Ratio and Proportion

### Practice set 7.1

- 1) i.  $\frac{9}{22}$                                       ii.  $\frac{4}{1}$   
      iii.  $\frac{7}{5}$                                       iv.  $\frac{5}{6}$   
 2) i.  $\frac{3}{1}$                                       ii.  $\frac{4}{15}$   
      iii.  $\frac{9}{5}$                                       iv.  $\frac{5}{11}$   
 3) i.  $\frac{6}{1}$       ii.  $\frac{1}{2}$       iii.  $\frac{10}{1}$   
      iv.  $\frac{9}{8}$       v.  $\frac{3}{1}$   
 4) a)  $\frac{5}{3}$       b)  $\frac{5}{8}$       c)  $\frac{3}{8}$



**Practice set 7.2**

- 1) ₹ 5695                      2) 420 km

**Exercise - 7**

- 1) i. a                      ii. d                      iii. c  
 2) i.  $\frac{3}{2}$                       ii.  $\frac{4}{9}$                       iii.  $\frac{1}{3}$   
 4)  $\frac{2}{5}$                       5)  $\frac{4}{3}$   
 6) Yashoda nursery  
 7) 8 kg

**8. Tringles and Square****Practice set 8.1**

| Name of Triangle | Vertices         | Sides                     | Angles                                     |
|------------------|------------------|---------------------------|--------------------------------------------|
| $\triangle ABC$  | points – A, B, C | side AB, side BC, side AC | $\angle ABC$ , $\angle BAC$ , $\angle ACB$ |
| $\triangle LMN$  | points – L, M, N | side LM, side MN, side LN | $\angle LMN$ , $\angle MNL$ , $\angle NLM$ |
| $\triangle XYZ$  | points – X, Y, Z | side XY, side YZ, side XZ | $\angle XYZ$ , $\angle YZX$ , $\angle YXZ$ |

- 2)  $\triangle ADE$ ,  $\triangle DEF$ ,  $\triangle DFB$ ,  $\triangle DEB$ ,  $\triangle ABE$ ,  $\triangle FBC$ ,  $\triangle CDE$ ,  $\triangle EFC$ ,  $\triangle CDA$   
 3) i.  $\triangle PQR$   
 ii. point – P, Q, R, F, E  
 iii. point – A, B, C  
 iv. point – G, D  
 v. point – P, Q, R, E, A, B, C, F

**Practice set 8.2**

- 1) i. Equilateral triangle  
 ii. Isosceles triangle  
 iii. Scalene triangle

- 2) i. Acute angled triangle  
 ii. Isosceles Triangle  
 iii. Right angled triangle  
 iv. Obtuse angled triangle  
 v. Equilateral triangle  
 vi. Scalene triangle

**Practice set 8.3**

- 1) i. side WZ  
 ii. side XY and side WZ  
 iii.  $\angle Y$   
 iv.  $\angle Y$  and  $\angle W$   
 v. diagonal XZ and diagonal YW  
 2)  $\angle P$  and  $\angle R$ ;  $\angle Q$  and  $\angle S$

**Exercise - 8**

- 1) i. Equilateral                      ii. Isosceles  
 iii. 90                      iv. diagonal  
 v. MN/ LO  
 2) i. false                      ii. true  
 iii. false  
 3) i.  $\square XWVU$ ,  $\square XUVW$ ,  
 $\square WVUX$ ,  $\square WXUV$ ,  
 $\square VUXW$ ,  $\square VWXU$ ,  
 $\square UXWV$ ,  $\square UVWX$   
 ii. (UV, XW) (XU, WV)  
 iii. UV and UX, VU and VW, WV and WX, XU and XW  
 iv. diagonal WU, diagonal VX  
 v.  $\angle U$  and  $\angle W$ ,  $\angle V$  and  $\angle X$   
 vi.  $\angle U$  and  $\angle X$ ,  $\angle W$  and  $\angle X$ ,  $\angle W$  and  $\angle V$ ,  $\angle U$  and  $\angle V$   
 4) 3, 6, 10, 10                      5) 150



## 9. Equations

### Practice set 9.1

- 2) i. 15      ii. 14      iii. 110  
iv. 6      v. 10

### Practice set 9.2

- 1) i. =      ii.  $\neq$       iii. =  
2) i. 12      ii. 2      iii. 31

### Practice set 9.3

- 1) i.  $x + 9$       ii.  $p \div 8$   
iii.  $n \times 0$       iv.  $a - 12$   
v.  $y \times 4$       vi.  $k \times 1$   
2) i.  $P = S - C$       iii.  $L = C - S$

### Practice set 9.4

- 1) i.  $4a = 24$       ii.  $37 - b = 12$   
iii.  $y - 8 = 15$       iv.  $\frac{p}{7} = 4$   
2) i.  $x = 2$       ii.  $p = 3$   
iii.  $y = 32$       iv.  $m = 3$   
3) i.  $x = 4$       ii.  $x = 17$   
iii.  $p = 9$       iv.  $k = 72$   
5) i. 114      ii. 82

### Exercise - 9

- 1) i. b      ii. a      iii. b  
iv. d      v. c      vi. b  
2) i. 5      ii. 11      iii. 16  
3) i. 15      ii. 8      iii. 4  
4) i. 13      ii. 8      iii. 49  
iv. 13  
5) 6  
6) Asma's age 16 years, Deepali's age 11 years.  
7) 910      8) 111

## 11. Decimal Fractions

### Practice set 11.1

- 1) i. 0.16      ii. 0.275  
iii. 0.06      iv. 2.69  
v. 0.015      vi. 23.5  
vii. 0.007      viii. 6.3  
2) i. 0.2      ii. 0.08  
iii. 0.01      iv. 0.4  
v. 6  
3) i.  $0.2 < 0.4$       ii.  $70.08 < 70.4$   
iii.  $6.6 > 6.58$       iv.  $7.4 > 7.35$   
v.  $0.76 < 0.8$

4)

| Ascending order           | Descending order          |
|---------------------------|---------------------------|
| i. 8.09, 8.59, 8.6, 8.8   | i. 8.8, 8.6, 8.59, 8.09   |
| ii. 7.66, 7.8, 8.06, 8.68 | ii. 8.68, 8.06, 7.8, 7.66 |

### Practice set 11.2

- 1) i. 27.91      ii. 17.09  
iii. 9.36      iv. 21.14  
2) i. 0.52      ii. 1.34  
iii. 3.12      iv. 0.8  
v. 2.3      vi. 3.63  
3) i.  $\frac{9}{10}$       ii.  $\frac{7}{100}$   
iii.  $\frac{191}{1000}$       iv.  $\frac{49}{1000}$   
v.  $\frac{436}{100}$       vi.  $\frac{15}{1000}$   
vii.  $\frac{456}{100}$       viii.  $\frac{584}{100}$



### Exercise - 11

- 1) i.  $\frac{8}{10}$       ii. 15      iii. 7  
iv. 0.8      v. 0.5
- 2) i.  $\frac{1}{5}$       ii.  $\frac{31}{50}$   
iii.  $\frac{121}{200}$       iv.  $\frac{17}{1000}$   
v.  $\frac{293}{50}$
- 3) i. 36.21      ii. 10.211  
iii. 23.03      iv. 16.5  
v. 2.65      vi. 2.11
- 4) 200, 40, 6, 0.1, 0.03, 0.005
- 5) 6.07 km.

### 12. Profit and Loss

#### Practice set 12.1

- 1) i) Profit ₹ 70      ii) Loss ₹ 175  
iii) Loss ₹ 190      iv) Profit ₹ 71
- 2) ₹ 750
- 3) Profit ₹ 96
- 4) i. ₹ 20,000      ii. ₹ 26,500  
iii. ₹ 6500      iv. ₹ 1500  
v. ₹ 5000      vi. ₹ 500

#### Practice set 12.2

- 1) Profit ₹ 49,000
- 2) ₹ 4300
- 3) ₹ 910
- 4) Profit ₹ 520

### Exercise - 12

- 1) i. c      ii. b
- 2) i. 61      ii. 3917      iii. 15
- 3) i. loss ₹ 630      ii. ₹ 1610  
iii. Profit ₹ 826

- 4) Profit ₹ 596
- 5) i. ₹ 88,808      ii. ₹ 1,03,173  
iii. ₹ 14,365

### 13. Percentage

#### Practice set 13.1

- 1) i. 71%      ii. 45%  
iii. 186%      iv. 19%
- 3) i.  $\frac{16}{25}$       ii.  $\frac{91}{100}$   
iii.  $\frac{7}{4}$       iv.  $\frac{1}{6}$   
v. 1
- 4) i. 40%      ii. 37.5%  
iii. 125%      iv. 6.25%  
v. 92%

#### Practice set 13.2

- 1) i. 100      ii. 24.79  
iii. 375      iv. 1044
- 2) i. 40%      ii. 50%  
iii. 25%      iv. 20%  
v. 86%
- 3) i. 96%      ii. 4%
- 4) ₹ 2,47,500
- 5) 14,625 sq.m.
- 6) ₹ 300

### Exercise - 13

- 1) i. b      ii. d
- 2) i. >      ii. =  
iii. =      iv. <
- 3) 25%
- 4) i. 13300      ii. 95%  
iii. 5%
- 5) 32%      6) 7830



## 15. Perimeter and Area

### Practice set 15.1

- (1) 60 m                      (2) 15 cm  
(3) 21 cm                    (4) 4 cm  
(5) 8 m  
(6) a) 9 cm  
      b) The sum of length and breadth will be 18 cm.

(7) 28 cm

### Practice set 15.2

- 1) i) 20 sq.cm.            ii) 14 sq.cm.  
2) i) 22 cm; 24 sq.cm.  
      ii) 11 m; 7 sq.cm.  
      iii) 19 m; 21 sq.cm.  
      iv) 29.6 cm; 50.35 sq.cm.  
3) i) 40 cm; 100 sq.cm.  
      ii) 56 m; 196 sq.cm.  
      iii) 22 m; 30.25 sq.m.  
4) 200 trees

### Exercise - 15

- 1) i) d  
      ii) b  
      iii) c  
2) i) 24 m.  
      ii) 18.5 cm  
3) Rectangle of largest area is (i)  
      Rectangle of smallest area (ii)  
4) 9 sq.cm.  
6) i) Balcony area : 75 sq.cm.  
      Bedroom area : 225 sq.cm.  
      Parking area : 180 sq.cm.  
      Kitchen area : 240 sq.cm.

Garden area : 105 sq.cm.

ii) 1225 sq.cm.

7) i) 840 m

ii) 800 m

iii) Anu

## 16. Baudhayana - Pythagoras Theorem

### Practice set 16.1

- 1) i)  $l(LN)=15$  unit  
      ii)  $l(DF)=17$  unit  
2) 15 meter  
3) 28 ft

### Practice set 16.2

- 1) iii) 15, 20, 25  
2) i) Yes  
      ii) No  
      iii) Yes

### Exercise - 16

- 1) i) c  
      ii) a  
      iii) c  
2) Length of side PR = 50 units.  
3) (6, 8, 10) is a Baudhayana - Pythagoras triplet.  
4) Like (3, 4, 5); (5, 12, 13); (8, 15, 17); (7, 24, 25) etc.

## 17. Bank and Simple Interest

### Practice set 17.1

- 1) 1) ₹ 6180                      2) ₹ 12500  
      3) ₹ 4260  
2) ₹ 8640





## Exercise - 17

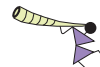
- 1) i) c  
ii) d
- 2) Interest = ₹ 2800
- 3) Interest = ₹ 2902.80
- 4) ₹ 20515
- 5) ₹ 62440
- 6) ₹ 14,300

## 18. Data Representation

## Practice set 18.2

- i) X axis - Types of plants  
Y axis - Number of plants
- ii) Jasmine, Jai, Hibiscus, Chrysanthemum
- iii) Chrysanthemum
- iv) by 40
- v) 240

## Important Notes

[illegible]

Hello Children!

If you feel uneasy about someone touching you inappropriately, you should not keep quiet. You must :

1. Not blame yourself.
2. Tell someone whom you trust.
3. You can also inform National Commission for Protection of Child Rights through the **POCSO e-box**.

When you get an unsafe touch, you may feel bad,  
confused and helpless.

You need not feel “bad” because it’s not your fault.



Press This Button

POCSO e-box available at [ncpcr@gov.in](mailto:ncpcr@gov.in)

If you are below 18 years of age, and are troubled or confused  
or abused or in distress or know some other child who is...

*Call 1098...because some numbers are good!  
They change lives!!!*



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